

# Constructing separator certificates for sparse global optimization

## Abstract

Sparse global optimization can admit small lower certificates even when a single spatial partition would be large. We address the construction of such certificates without knowing a global minimizer. We study certificates formed by convex bag models and affine minorants on separator cells of a tree decomposition. Under global quadratic growth at a unique minimizer, Lipschitz bag gradients, and quadratic local-model error, we give an algorithm that repeatedly rebuilds graded partitions around a point recovered from its current certificate. A bound on aggregate copy disagreement gives contraction on arbitrary branching trees. For fixed structural parameters, conditioning, and numerical scales, the final partition has  $O(N(1 + \log_+(N/\varepsilon)))$  boxes and cells, and construction uses  $O(N(1 + \log_+(N/\varepsilon))^3)$  local convex minimizations, where  $\log_+ z = \max\{0, \log z\}$ . The minimizer may be on the boundary, and no nonconvex local search is required. Certified inexact solves and gradient errors preserve these orders. For rational polynomial objectives of fixed degree, affine Taylor models give an exact rational implementation whose bit cost is polynomial in the input length and requested precision for fixed structural and conditioning parameters. We also characterize separator inconsistency through an exact distance-to-band identity and prove limitations of pointwise localization and unadjusted slopes. For contracting nonlinear dynamics, graph strips, adjoint-adjusted slopes, and forward repair yield analogous guarantees; a finite-precision implementation returns a compressed exactly feasible trajectory and certified objective bounds.

**Keywords:** global optimization; tree decomposition; separator certificates; quadratic growth; rational arithmetic; nonlinear optimal control.

## 1 Introduction

Consider a sum of functions, each depending on a small subset of the variables. A tree decomposition can separate most of the variables into small overlapping bags. It is natural to use this structure in a global optimization proof: bound each bag locally, communicate lower bounds on the overlaps, and combine the bounds along the tree. The difficulty is that the communicated functions are generally nonconvex, nonsmooth, or expensive to represent. Replacing them by constants loses first-order information. Replacing them by affine functions on small cells preserves more information, but raises a different question: how many cells are needed, and how can an algorithm find them?

This paper studies a finite certificate with three ingredients: convex lower models on bag boxes, affine minorants on separator cells, and local inequalities that can be checked independently. A lower certificate and a feasible upper value certify an absolute objective gap. Our main result constructs these objects by *regridding*: at each stage, it discards the previous partitions and builds fresh graded partitions around a consistent point recovered from the previous lower-bound calculation. The new partitions are finer near that point and coarser farther away. Their separator slopes are subtree sums of bag gradients evaluated at the same point.

The assumptions are substantive. The objective has global quadratic growth about a unique minimizer, each bag gradient is Lipschitz, and each local lower model has an error bounded by

a constant times the square of the box width. The structural parameters are the maximum bag dimension and the maximum number of bags containing a coordinate. The constants also depend on the ratios of curvature and model error to quadratic growth. They do not depend on the branching degree or the depth of the decomposition. The algorithm does not know the minimizer and does not use stationarity at it. Boundary minimizers are included.

## 1.1 Contributions and computational guarantees

The principal contribution is a construction bound on arbitrary tree decompositions. A configuration attaining the lower bound may assign different values to copies of the same coordinate. Those disagreements cannot in general be controlled by the largest distance of a bag from the current center. We instead bound their total squared norm. This gives a contraction of the recovered consistent point toward the minimizer and also bounds its objective gap. For a sufficient fixed grading ratio, the number of final boxes and cells is linear in the number  $N$  of bags and logarithmic in accuracy, with an exponential dependence on bag dimension.

The distinction between a certificate and its construction is essential. Let  $p$  be the maximum bag dimension,  $q$  the maximum separator dimension, and  $\theta$  the grading ratio. Let  $J$  denote the number of scales needed to reach the requested accuracy. The explicit definition of  $J$  appears in [theorem 4.5](#); for fixed instance constants,  $J = O(1 + \log_+(N/\varepsilon))$ . The main quantities are

| Quantity                                | Bound for a sufficient fixed $\theta$ |
|-----------------------------------------|---------------------------------------|
| Final boxes and cells                   | $O(N(4/\theta)^p(J+1))$               |
| Boxes and cells created over all stages | $O(N(4/\theta)^p(J+1)^2)$             |
| Local convex minimizations              | $O(N3^q(4/\theta)^p(J+1)^3)$          |

These are counts of specified objects and oracle calls. They do not bound the internal work of an arbitrary convex-minimization oracle. The paper provides two ways to make this distinction operational. First, certified inexact solves are allowed: local errors are summed along a selected configuration rather than over every subproblem. Second, rational polynomial objectives of fixed degree admit affine Taylor lower models, for which the local problems reduce to exact endpoint minimization. This gives explicit bit and output bounds. The resulting dependence on conditioning is numerical, so the theorem is not a polynomial-time claim for arbitrary rational input with unbounded conditioning.

The output is a global lower proof together with a feasible upper value, rather than only an approximate solution. In the polynomial realization, an independent checker can reconstruct every local affine model and verify the gap by rational arithmetic. It does not need to verify the growth constant used to analyze construction time. This separation is useful when a computed optimum must be checked independently, or when a sparse problem must be solved to high precision with an explicit proof-size bound.

A budgeted search removes the need to know the sufficient grading ratio. Every trial remains sound, and a successful sufficient trial gives a bound on the total search work. A trial that stops earlier may use a different ratio and a different scale. Its validity and output budget are guaranteed; the sharper final partition bound above belongs to the sufficient fixed-ratio run. We keep these statements separate.

The supporting theory has two purposes. We first quantify what is lost when different bags use inconsistent copies of a separator coordinate. For one separator, the best gap in a permitted linear class of shifts equals twice the uniform distance from that class to the admissible band between the child value function and the opposite-side lower requirement. Independent cell pieces reduce the gap

to the largest cell bracket. Tree versions give lower and upper bounds in terms of the local deficits. The available conditional margin discounts approximation error away from optimal separator values. This explains why uniform approximation of every conditional value function can be unnecessarily costly. A scalar covering result makes this observation quantitative.

We then extend the construction to scalar contracting nonlinear dynamics. Box partitions are intersected with affine strips containing the local graphs. Forward simulation repairs a relaxed configuration into an exactly feasible trajectory. Ordinary objective-gradient slopes are not enough under coupling constraints; we use slopes adjusted by a backward adjoint recursion. This recursion cancels state derivatives at arbitrary ambient centers, which also permits finite-precision resets. In rational polynomial dynamics, exact expansion of a trajectory can have very large bit length. A recurrence representation, together with certified enclosures and an objective bound, supplies a smaller exactly feasible output.

## 1.2 Relation to prior work

Separator messages, cost shifting, and optimization over a tree are established ideas. In finite-state graphical models, agreement on trees leads to linear-programming and reparameterization interpretations [27]; AND/OR search exploits separator-conditioned subproblems and cached states [18]. Weighted constraint satisfaction combines tree decomposition with local consistency and optimized cost shifts [4, 6]. For continuous pairwise models, Wald and Globerson [28] impose agreement on selected expectations and derive a dual consisting of cost-shifted local minima. Their exactness result for convex-decomposable pairwise models has a different hypothesis from our growth-based construction. The restricted-split duality in [theorem 3.2](#) belongs to this line of work; we give its compact minimax proof for the function classes used here. Unrestricted tree gluing is also an established mechanism in sparse polynomial optimization [17].

Positive-margin splitting under running intersection is central to sparse polynomial representations. Grimm, Netzer, and Schweighofer [10] construct such splits qualitatively; Korda, Magron, and Ríos-Zertuche [15] quantify polynomial approximation of separator functions. Local polynomial shifts also appear in tightness criteria for sparse moment relaxations [23]. Factor-two approximation bounds have precedents in approximate linear programming [5], and moment matching gives a zero-width approximation duality [14]. Approximation of interval-valued bands through oriented distance is studied by Ginchev and Hoffmann [8]. For a real interval band, their objective is  $\sup_s \max\{L(s) - \chi(s), \chi(s) - U(s)\}$ . When the band pinches, this is the uniform distance from  $\chi$  to the band. Thus this approximation objective is established. We do not claim band approximation, qualitative splitting, or the factor-two approximation principle as new. Our band results identify the further equality between the optimized restricted dynamic-programming gap and twice this distance, and distinguish separate-edge approximation from one jointly exact tree split. The graded split and scalar covering proof use the margin left in these bands.

Continuous decomposition methods provide closer algorithmic comparisons. Berenguel et al. [2] study interval branch-and-bound for additively separable functions with common variables. Their shared-variable pruning rule combines interval lower bounds from retained subboxes whose common-variable intervals overlap the current interval [2, Proposition 1 and Theorem 2]. This supplies a direct antecedent for compatible-box decomposition bounds. Our construction communicates affine separator minorants on a tree and bounds the size and work of the resulting global certificate. Bienstock and Muñoz [3] construct bounded-treewidth linear-programming formulations for polynomial optimization, with explicit dependence on inverse accuracy. Zhang and Sun [29] give accuracy-dependent iteration bounds for multistage nonconvex decomposition under their state-covering and oracle assumptions. MUSE-BB [16] uses decomposable scenario subproblems and

multisection branching in one spatial search tree. These methods establish the value of decomposition, but their stated guarantees differ from our final affine-certificate size and construction count under point growth. Robertson, Cheng, and Scott [24] analyze convergence orders of reduced-space value-function relaxations for nonconvex stochastic programs.

Adaptive grids are likewise established. Munos and Moore [21] refine state-space cells and rebuild discretized dynamic programs in optimal control. Nagarajan et al. [22] combine adaptive multivariate partitioning, piecewise relaxations, and bound tightening for global optimization. Our contribution is not adaptivity itself. It is the aggregate disagreement estimate that makes a fresh graded partition contract around a point obtained from an affine separator certificate, on an arbitrary supplied tree. The logarithmic accuracy dependence relies on global quadratic growth and the displayed structural and conditioning parameters. Logarithmic accuracy dependence also occurs in optimistic optimization under suitable local-growth and partition assumptions [20]. That guarantee concerns simple regret. Certified Lipschitz optimization has instance-dependent value-oracle bounds [1]. Our certificate and work counts use different information: bag gradients, local lower models, and local convex minimizations. These oracle models should not be compared as if they supplied the same operations.

Inexact dynamic programming and inexact first-order methods have their own error-propagation theory [7, 19]. Guigues develops deterministic and stochastic inexact decomposition from approximate primal and dual subproblem solves [11, 12]. These related analyses track stage errors and permit increasingly accurate solves; they already establish inexact decomposition as a method. Reliable rational lower bounds from numerical linear programming are developed by Steffy and Wolter [26], and limited-precision LP oracles can be used to recover exact solutions [9]. We use a specific certified contract: an exact local feasible point, a valid lower bound, an error budget proportional to squared cell width, and an aggregate separator-slope error. Its residual identity charges one local error per selected bag. For polynomial box objectives, a separate affine Taylor construction eliminates the general convex oracle: every local affine problem is solved at interval endpoints, and the resulting rational certificate has a compact representation. For fixed degree, bag dimension, occurrence, and numerical conditioning, the bit cost is polynomial in input length and precision. It is not uniformly polynomial in the curvature-to-growth ratio. This realization selects a particular aggregate bag model; it does not implement every possible factorwise relaxation.

Backward adjoints are standard in nonlinear optimal control [13]. The dynamics extension uses them to cancel the first-order objective error of forward feasibility repair in a separator certificate. The cancellation holds at ambient-box centers as well as feasible centers. Contracting graph maps then permit short certified enclosures and repeated center resets. The assumptions include full-box invariance, feasible point growth, and sound derivative bounds; this is a specific constrained extension rather than a theorem for arbitrary nonlinear constraints.

To the best of our knowledge, the results reviewed above do not give the combination proved here: construction of affine separator certificates without a known optimizer on arbitrary tree decompositions, the final logarithmic partition count and explicit incidence-based oracle count under point growth, certified local error budgets for that construction, and the stated rational implementations. The claims concern this combination and these precise contracts. They do not assert novelty of tree dynamic programming, cost shifting, adaptive partitioning, adjoints, or exact rational linear programming.

### 1.3 Organization

Section 2 defines the problem, the certificate, and its dynamic program. Section 3 studies separator consistency and its approximation cost. Section 4 proves the main construction bound; sections 5

and 6 treat certified errors and rational arithmetic. Sections 7 and 8 give the nonlinear dynamics extension and its output representation. Section 9 proves examples that delimit the assumptions. The appendices give complementary covering results with their additional hypotheses stated explicitly.

## 2 Problems and certificates

### 2.1 The problem and its decomposition

Let  $X = \prod_{i=1}^n [L_i, U_i]$  be a compact box and consider

$$f^* = \min_{x \in X} F(x), \quad F(x) = \sum_{t \in T} a_t(x_{V_t}). \quad (1)$$

The set  $T$  is the vertex set of a rooted tree, with root  $r$  and bags  $V_t \subseteq \{1, \dots, n\}$ . The bags cover the coordinates. For each coordinate  $i$ , the bags containing  $i$  induce a connected subtree. This is the running intersection property. A given factorization of  $F$  can be put in this form by assigning each factor to one bag containing its scope. The analysis preserves this assignment; it does not merge bags or factors.

Write  $\pi(t)$  for the parent of  $t \neq r$ ,  $\text{ch}(t)$  for its children, and  $\text{sub}(t)$  for its rooted subtree. Define

$$S_t = V_t \cap V_{\pi(t)}, \quad R_t = V_t \setminus S_t \quad (t \neq r), \quad S_r = \emptyset, \quad R_r = V_r.$$

The relevant structural parameters are

$$N = |T|, \quad p = \max_t |V_t|, \quad q = \max_{t \neq r} |S_t|, \quad k = \max_i |\{t : i \in V_t\}|.$$

We take  $q = 0$  when  $N = 1$ . In particular,  $q \leq p$ ; repeated or contained bags are permitted. The width of this decomposition is  $w = p - 1$ ; the graph's treewidth may be smaller. A bound in terms of  $p$  alone does not control  $k$ .

For a coordinate set  $I$ ,  $X_I$  denotes the projection of  $X$  onto  $I$ . For a box  $B$ ,  $B_I$  has the same meaning, and  $\text{width}(B)$  is its largest side length. Fixed coordinates are substituted into the objective and removed before constructing partitions. All remaining intervals have positive length. A zero-dimensional projection has one point, one cell, and width zero. If all coordinates are fixed, evaluation solves the problem. Below  $n \geq 1$ ,  $p \geq 1$ , and  $s_0 = \max_i (U_i - L_i) > 0$ .

Initially the functions  $a_t$  need only be continuous. Their conditional value functions are

$$\phi_t(s) = \min \left\{ \sum_{u \in \text{sub}(t)} a_u(x_{V_u}) : x \in X_{\cup_{u \in \text{sub}(t)} V_u}, \quad x_{S_t} = s \right\}.$$

Running intersection and the product domain give the recursion

$$\phi_t(s) = \min_{y \in X_{R_t}} \left[ a_t(s, y) + \sum_{u \in \text{ch}(t)} \phi_u((s, y)_{S_u}) \right], \quad \phi_r = f^*. \quad (2)$$

Every minimum is attained. Induction in this recursion also shows that  $\phi_t$  is continuous: a continuous function minimized over a fixed compact box has a continuous minimum. No differentiability of  $\phi_t$  is assumed.

## 2.2 Local models and growth

For every bag box  $B \subseteq X_{V_t}$ , let  $m_{t,B}$  be a supplied continuous convex lower model of  $a_t$  on  $B$ . The quantitative results use the uniform contract

$$0 \leq a_t(v) - m_{t,B}(v) \leq \frac{A}{4} \text{width}(B)^2 \quad (v \in B), \quad (3)$$

with  $A \geq 0$ . This is a bag-level assumption. Factorwise quadratic-error models imply it after summing their error bounds, but vertex exactness is not required. The arithmetic implementation below uses affine Taylor models that satisfy precisely this contract.

We also assume that  $a_t$  is continuously differentiable on a neighborhood of  $X_{V_t}$  and

$$\|\nabla a_t(v) - \nabla a_t(v')\| \leq M \|v - v'\| \quad (v, v' \in X_{V_t}), \quad (4)$$

where  $M \geq 0$  and all unmarked norms are Euclidean. The main convergence assumption is global quadratic growth about one minimizer:

$$F(x) - f^* \geq g \|x - x^*\|^2 \quad (x \in X), \quad g > 0. \quad (5)$$

It implies that  $x^*$  is unique. The algorithm does not need  $x^*$ , and a search variant does not need the numerical value of  $g$ . The minimizer may lie on the boundary of  $X$ . Quadratic growth is an assumption on the whole box; local positive curvature alone would not give the stated global complexity bound without control of other nearly optimal regions. We write  $\log_+ z = \max\{0, \log z\}$  for  $z > 0$ .

## 2.3 A finite lower certificate

**Definition 2.1.** A separator certificate consists of the following objects. For each bag  $t$ , a finite partition  $\mathcal{L}_t$  of  $X_{V_t}$  into closed boxes with disjoint interiors is specified. For each  $t \neq r$ , a similar partition  $\mathcal{P}_t$  of  $X_{S_t}$  is specified, with affine functions

$$l_{t,D}(s) = \lambda_{t,D}^\top s + \beta_{t,D} \quad (D \in \mathcal{P}_t).$$

For each  $B \in \mathcal{L}_t$  and  $u \in \text{ch}(t)$ , a continuous convex function  $\psi_{u,B}$  on  $B_{S_u}$  satisfies

$$\psi_{u,B}(s) \leq l_{u,D'}(s) \quad \text{for every } D' \in \mathcal{P}_u, s \in B_{S_u} \cap D'. \quad (6)$$

With  $\text{Rel}_{t,B}(v) = m_{t,B}(v) + \sum_{u \in \text{ch}(t)} \psi_{u,B}(v_{S_u})$ , the local inequalities are

$$\text{Rel}_{t,B}(v) \geq l_{t,D}(v_{S_t}) \quad (v \in B, v_{S_t} \in D), \quad t \neq r, D \in \mathcal{P}_t. \quad (7)$$

At the root they read  $\text{Rel}_{r,B}(v) \geq \text{LB}$  for every  $B \in \mathcal{L}_r$  and  $v \in B$ . The number of boxes and cells is

$$\mathcal{S} = \sum_t |\mathcal{L}_t| + \sum_{t \neq r} |\mathcal{P}_t|.$$

A certificate together with a feasible value  $\text{UB} \geq f^*$  proves absolute accuracy  $\varepsilon$  if  $\text{UB} - \text{LB} \leq \varepsilon$ .

All nonempty intersections in this definition are included, even if they are faces of smaller dimension. This convention avoids boundary choices in the later configuration identity. The number  $\mathcal{S}$  measures partition size, not the number of local checks or the number of bits. Representations of the models and local proof data must be counted separately. A local check minimizes a continuous convex function over  $B \cap \{v : v_{S_t} \in D\}$ , a compact box that may have fixed coordinates.

**Proposition 2.2** (Validity). *Every certificate satisfies  $l_{t,D}(s) \leq \phi_t(s)$  for  $s \in D$  and  $\text{LB} \leq f^*$ .*

*Proof.* Proceed from the leaves of the tree to the root. Fix  $s \in D$  and choose  $v = (s, y)$  attaining (2). Some  $B \in \mathcal{L}_t$  contains  $v$ . For each child  $u$ , choose a cell  $D' \in \mathcal{P}_u$  containing  $v_{S_u}$ . The induction hypothesis and (6) imply  $\psi_{u,B}(v_{S_u}) \leq \phi_u(v_{S_u})$ . Therefore

$$l_{t,D}(s) \leq \text{Rel}_{t,B}(v) \leq a_t(v) + \sum_u \phi_u(v_{S_u}) = \phi_t(s).$$

The same argument at the root gives  $\text{LB} \leq \phi_r = f^*$ .  $\square$

**Proposition 2.3** (Cancellation along a consistent point). *For any  $x \in X$ , choose a bag leaf  $B_t$  containing  $x_{V_t}$  at every bag. Then*

$$\text{LB} \leq \sum_t m_{t,B_t}(x_{V_t}). \quad (8)$$

*Proof.* Choose any  $D_t \in \mathcal{P}_t$  containing  $x_{S_t}$ . The local inequality at  $t$  and the child inequality at its parent apply to these choices, because both projected boxes contain  $x_{S_t}$ . Substitute the local inequality into the parent inequality, starting at the root. Every nonroot affine minorant cancels once, leaving (8).  $\square$

## 2.4 A dynamic program for fixed slopes

The construction uses one slope  $\lambda_t$  for all cells of a separator. For  $B \in \mathcal{L}_t$ , set

$$\psi_{u,B}(s) = \lambda_u^\top s + \min_{D' \in \mathcal{P}_u: D' \cap B_{S_u} \neq \emptyset} \beta_{u,D'}. \quad (9)$$

This affine function satisfies (6). In postorder, compute the largest permissible intercepts

$$\beta_{t,D} = \min_{B \in \mathcal{L}_t: B_{S_t} \cap D \neq \emptyset} \min_{v \in B: v_{S_t} \in D} \left[ m_{t,B}(v) + \sum_{u \in \text{ch}(t)} \psi_{u,B}(v_{S_u}) - \lambda_t^\top v_{S_t} \right], \quad (10)$$

and at the root compute  $\text{LB} = \min_{B \in \mathcal{L}_r} \min_{v \in B} \text{Rel}_{r,B}(v)$ . These are convex minimization problems. They have minimizers because their domains are nonempty compact boxes and their objectives are continuous. The calculation produces a certificate for any choice of slopes and partitions.

A *configuration* chooses  $B_t \in \mathcal{L}_t$  and  $z^t \in B_t$  for every bag, and  $D_t \in \mathcal{P}_t$  for every nonroot bag, with

$$z_{S_t}^t \in D_t, \quad D_t \cap (B_{\pi(t)})_{S_t} \neq \emptyset.$$

It need not equate the two copies of a separator coordinate.

**Proposition 2.4** (Unfolding the dynamic program). *The fixed-slope program satisfies*

$$\text{LB} = \min_{\text{configurations}} \Phi, \quad \Phi = \sum_t m_{t,B_t}(z^t) + \sum_{t \neq r} \lambda_t^\top (z_{S_t}^{\pi(t)} - z_{S_t}^t). \quad (11)$$

*A minimizing configuration can be recovered from stored minimizing choices and local minimizers.*

*Proof.* In (10), substitute (9). For each child the minimizing cell is constrained only to meet the projection of  $B$ , independently of the parent's point  $v$ . Substituting the child's intercept introduces its own leaf and point, constrained by that cell. Continue down the tree. A sum of independent minima equals the minimum over their joint choices. Each nonroot slope occurs once at the parent point and once with a minus sign at its own point. The resulting expression is exactly (11). All choices are among finitely many compact domains, so the minimum is attained. Storing the choices during the postorder pass and following them from the root recovers a minimizing configuration.  $\square$

## 2.5 Slopes computed at a current center

Let  $\text{top}(i)$  be the bag closest to the root among those containing  $i$ . Running intersection makes it unique. A configuration defines a consistent point by  $x_i = z_i^{\text{top}(i)}$ . This point belongs to  $X$ . For any center  $c \in X$ , define

$$\lambda_{t,i}(c) = \sum_{u \in \text{sub}(t): i \in V_u} \partial_i a_u(c_{V_u}) \quad (i \in S_t). \quad (12)$$

These are subtree sums of ordinary bag gradients, not derivatives of conditional value functions. They can be computed by a postorder pass over the coordinate occurrences.

**Lemma 2.5** (First-order cancellation). *For every center and every configuration,*

$$\sum_t \nabla a_t(c_{V_t})^\top (z^t - x_{V_t}) = \sum_{t \neq r} \lambda_t(c)^\top (z_{S_t}^t - z_{S_t}^{\pi(t)}). \quad (13)$$

Consequently, if  $e_t(v) = a_t(v) - m_{t,B_t}(v)$  and the slopes are (12), then

$$\Phi = F(x) + \sum_t [a_t(z^t) - a_t(x_{V_t}) - \nabla a_t(c_{V_t})^\top (z^t - x_{V_t})] - \sum_t e_t(z^t). \quad (14)$$

*Proof.* Fix  $i$ . On its occurrence subtree,  $z_i^t - x_i$  is the sum of the successive differences  $z_i^u - z_i^{\pi(u)}$  on the path from  $\text{top}(i)$  to  $t$ . Multiply by  $\partial_i a_t(c_{V_t})$  and sum over the occurrences  $t$ . The coefficient of a given edge difference is the sum of these derivatives over the occurrences below that edge, which is  $\lambda_{u,i}(c)$ . Summing over coordinates proves (13). Substitution into (11) gives (14).  $\square$

The cancellation holds at every center in the box. It uses neither stationarity nor knowledge of  $x^*$ . This is the identity that allows a certificate to guide the choice of the next center.

## 3 Separator consistency and restricted splits

An affine separator minorant is a restricted cost shift between two bags. This section identifies the loss caused by restricting those shifts. The bag minima in this section are exact; an error bound for relaxed bags is given at the end. These results explain the consistency part of a certificate. The construction and verification of finite certificates also require the local models and compatibility checks of [definition 2.1](#).

**Definition 3.1** (Separator splits). Let  $N$  be the number of bags. Identify a nonroot bag  $t$  with its edge to its parent  $\pi(t)$ . A split is a family of bounded functions  $\chi_t : X_{S_t} \rightarrow \mathbb{R}$ ,  $t \neq r$ . Its shifted bag functions and bound are

$$a_t^\chi(z) = a_t(z) + \sum_{u \in \text{ch}(t)} \chi_u(z_{S_u}) - \chi_t(z_{S_t}), \quad \rho(\chi) = \sum_t \inf_{z \in X_{V_t}} a_t^\chi(z), \quad (15)$$

where the last term in  $a_t^\chi$  is omitted. For linear spaces  $\Phi_t$  of bounded separator functions containing the constants, put  $\Phi = \prod_{t \neq r} \Phi_t$  and

$$\rho(\Phi) = \sup_{\chi \in \Phi} \rho(\chi), \quad \text{gap}(\Phi) = f^* - \rho(\Phi).$$

Write  $\text{gap}(\chi) = f^* - \rho(\chi)$  for a fixed split. Constants in a split telescope, and  $\sum_t a_t^\chi(x_{V_t}) = F(x)$ . Thus every split gives a valid bound and every gap is nonnegative.

For an edge  $t$ , let  $A_t = \text{sub}(t)$  and let  $B_t$  be the other bags. Define the value functions of the two sides by

$$U_t(s) = \min_{x_{S_t}=s} \sum_{v \in A_t} a_v(x_{V_v}), \quad V_t(s) = \min_{x_{S_t}=s} \sum_{v \in B_t} a_v(x_{V_v}).$$

The child-side function  $U_t$  is the conditional value  $\phi_t$  of [equation \(2\)](#). The minimizations use the variables appearing on the respective side. Running intersection makes their private variables disjoint. Hence

$$\begin{aligned} L_t &= f^* - V_t, & w_t &= U_t - L_t \geq 0, & f^* &= \min_s (U_t(s) + V_t(s)), \\ \mathcal{B}_t &= \{\psi : X_{S_t} \rightarrow \mathbb{R} \text{ bounded} : L_t \leq \psi \leq U_t\}. \end{aligned} \tag{16}$$

The functions  $U_t, V_t$  are continuous: each is a minimum of a continuous function over a fixed compact product of private-variable intervals. The margin  $w_t(s)$  also equals  $\min\{F(x) - f^* : x_{S_t} = s\}$ , and therefore vanishes at every optimal separator value. Denote by  $\mathcal{E}$  the set of exact bounded splits, that is, the splits with  $\rho(\psi) = f^*$ .

The dynamic-programming split  $\psi_t = U_t$  belongs to  $\mathcal{E}$ . Indeed its nonroot shifted bag is  $a_t + \sum_{u \in \text{ch}(t)} U_u - U_t$ , which is nonnegative and has infimum zero by the defining recursion for  $U_t$ . The root shifted bag has infimum  $f^*$ . The same argument works for bounded discontinuous bag functions if minima are replaced by infima. This observation will allow us to absorb bounded splits when considering either side of an edge.

### 3.1 Duality and bag envelopes

The dual interpretation of restricted splits is agreement of neighboring bag measures on a chosen family of test functions. We give the compact form needed here. The proof uses compact minimax [\[25\]](#) and allows compact convex sets of normalized functionals.

**Theorem 3.2** (Restricted split duality). *Suppose  $\Phi_t \subset C(X_{S_t})$  for every edge. For each bag let  $\mathcal{M}_t$  be a nonempty weak-star compact convex subset of  $C(X_{V_t})^*$  whose members satisfy  $\ell(1) = 1$ . Then*

$$\begin{aligned} & \sup_{\chi \in \Phi} \sum_t \min_{\ell \in \mathcal{M}_t} \ell(a_t^\chi) \\ &= \min \left\{ \sum_t \ell_t(a_t) : \ell_t \in \mathcal{M}_t, \quad \ell_t(h) = \ell_{\pi(t)}(h) \text{ for all } h \in \Phi_t, \quad t \neq r \right\}. \end{aligned} \tag{17}$$

*Separator functions in the constraints are pulled back to the two bags. When the constraint set is nonempty, its minimum is attained. When it is empty, both sides have value  $+\infty$ . For  $\mathcal{M}_t$  equal to the probability measures on  $X_{V_t}$ , the left side is  $\rho(\Phi)$ .*

*Proof.* On the compact convex product  $\mathcal{M} = \prod_t \mathcal{M}_t$ , consider

$$Q(\ell, \chi) = \sum_t \ell_t(a_t) + \sum_{t \neq r} (\ell_{\pi(t)}(\chi_t) - \ell_t(\chi_t)).$$

For a fixed split this is continuous affine in  $\ell$ . For fixed  $\ell$  it is continuous linear in the separator functions equipped with their uniform norms. The compact minimax theorem gives  $\sup_\chi \min_\ell Q = \min_\ell \sup_\chi Q$ . The supremum of the mismatch terms is zero if all the stated constraints hold, and is  $+\infty$  otherwise, because the spaces are linear. The consistent families form a closed subset of  $\mathcal{M}$ , proving attainment when they exist. For probability measures, minimizing the expectation of a continuous bag function is the same as minimizing the function itself.  $\square$

Containment of all point evaluations in each  $\mathcal{M}_t$  is sufficient for the functional bound to be a valid relaxation of the bag minima. Normalization alone does not ensure this; the general identity in [theorem 3.2](#) is distinct from validity of a chosen bag relaxation.

If all separator tests are allowed, neighboring separator marginals are equal. Such bag measures have a joint extension on the tree. To see this, sample the root bag from its measure. For each child, sample its remaining variables from a conditional law given the separator. Conditional laws exist on compact metric spaces. Running intersection ensures that those remaining variables have not yet been assigned and that the separator already has the required law. Induction gives every prescribed bag marginal. The expected objective of this joint law is at least  $f^*$ , confirming exactness of unrestricted splitting.

**Proposition 3.3** (Closed bag envelopes). *For a bounded function  $f$  on a compact box, the infimum of  $f$  equals the minimum of its closed convex envelope. Let  $h_t^\chi$  be the closed convex envelope of  $a_t^\chi$  on  $X_{V_t}$ . For any spaces of bounded split functions containing constants,*

$$\sup_{\chi \in \Phi} \min_{x \in X} \sum_t h_t^\chi(x_{V_t}) = \rho(\Phi + \text{Aff}), \quad (18)$$

where  $\text{Aff}$  consists of affine separator functions. In particular,  $\rho(\text{Aff}) = \min_{x \in X} \sum_t \text{vex}_{X_{V_t}} a_t(x_{V_t})$ .

*Proof.* Fix a split and write  $f_t = a_t^\chi$ . The set

$$K_t = \text{cl conv}\{(z, f_t(z)) : z \in X_{V_t}\}$$

is compact. Its lower fiber boundary  $h_t(z) = \min\{v : (z, v) \in K_t\}$  is lower semicontinuous, convex, and below  $f_t$ . Every closed convex minorant of  $f_t$  has a closed convex epigraph containing  $K_t$ , so it is below  $h_t$ . Thus  $h_t = h_t^\chi$ . The infimum of a linear functional over a set equals its infimum over the closed convex hull. Consequently, for every affine  $b$ ,

$$\min_{(z,v) \in K_t} (v + b(z)) = \inf_z (f_t(z) + b(z)).$$

The case  $b = 0$  proves the individual bag statement.

Minimize  $\sum_t v_t$  over  $(z_t, v_t) \in \prod_t K_t$  with equal neighboring separator copies. Running intersection identifies these consistent copies with one global  $x$ , and minimization in the fibers gives  $\min_x \sum_t h_t(x_{V_t})$ . Apply compact minimax to this compact product and unrestricted affine separator multipliers. The supremum over the multipliers enforces equality of the separator copies. The preceding linear-functional identity then gives

$$\min_x \sum_t h_t(x_{V_t}) = \sup_{b \in \text{Aff}} \sum_t \inf a_t^{\chi+b}.$$

Taking the supremum over  $\chi$  proves [equation \(18\)](#). The argument uses boundedness, not continuity, of the chosen split. For the last assertion take  $\Phi$  to consist of constants.  $\square$

**Proposition 3.4** (Attainment and affine exactness). *If every  $\Phi_t$  is a finite-dimensional subspace of  $C(X_{S_t})$ , then  $\rho(\Phi)$  is attained. Consequently,  $\text{gap}(\text{Aff}) = 0$  if and only if an affine split has a global point whose bag projections minimize all its shifted bags. For an exact split, every global minimizer has this property.*

*Proof.* Quotient out constant edge functions, which leave the split objective unchanged. If the quotient is zero-dimensional, attainment is immediate. Otherwise, for a direction  $d$  in the remaining finite-dimensional space, put  $D_t = \sum_{u \in \text{ch}(t)} d_u - d_t$ , omitting  $d_t$  at the root. Since  $\sum_t D_t(x_{V_t}) = 0$ ,

$\sum_t \min D_t \leq 0$ . If equality holds, all the nonnegative differences  $D_t(x_{V_t}) - \min D_t$  vanish for every global  $x$ . Every bag point extends to a point of the box, so every  $D_t$  is constant. Starting at the leaves then shows that every  $d_t$  is constant. Thus the sum is strictly negative for every nonzero quotient direction. It is continuous on the quotient unit sphere, so it is at most  $-\kappa$  there for some  $\kappa > 0$ . For a direction of norm  $R$ ,

$$\rho(d) \leq \sum_t \sup a_t + \sum_t \inf D_t \leq \sum_t \sup a_t - \kappa R.$$

The split objective is continuous, because changing the split uniformly changes every bag infimum by at most the sum of the incident uniform changes. Its upper level sets are therefore compact on the quotient, which proves attainment.

If one point minimizes every shifted bag, the sum of these minima is its objective value. Validity forces that value to be  $f^*$ . Conversely, for an exact split and a global minimizer  $x^*$ , the nonnegative differences  $a_t^\chi(x_{V_t}^*) - \inf a_t^\chi$  sum to zero, so every difference vanishes.  $\square$

### 3.2 One separator and the need for joint control

For bounded functions, distances below use the uniform norm. For a function class  $\Phi$  and a set  $\mathcal{B}$  write  $\text{dist}_\infty(\Phi, \mathcal{B}) = \inf_{\chi \in \Phi, \psi \in \mathcal{B}} \|\chi - \psi\|_\infty$ , and write  $\text{dist}_\infty(f, \Phi) = \inf_{\chi \in \Phi} \|f - \chi\|_\infty$ .

**Theorem 3.5** (The one-separator band identity). *Consider two sides joined by one separator, with value functions  $U, V$  and  $L = f^* - V$ . Let  $\mathcal{B} = \{L \leq \psi \leq U\}$  and let  $\Phi$  be a linear space of bounded separator functions containing constants. Then*

$$\text{gap}(\Phi) = \inf_{\chi \in \Phi} \{\sup(\chi - U) + \sup(L - \chi)\} = 2 \text{dist}_\infty(\Phi, \mathcal{B}). \quad (19)$$

*For continuous  $\Phi, U, L$ , the band in the distance can be restricted to continuous functions. If the separator has a finite single-valued partition  $\mathcal{P}$  and  $\Phi$  allows independent pieces  $\chi_D \in \Phi_D$  on each cell, where every  $\Phi_D$  is a linear space containing constants, then*

$$\text{gap}(\Phi) = \max_{D \in \mathcal{P}} g_D, \quad g_D = \inf_{\chi_D \in \Phi_D} \left\{ \sup_D(\chi_D - U) + \sup_D(L - \chi_D) \right\}. \quad (20)$$

*Individual cell brackets  $g_D$  may be negative.*

*Proof.* The two side infima for a split  $\chi$  are  $\inf(U - \chi) = -\sup(\chi - U)$  and  $\inf(V + \chi) = f^* - \sup(L - \chi)$ , proving the bracket expression. Put  $a = \sup(\chi - U)$  and  $b = \sup(L - \chi)$ . We have  $a + b \geq -\inf(U - L) = 0$ . Shift  $\chi$  by  $(b - a)/2$ , so both suprema become  $(a + b)/2$ . Clipping the shifted function into the band gives  $\psi = \min\{\max\{\chi, L\}, U\}$  and  $\|\chi - \psi\|_\infty \leq (a + b)/2$ . This proves the lower inequality for the distance identity; the clip is continuous when its three input functions are continuous.

Conversely, for  $\psi \in \mathcal{B}$ ,

$$\sup(\chi - U) + \sup(L - \chi) \leq \sup(\chi - \psi) + \sup(\psi - \chi) = \text{osc}(\chi - \psi).$$

For any bounded function  $q$ ,  $\inf_{c \in \mathbb{R}} \|q - c\|_\infty = \text{osc}(q)/2$ . Constants belong to the class, so optimizing gives the other inequality.

A global bracket is at least the bracket of its restriction to each cell. For the reverse inequality, choose a piece on each cell within  $\delta$  of  $g_D$ , and shift that piece by a constant so its two suprema are equal. The resulting global bracket is at most  $\max_D g_D + \delta$ . Let  $\delta$  decrease to zero.  $\square$

Thus a restricted split need not approximate a particular value function: it needs to approximate some function between  $L$  and  $U$ . At separator values where the margin is positive, this band provides room for error. For a zero-width band, the gap is twice the usual uniform approximation error. With piecewise constants, the cell bracket is  $\sup_D L - \inf_D U$ .

**Theorem 3.6** (Tree sandwich). *For the tree and split classes of [definition 3.1](#) with  $N \geq 2$ ,*

$$2 \max_{t \neq r} \text{dist}_\infty(\Phi_t, \mathcal{B}_t) \leq \text{gap}(\Phi) \leq 2 \inf_{\psi \in \mathcal{E}} \sum_{t \neq r} \text{dist}_\infty(\psi_t, \Phi_t) \leq 2 \sum_{t \neq r} \text{dist}_\infty(U_t, \Phi_t). \quad (21)$$

*The infimum in the middle is over one joint exact split.*

*Proof.* For  $\psi \in \mathcal{E}$  write  $m_t = \inf a_t^\psi$ , so  $\sum_t m_t = f^*$ . If  $q_t = \chi_t - \psi_t$  has infimum  $\alpha_t$  and supremum  $\beta_t$ , then

$$\inf a_t^\chi \geq m_t + \sum_{u \in \text{ch}(t)} \alpha_u - \beta_t,$$

with no last term at the root. Summing gives  $\text{gap}(\chi) \leq \sum_{t \neq r} \text{osc}(q_t)$ . Constants in each class allow independent centering, so optimizing the oscillations gives the middle bound. The last bound uses the exact split  $U$ .

For the lower bound fix one edge and enlarge every other class to all bounded functions. For a fixed function on the chosen edge, exact dynamic programming on each side reduces its optimized bound to  $\inf(U_t - \chi_t) + \inf(V_t + \chi_t)$ . This remains valid for the bounded modified bag functions, by the infimum-based recursion given above. Enlarging classes can only increase the bound. Apply [theorem 3.5](#) to this two-side problem and then maximize over the chosen edge.  $\square$

**Example 3.7** (Original bands do not control a joint affine split). On  $[-1, 1]^2$  take a three-bag path, rooted at an endpoint, with functions

$$a_r(s_1) = 2\beta s_1^2, \quad a_m(s_1, s_2) = \kappa(s_1 - s_2)^2 - \beta s_2^2, \quad a_l(s_2) = 2\beta s_2^2,$$

where  $\beta > 0$  and  $\kappa \geq 2\beta$ . Here  $f^* = 0$  and both original bands contain zero, but  $\text{gap}(\text{Aff}) = \beta$ . Indeed the first band's endpoints are

$$U_1(s_1) = \frac{\kappa\beta}{\kappa + \beta} s_1^2, \quad L_1(s_1) = -2\beta s_1^2,$$

and the second has  $U_2(s_2) = 2\beta s_2^2$  and

$$L_2(s_2) = -\left(\frac{2\beta\kappa}{2\beta + \kappa} - \beta\right) s_2^2 \leq 0.$$

These follow by minimizing quadratics whose unconstrained minimizers lie in the box. After removing telescoping constants, an affine split is  $\chi_i = \lambda_i s_i$ . Evaluating the middle shifted bag at  $(s_1, s_2) = (1, 1)$  and  $(-1, -1)$  bounds its infimum by  $-\beta - |\lambda_2 - \lambda_1|$ . The other two infima are at most zero by evaluation at zero. Hence  $\rho(\text{Aff}) \leq -\beta$ . The zero split attains  $-\beta$ , proving the claim. In particular, refinement driven only by the affine brackets of the original edge bands can stop without reaching the desired joint gap.

### 3.3 Graded exact splits and discounted bag errors

A particular joint exact split retains a controlled part of every edge band. This gives a margin-sensitive bound in every separator dimension.

**Theorem 3.8** (Graded exact splits). *Assume  $n_T = N - 1 \geq 1$ . For each nonroot bag  $t$ , let  $b_t$  be the number of edges strictly inside  $\text{sub}(t)$ , and put*

$$\tau = \frac{1}{2n_T}, \quad \theta_t = (2b_t + 1)\tau, \quad \psi_t = U_t - \theta_t w_t. \quad (22)$$

Then  $\psi$  is exact. For all bounded perturbations  $q_t$ ,

$$\text{gap}(\psi + q) \leq \sum_{t \neq r} \{\sup(q_t - \tau w_t) + \sup(-q_t - \tau w_t)\}. \quad (23)$$

Define the narrower band  $\mathcal{C}_t = \{\sigma : |\sigma - \psi_t| \leq \tau w_t\}$ . It lies in  $\mathcal{B}_t$ , and  $\text{gap}(\Phi) \leq 2 \sum_{t \neq r} \text{dist}_\infty(\Phi_t, \mathcal{C}_t)$ . For independent cell classes on partitions  $\mathcal{P}_t$ , with every  $\Phi_{t,D}$  a linear space containing constants, this gives

$$\text{gap}(\Phi) \leq \sum_{t \neq r} \max_{D \in \mathcal{P}_t} g_{t,D}, \quad g_{t,D} = \inf_{\ell \in \Phi_{t,D}} \left\{ \sup_D(\ell - \psi_t - \tau w_t) + \sup_D(\psi_t - \tau w_t - \ell) \right\}. \quad (24)$$

*Proof.* Define the dynamic-programming bag margins

$$g_t = a_t + \sum_{u \in \text{ch}(t)} U_u - U_t \quad (t \neq r), \quad g_r = a_r + \sum_{u \in \text{ch}(r)} U_u - f^*,$$

and the bag-projection margin

$$W_t(z) = \min\{F(x) - f^* : x_{V_t} = z\}. \quad (25)$$

Fixing a bag separates its child subtrees and its parent side. Therefore  $W_t = g_t + w_t$  for  $t \neq r$ , and  $W_r = g_r$ . Furthermore  $W_t(z)$  dominates every incident separator margin evaluated at its corresponding projection, because fixing the bag fixes each such separator.

Write  $\chi_t = U_t - e_t$ , where  $e_t = \theta_t w_t - q_t$ . The gap is the sum of bag deficits

$$D_t = \sup_z \left\{ \sum_{u \in \text{ch}(t)} e_u(z_{S_u}) - e_t(z_{S_t}) - g_t(z) \right\},$$

omitting  $e_t$  at the root. For a nonroot bag the expression inside this supremum is

$$\begin{aligned} & \sum_{u \in \text{ch}(t)} (-q_u - \tau w_u) + (q_t - \tau w_t) \\ & + \sum_{u \in \text{ch}(t)} (\theta_u + \tau) w_u + (1 - \theta_t + \tau) w_t - W_t. \end{aligned}$$

All coefficients of margins in the second line are nonnegative. The identity  $b_t = \sum_{u \in \text{ch}(t)} (b_u + 1)$  implies  $\theta_t = \sum_u (\theta_u + \tau) + \tau$ , so their sum is one. Since each margin is at most  $W_t$ , that line is nonpositive. At the root,  $\sum_u (\theta_u + \tau) = 1$ , giving the same conclusion with no own-separator term. Bound the remaining terms by separate suprema and sum over the bags. Each edge appears twice, proving [equation \(23\)](#).

Taking  $q = 0$  makes the right side zero because every margin vanishes at an optimal separator value. Thus  $\psi$  is exact. We have  $\tau \leq \theta_t \leq 1 - \tau$ , so  $\mathcal{C}_t \subset \mathcal{B}_t$ . For  $\sigma_t \in \mathcal{C}_t$  and  $\chi_t \in \Phi_t$ , the edge bracket in [equation \(23\)](#) is at most  $\text{osc}(\chi_t - \sigma_t)$ . Centering by constants and optimizing gives the distance bound. On a finite cell partition, choose near-minimizing pieces and balance their two suprema independently, as in [theorem 3.5](#); the edge bracket is then at most its largest cell bracket. This proves [equation \(24\)](#).  $\square$

For one edge,  $\theta = \tau = 1/2$  and  $\mathcal{C}$  is the full band, so [equation \(23\)](#) is the exact one-edge bracket. For many edges, the grading provides room on both sides of every split. The scalar construction in [appendix A](#) uses this room to bound separator cell counts by coverings of projected near-optimal sets. It assumes exact bag minima and scalar separators; it does not give a general bound on the number of certificate leaves.

**Proposition 3.9** (Discounted bag errors). *Let  $n_T = N - 1 \geq 1$ , and let  $b_t$  be the number of edges strictly inside  $\text{sub}(t)$  as in [theorem 3.8](#). Let  $\tilde{a}_t = a_t - \text{err}_t$ , where  $\text{err}_t : X_{V_t} \rightarrow \mathbb{R}$  is bounded and nonnegative, and let  $\tilde{\rho}$  be its split bound with exact bag infima. Using the original margins  $w_t, W_t$ , put*

$$\tau' = \frac{1}{3n_T + 1}, \quad \theta'_t = (3b_t + 2)\tau', \quad \psi'_t = U_t - \theta'_t w_t.$$

Then for all bounded perturbations  $q$ ,

$$f^* - \tilde{\rho}(\psi' + q) \leq \sum_{t \neq r} \{ \sup(q_t - \tau' w_t) + \sup(-q_t - \tau' w_t) \} + \sum_t \sup_{z \in X_{V_t}} (\text{err}_t(z) - \tau' W_t(z)). \quad (26)$$

*Proof.* The bag deficit in the preceding proof gains  $\text{err}_t$ . Reserve the term  $\text{err}_t - \tau' W_t$  and repeat the decomposition with  $\tau', \theta'_t$ . For a nonroot bag, the remaining margin coefficients are  $\theta'_u + \tau'$  for its children and  $1 - \theta'_t + \tau'$  for its own separator. They are nonnegative and their sum is  $1 - \tau'$ , because  $\theta'_t = \sum_u (\theta'_u + \tau') + 2\tau'$ . They multiply margins bounded by  $W_t$ , and are accompanied by  $-(1 - \tau')W_t$ . Their total is therefore nonpositive. At the root  $\sum_u (\theta'_u + \tau') = 3n_T \tau' = 1 - \tau'$ , which gives the same inequality. Sum the separate suprema as before.  $\square$

*Remark 3.10* (Computational scope). The value functions and projection margins are global quantities. The preceding approximation bounds do not supply inexpensive oracles for them. Independent separator pieces also require evaluating a bag on the common refinement of its incident partitions. The product of the incident cell counts is an upper bound on the number of such regions; overlapping separator coordinates can reduce it. Consequently a small number of separator coefficients alone is not a bound on computational work. The following sections construct and verify finite affine certificates directly, with their stated local oracles and error budgets.

## 4 Constructing certificates by regriding

We construct a certificate without knowing the minimizer. At each stage we build new partitions around the point obtained at the preceding stage and use subtree slopes evaluated at that same point. The partitions need not refine one another. The proof controls the sum of squared copy errors; it does not require every reconstructed coordinate to lie within the current mesh width of the minimizer.

Throughout this section, the models satisfy [\(3\)](#), the bag gradients are  $M$ -Lipschitz in the Euclidean norm, and [\(5\)](#) holds with  $g > 0$ . Models are continuous and convex on their closed boxes. An exact local optimization oracle returns both the minimum value and an attaining point

on each nonempty box intersection. Bag values and gradients are also available exactly. These assumptions specify an oracle model; the number of oracle calls does not bound their internal cost. Fixed coordinates of the original domain have been removed, as in the model section.

#### 4.1 Graded shell partitions

For a point  $c \in \mathbb{R}^d$  and radius  $h > 0$ , put  $Q_\infty(c, h) = \{v : \|v - c\|_\infty \leq h\}$ . Distances from a set are infimum distances, and widths are the largest coordinate side length.

**Lemma 4.1** (Shell construction). *Let  $X'$  be a nondegenerate box in  $\mathbb{R}^d$  whose sides are at most  $s_0$ , let  $c \in X'$ ,  $h > 0$ , and let  $\theta = 2^{-\mu}$  with an integer  $\mu \geq 1$ . Set  $H = \max\{0, \lceil \log_2(s_0/h) \rceil\}$ . There is a finite closed-box partition  $\Pi(c; h, \theta)$  of  $X'$  with disjoint interiors, at most  $(H + 1)(4/\theta)^d$  members, and*

$$\text{width}(Y) \leq \max\{h, \theta \text{dist}_\infty(Y, c)\} \leq h + \theta \text{dist}_\infty(Y, c) \quad (Y \in \Pi(c; h, \theta)). \quad (27)$$

*Each member retains a shell-level index between 0 and  $H$  and is a clipped grid cube. The level-0 grid has at most two intervals in each coordinate. Each other level has at most  $4/\theta$  intervals in each coordinate. In dimension zero use the single cell, of width zero.*

*Proof.* Cut  $Q_\infty(c, h)$  at the coordinate hyperplanes through  $c$  into its  $2^d$  cubes of side  $h$ ; these form level 0. For  $\ell = 1, \dots, H$ , divide  $Q_\infty(c, 2^\ell h)$  into a grid of side  $g_\ell = \theta 2^{\ell-1} h$ , anchored at its lower corner. There are  $4/\theta$  positions per coordinate. Because  $1/\theta$  is an integer, the boundary of  $Q_\infty(c, 2^{\ell-1} h)$  lies on grid hyperplanes. Retain the cubes that are not contained in that inner cube. Their interiors are outside the inner cube, so their distance from  $c$  is at least  $2^{\ell-1} h$ , and their width is  $g_\ell \leq \theta \text{dist}_\infty(Y, c)$ .

The levels tile the outer cube with disjoint interiors. It contains  $X'$  because  $c \in X'$  and  $2^H h \geq s_0$ . Intersect all cubes with  $X'$  and discard intersections having empty interior in  $\mathbb{R}^d$ . Clipping decreases widths and increases distances. It preserves coverage: every point of  $X'$  is a limit of interior points of  $X'$  off the finitely many grid hyperplanes, and one retained closed intersection contains a subsequence and hence its limit. Thus the discarded pieces do not remove a boundary point from the union. Each level contributes at most  $(4/\theta)^d$  members, including the central level, proving the claim.  $\square$

The fixed-coordinate reduction is necessary here. If  $X'$  had a fixed coordinate, every clipped cube would have empty ambient interior. One can instead work in its affine hull; removing fixed coordinates gives the same construction with a simpler dimension convention.

#### 4.2 The algorithm and its constants

Write

$$\begin{aligned} C_0 &= k(k-1)p, & B_0 &= \frac{MC_0}{2} + \frac{A}{4}, & \eta &= \frac{g}{20}, \\ C &= 12B_0 + \frac{6kC_0M^2}{\eta}, & B &= \max\left\{p, \frac{2C}{g}\right\}. \end{aligned} \quad (28)$$

Choose a dyadic  $\theta = 2^{-\mu} \leq 1/2$  satisfying

$$12C_0\theta^2 \leq 1, \quad \sqrt{12}k\sqrt{C_0}M\theta \leq \frac{\eta}{4}, \quad 12kB_0\theta^2 \leq \frac{\eta}{4}. \quad (29)$$

A restriction with a zero coefficient imposes no condition. In particular, these formulas allow  $k = 1$ ,  $M = 0$ , and  $A = 0$ .

Start from any  $x^{(-1)} \in X$ , with incumbent value  $\text{UB} = F(x^{(-1)})$ . At stage  $j = 0, 1, \dots$ , set  $h_j = s_0 2^{-j}$  and  $c = x^{(j-1)}$ , and perform the following steps.

1. Build the fresh partitions  $\mathcal{L}_t = \Pi(c_{V_t}; h_j, \theta)$  and, for  $t \neq r$ ,  $\mathcal{P}_t = \Pi(c_{S_t}; h_j, \theta)$ .
2. Use the common slopes  $\lambda_t(c)$  from (12). Compute the maximal intercepts and root bound  $\text{LB}_j$  by the dynamic program of proposition 2.4. Store the minimizing local points and child-cell choices.
3. Backtrack a minimizing configuration and set  $x_i^{(j)} = z_i^{\text{top}(i)}$ . Update the incumbent if  $F(x^{(j)}) < \text{UB}$ , retaining this feasible point and its value.
4. Stop if  $\text{UB} - \text{LB}_j \leq \varepsilon$ , returning the current certificate and the retained incumbent.

The next center is always the newly reconstructed point, even when an earlier point remains the incumbent. All local subproblems have a nonempty compact domain and continuous convex objective, so their minimum is attained. Finite minima and backtracking therefore exist. Ties can be resolved arbitrarily.

Certificate validity holds for every dyadic  $\theta$ , whether or not (29) holds. Those conditions prove the construction and size bounds, whereas the stopping test uses only valid lower and feasible upper bounds.

### 4.3 Aggregate copy drift

Fix a stage and an arbitrary configuration. Let  $x$  be its consistent point, and define

$$\begin{aligned} b_t &= \text{width}(B_t), & d_t &= \text{width}(D_t) \quad (t \neq r), \\ Q &= \sum_t b_t^2 + \sum_{t \neq r} d_t^2, & E &= \sum_t \|z^t - x_{V_t}\|_2^2, & R &= \|x - c\|_2. \end{aligned} \quad (30)$$

Here  $d_t$  is a width, not a dimension.

**Lemma 4.2** (Occurrence-based drift). *Every configuration satisfies  $E \leq C_0 Q$ .*

*Proof.* For each variable  $i$ , its occurrence bags form a rooted subtree  $T_i$  of size  $m_i \leq k$ . Its root is  $\text{top}(i)$ . On an edge of this subtree, the child copy belongs to  $D_t$  and the parent copy belongs to  $(B_{\pi(t)})_{S_t}$ . These boxes meet. Joining the two copies through an intersection point gives

$$\left| z_i^t - z_i^{\pi(t)} \right| \leq b_{\pi(t)} + d_t.$$

The parent copy need not itself belong to  $D_t$ .

Make a matrix whose rows are the occurrences in  $T_i$  and whose columns are the bag and separator widths on  $T_i$ . The row for  $t$  has a 1 at the parent-bag width and child-cell width for every edge on the path from  $\text{top}(i)$  to  $t$ , and zero elsewhere. It maps these nonnegative widths to upper bounds on  $|z_i^t - x_i|$ . Each row has  $2 \text{depth}_i(t)$  ones. Its squared operator norm is at most its squared Frobenius norm, which is

$$2 \sum_{t \in T_i} \text{depth}_i(t) \leq m_i(m_i - 1) \leq k(k - 1).$$

To see the depth bound, order the vertices with parents preceding children; the depth of the vertex in position  $a$  is at most  $a - 1$ . Sum the resulting coordinate inequalities. A bag-width square is counted at most  $|V_t|$  times, and a separator-width square at most  $|S_t|$  times. Both dimensions are at most  $p$ , so

$$E \leq k(k - 1) \left( \sum_t |V_t| b_t^2 + \sum_{t \neq r} |S_t| d_t^2 \right) \leq k(k - 1)pQ.$$

No tree depth or degree is used. □

**Lemma 4.3** (Grading absorption). *If the first condition in (29) holds, then*

$$Q \leq 12Nh_j^2 + 12k\theta^2 R^2. \quad (31)$$

*Proof.* Apply (27) to a bag leaf containing  $z^t$ :

$$b_t \leq h_j + \theta \|x_{V_t} - c_{V_t}\|_2 + \theta \|z^t - x_{V_t}\|_2.$$

The same inequality holds for  $d_t$  with the vectors restricted to  $S_t$ . A variable with  $m_i$  occurrences belongs to exactly  $m_i - 1$  separators. Thus the sum of the bag and separator squared center differences is at most  $(2k - 1)R^2$ .

The copy error vanishes on  $V_t \setminus S_t$ : every such coordinate has its top occurrence at  $t$ . Consequently its bag-plus-separator sum of squares is exactly  $2E$ . There are at most  $2N - 1$  boxes. Using  $(u + v + w)^2 \leq 3(u^2 + v^2 + w^2)$  gives

$$Q \leq 6Nh_j^2 + 3(2k - 1)\theta^2 R^2 + 6\theta^2 E.$$

By lemma 4.2, the last term is at most  $6\theta^2 C_0 Q \leq Q/2$ . Absorb it and use  $2k - 1 \leq 2k$ . When  $C_0 = 0$ , the last term is zero and the same bound holds directly.  $\square$

#### 4.4 The error relative to the current center

**Lemma 4.4** (Uniform configuration error). *Under (29), every configuration, with slopes  $\lambda_t(c)$ , satisfies*

$$|F(x) - \Phi| \leq \eta \|x - c\|_2^2 + CNh_j^2. \quad (32)$$

*Proof.* Let  $\text{err}_t = a_t(z^t) - m_{t,B_t}(z^t)$ , so  $0 \leq \text{err}_t \leq Ab_t^2/4$ . The telescoping identity of lemma 2.5, applied at  $c$ , gives

$$\Phi = F(x) + \sum_t T_t - \sum_t \text{err}_t, \quad T_t = a_t(z^t) - a_t(x_{V_t}) - \langle \nabla a_t(c_{V_t}), z^t - x_{V_t} \rangle.$$

The Lipschitz-gradient Taylor bound and Cauchy–Schwarz imply

$$|T_t| \leq M \|x_{V_t} - c_{V_t}\|_2 \|z^t - x_{V_t}\|_2 + \frac{M}{2} \|z^t - x_{V_t}\|_2^2.$$

Since each variable occurs in at most  $k$  bags,

$$\begin{aligned} |F(x) - \Phi| &\leq M\sqrt{k} R\sqrt{E} + \frac{M}{2} E + \frac{A}{4} Q \\ &\leq M\sqrt{kC_0} R\sqrt{Q} + B_0 Q. \end{aligned}$$

Substitute (31) and use  $\sqrt{u + v} \leq \sqrt{u} + \sqrt{v}$  to obtain

$$\begin{aligned} |F(x) - \Phi| &\leq \sqrt{12kC_0} M\sqrt{N} h_j R + 12B_0 N h_j^2 \\ &\quad + \left( \sqrt{12} k \sqrt{C_0} M \theta + 12k B_0 \theta^2 \right) R^2. \end{aligned}$$

Young's inequality bounds the first term by  $\eta R^2/2 + 6kC_0 M^2 N h_j^2/\eta$ . The last two conditions in (29) bound the coefficient in parentheses by  $\eta/2$ . This proves the claim.  $\square$

The gradient at  $c$  need not vanish. Its linear copy terms cancel algebraically. This is why the estimate applies to boundary minimizers without a stationarity hypothesis or differentiability of subtree value functions.

## 4.5 Contraction, termination, and box counts

**Theorem 4.5** (Constructive regidding). *For every  $\varepsilon > 0$ , the algorithm with a grading ratio satisfying (29) returns a valid certificate and feasible incumbent with  $\text{UB} - \text{LB} \leq \varepsilon$ . With the constants in (28), put*

$$J = \max \left\{ 0, \left\lceil \log_2 \left( s_0 \sqrt{gBN/\varepsilon} \right) \right\rceil \right\}. \quad (33)$$

*The algorithm stops by stage  $J$ , and each completed stage satisfies*

$$\|x^{(j)} - x^*\|_2^2 \leq BNh_j^2, \quad 0 \leq \text{UB} - \text{LB}_j \leq gBNh_j^2. \quad (34)$$

*Writing  $m = 4/\theta$ , its final number of leaves plus cells is at most  $2Nm^p(J+1)$ , and the total number created through stopping is at most  $Nm^p(J+1)(J+2)$ .*

*Proof.* A consistent configuration at  $x^*$  exists: choose leaves and cells containing its restrictions. It has value at most  $f^*$  because the models are lower bounds and all copy differences vanish. By [proposition 2.4](#), a minimizing configuration has  $\Phi = \text{LB}_j \leq f^*$ . Put  $e_j = \|x^{(j)} - x^*\|_2^2$ . Quadratic growth and (32) give

$$ge_j \leq F(x^{(j)}) - f^* \leq F(x^{(j)}) - \text{LB}_j \leq \frac{g}{20} \|x^{(j)} - x^{(j-1)}\|_2^2 + CNh_j^2.$$

The squared triangle inequality implies

$$e_j \leq \frac{e_{j-1}}{9} + \frac{10C}{9g} Nh_j^2. \quad (35)$$

For  $j \geq 1$ , the inductive bound  $e_{j-1} \leq BNh_{j-1}^2$  and  $h_{j-1} = 2h_j$  give  $e_j \leq (4B/9 + 10C/(9g))Nh_j^2 \leq BNh_j^2$ , since  $B \geq 2C/g$ . At stage zero, coverage of the variables gives  $n \leq pN$ , and  $e_{-1} \leq ns_0^2 \leq BNh_0^2$ . Thus the base coefficient is at most  $B/9 + 10C/(9g) \leq 2B/3$ , which also proves the claim.

For  $j \geq 1$ , the squared center difference is at most  $2(e_j + e_{j-1}) \leq 10BNh_j^2$ . At stage zero it is at most  $4BNh_0^2$ . The incumbent includes the current point, so

$$0 \leq \text{UB} - \text{LB}_j \leq F(x^{(j)}) - \text{LB}_j \leq (gB/2 + C)Nh_j^2 \leq gBNh_j^2.$$

No monotonicity of the lower bounds was used. The definition of  $J$  ensures  $gBNh_J^2 \leq \varepsilon$ , proving termination and the gap guarantee.

At stage  $j$ , the shell level count is  $j+1$ , because  $h_j = s_0 2^{-j}$ . [Lemma 4.1](#) gives at most  $2Nm^p(j+1)$  leaves and cells, including zero-dimensional cells. Summing this bound over  $j = 0, \dots, J$  proves the cumulative count. Earlier partitions can be discarded after retaining the new center and the incumbent.  $\square$

For the largest admissible dyadic ratio, with  $\kappa = M/g$  and  $a = A/g$ , direct solution of (29) gives

$$\frac{1}{\theta} = O \left( 1 + \sqrt{C_0} + k\sqrt{C_0}\kappa + \sqrt{k(C_0\kappa + a)} \right). \quad (36)$$

Also  $C/g = 6C_0\kappa + 3a + 120kC_0\kappa^2$ . These are numerical conditioning factors. Bounded width does not imply bounded occurrence: in an edge-bag decomposition of a star, the central variable occurs in all bags. The theorem removes a separate degree parameter while retaining its stated dependence on  $k$ .

## 4.6 Touching incidences and local work

**Proposition 4.6** (Shell-incidence count). *Let  $q = \max_{t \neq r} |S_t|$ , with  $q = 0$  when there are no nonroot bags. At stage  $j$ , both the own-separator incidences and the parent-leaf/child-cell incidences can be processed with at most  $N3^q m^p (j+1)^2$  incidences of each type. The exact local optimization-call count through stage  $J$  is at most*

$$N3^q m^p \frac{(J+1)(J+2)(2J+3)}{6}. \quad (37)$$

*With bag evaluations and local optimizations counted as oracle calls, coordinate processing and affine arithmetic require  $O(pN3^q m^p (J+1)^3)$  counted operations through stage  $J$ .*

*Proof.* Consider a bag partition in dimension  $d \leq p$  and a separator partition in dimension  $q' \leq q$ , projected into that bag. For one pair of levels, each partition is a subset of an unclipped grid with at most  $m$  positions per coordinate. Compare the unclipped side lengths. If the bag grid is finer, an interval of its side length meets at most three intervals of the separator grid, including endpoint contacts. Each of at most  $m^d$  bag cubes therefore meets at most  $3^{q'}$  separator cubes.

If the separator grid is finer, each separator cube meets at most  $3^{q'}$  projected bag-grid positions. Each such position has at most  $m^{d-q'}$  private-coordinate fibers, and there are at most  $m^{q'}$  separator cubes. This again gives at most  $3^{q'} m^d$  incidences. Clipping cannot create an intersection between previously disjoint cubes. The central grid satisfies the same position bound. Dimension-zero separators have one cell and satisfy the count directly.

There are  $(j+1)^2$  level pairs. Summing the own-separator incidences over nonroot bags and including one minimum per root leaf gives at most  $N3^q m^p (j+1)^2$  local programs. Summing the parent-leaf/child-cell incidences over the  $N-1$  tree edges gives at most  $(N-1)3^q m^p (j+1)^2$  child incidences. These determine the minimum compatible child intercept for each parent leaf. Child choices are separate finite minima, so there is no product over the children.

For completeness, incidence enumeration need not use a dense all-pairs table. In the first case, enumerate the finer bag grid and compute, in each separator coordinate, the at most three coarse intervals that meet it. In the second case, enumerate separator cubes, their at most three coarse positions per coordinate, and the private bag fibers. Filter the resulting pairs by the exact shell membership and clipped-box tests. The enumeration costs  $O(p)$  per candidate; its candidate count has the same bound just proved. It can be streamed while updating minima.

Slopes are obtained in  $O(Np)$  arithmetic per stage by a bottom-up accumulation of bag gradients. For each leaf, the sum of child slopes is an affine coefficient vector; the corresponding child-intercept sum can be accumulated from the incidence minima. The total number of parent-leaf/child combinations is at most  $(N-1)m^p(j+1)$ , using the uniform leaf bound. Thus this additional work is covered by the stated count. Finally,  $\sum_{j=0}^J (j+1)^2 = (J+1)(J+2)(2J+3)/6$ , proving (37) and the cumulative work bound.  $\square$

The actual separator dimension  $q$  can equal  $p$  when adjacent bags coincide. If no bag is contained in its parent, then  $q \leq p-1$ . We retain the supplied decomposition and assignment; merging bags could change the bag-gradient and model-error constants.

Counting bag evaluations as oracle calls is material. The number of factors or terms assigned to a bag is not bounded by  $N$  and  $p$ . Their representation, model-generation, evaluation, and numerical certification costs must enter an implementation bound. Moreover, the leaf-plus-cell measure does not include separate proof objects for every local incidence or the bit lengths of numerical entries. The following arithmetic section supplies such bounds for one explicit representation.

## 4.7 Searching without growth or error constants

**Proposition 4.7** (Budgeted parameter search). *Suppose all requested oracle calls terminate. An algorithm that does not use  $g$ ,  $M$ , or  $A$  to select its grading ratio can still return a valid certificate and incumbent of gap at most  $\varepsilon$ . Fix any sufficient index  $\mu_*$  and a bound  $W \geq \max\{2, 2^{\mu_*}\}$  on the counted work of its run through stopping. Then a simple search uses  $O(W \log(2 + W))$  counted operations before returning. Oracle-internal costs remain excluded.*

*Proof.* For rounds  $r = 1, 2, \dots$ , start a fresh run for each  $\mu = 1, \dots, r$  from the same initial point. Give each run a budget of  $2^r$  counted operations, covering box construction, incidence enumeration, scalar comparisons and arithmetic, bag-value and gradient evaluations, and local optimization calls. Enforce the budget before each counted operation. There is no separate stage cap. Stop at the first completed stage that passes its own gap test.

For every trial, validity supplies  $\text{LB} \leq f^*$  and the retained feasible incumbent supplies  $f^* \leq \text{UB}$ . A successful test therefore proves the required guarantee independently of the trial's grading ratio. A sufficiently large  $\mu_*$  satisfies (29); by [theorem 4.5](#) its run terminates in finite counted work. At round  $R = \lceil \log_2 W \rceil$ , one has  $R \geq \mu_*$  and  $2^R \geq W$ , so that round permits the entire sufficient run. The total trial budgets through round  $R$  are

$$\sum_{r=1}^R r 2^r = (R-1)2^{R+1} + 2 \leq 2R2^R = O(W \log(2 + W)).$$

Every round makes finite progress because it contains finitely many budgeted operations and each requested call terminates.  $\square$

One can take  $W$  to be a constant multiple of the work bound in [proposition 4.6](#) for the sufficient run; its factor  $m^p$  ensures  $W \geq 2^{\mu_*}$ . The returned trial may have a different grading ratio and stopping stage. Its gap guarantee is unchanged, and its actual partitions satisfy the shell count for those returned parameters. The sharper final-box bound in [theorem 4.5](#) refers to a run with a fixed sufficient ratio; parameter search does not identify its returned trial with that run. The returned certificate has at most  $O(W)$  boxes under the work-budget convention, since their emission is charged in its last trial budget. Total work and final output are distinct measures.

## 5 Certified inexact oracles

The regridding argument survives certified errors in local optimization, separator slopes, and incumbent evaluation. Local errors accumulate along the selected configuration, once per bag. They do not accumulate over all the convex problems solved to construct the certificate. Throughout this section, the models satisfy (3), the bag gradients are  $M$ -Lipschitz, and quadratic growth holds with constant  $g > 0$  as in (5). We use  $C_0, B_0, \eta, C$  from (28), with the model error constant denoted by  $A$ .

### 5.1 Certified local problems and selected residuals

Fix one stage and one exactly represented slope  $\hat{\lambda}_t$  per nonroot separator. The partitions are fixed during the bottom-up pass. Once the child intercepts  $\hat{\beta}_{u,D'}$  are available, put

$$b_{u,B} = \min\{\hat{\beta}_{u,D'} : D' \in \mathcal{P}_u, D' \cap B_{S_u} \neq \emptyset\}$$

for each parent leaf  $B$ , and store a minimizing child cell. For a nonroot bag, each touching pair  $(B, D)$  has the nonempty compact domain  $C_{t,B,D} = \{v \in B : v_{S_t} \in D\}$ . Its local objective is

$$q_{t,B,D}(v) = m_{t,B}(v) + \sum_{u \in \text{ch}(t)} [\hat{\lambda}_u^T v_{S_u} + b_{u,B}] - \hat{\lambda}_t^T v_{S_t}. \quad (38)$$

At the root the domain is  $B$  and the final slope term is absent. A certified local oracle returns lower and upper numbers  $\ell_{t,B,D}$ ,  $u_{t,B,D}$  and a point  $v_{t,B,D}$  satisfying

$$v_{t,B,D} \in C_{t,B,D}, \quad \ell_{t,B,D} \leq \min_{C_{t,B,D}} q_{t,B,D}, \quad q_{t,B,D}(v_{t,B,D}) \leq u_{t,B,D} \leq \ell_{t,B,D} + \delta_{t,j}. \quad (39)$$

Here  $\delta_{t,j} \geq 0$  is a budget shared by the tasks of bag  $t$  at stage  $j$ . Feasibility and the inequalities are exact statements, not numerical tolerances. Rational bounds and points are allowed; other exact representations are also allowed when the required comparisons and arithmetic are available.

Set

$$\hat{\beta}_{t,D} = \min_{B: B_{S_t} \cap D \neq \emptyset} \ell_{t,B,D}, \quad \text{LB}_j = \min_{B \in \mathcal{L}_r} \ell_{r,B}, \quad (40)$$

and store a minimizing leaf and its oracle point for each table entry. Backtracking starts at a minimizing root leaf. In each selected leaf it uses the stored point and the child cells minimizing  $b_{u,B}$ , then continues with those child entries. All intersections, including faces, are included in this construction.

**Proposition 5.1** (Validity and exact selected residuals). *The preceding construction gives a certificate in [definition 2.1](#), with cell functions  $\hat{l}_{t,D}(s) = \hat{\lambda}_t^T s + \hat{\beta}_{t,D}$  and child functions  $\hat{\psi}_{u,B}(s) = \hat{\lambda}_u^T s + b_{u,B}$ . In particular,  $\text{LB}_j \leq f^*$  for arbitrary slopes and nonnegative budgets. Backtracking gives a configuration, whose value is*

$$\hat{\Phi} = \sum_t m_{t,B_t}(z^t) + \sum_{t \neq r} \hat{\lambda}_t^T (z_{S_t}^{\pi(t)} - z_{S_t}^t). \quad (41)$$

If  $\ell_t$  and  $q_t$  denote its selected local bound and objective, then

$$\hat{\Phi} - \text{LB}_j = \sum_t \rho_t, \quad \rho_t = q_t(z^t) - \ell_t \in [0, \delta_{t,j}]. \quad (42)$$

Consequently,

$$0 \leq \hat{\Phi} - \text{LB}_j \leq \sum_t \delta_{t,j}. \quad (43)$$

*Proof.* For every child cell meeting  $B_{S_u}$ , the definition gives  $b_{u,B} \leq \hat{\beta}_{u,D'}$ . Thus the child function lies below that cell function throughout their intersection, as required by the child-minorant inequality. For an own-separator pair, every feasible  $v$  satisfies  $q_{t,B,D}(v) \geq \ell_{t,B,D} \geq \hat{\beta}_{t,D}$ . Adding the incoming affine term gives the local certificate inequality. The root inequality follows from  $q_{r,B}(v) \geq \ell_{r,B} \geq \text{LB}_j$ . Certificate validity now follows from [proposition 2.2](#); no maximality of the intercepts is needed.

Each backtracked point belongs to its selected leaf and own cell, and each selected child cell meets the selected parent projection. These are precisely the configuration conditions in [\(11\)](#). For every selected nonroot task,  $\ell_t = \hat{\beta}_{t,D_t}$ , while the selected root bound equals  $\text{LB}_j$ . Also  $b_{t,B_{\pi(t)}} = \hat{\beta}_{t,D_t}$  by the stored child choice. Summing [\(38\)](#) over selected bags therefore gives

$$\sum_t q_t(z^t) = \hat{\Phi} + \sum_{t \neq r} \hat{\beta}_{t,D_t}, \quad \sum_t \ell_t = \text{LB}_j + \sum_{t \neq r} \hat{\beta}_{t,D_t}.$$

Subtracting proves the identity. Finally,  $\ell_t \leq q_t(z^t) \leq u_t \leq \ell_t + \delta_{t,j}$  by [\(39\)](#). This proves the residual intervals and their sum bound.  $\square$

Thus errors already inherited through child intercepts are counted once by (42). No factor for the number of leaves, cells, touching pairs, or oracle calls enters this bound. In an implementation, the inherited constants  $b_{u,B}$  can be omitted from the numerical optimization objective and added back exactly to both returned bounds.

## 5.2 Approximate slopes and bag gradients

For a configuration at center  $c$ , use the regridding quantities

$$b_t = \text{width}(B_t), \quad d_t = \text{width}(D_t), \quad Q = \sum_t b_t^2 + \sum_{t \neq r} d_t^2, \quad R = \|x - c\|,$$

where  $x_i = z_i^{\text{top}(i)}$ . Let  $\Phi$  be its value with the exact current-center slopes  $\lambda_t(c)$  of (12), and set

$$C_1 = 2p \max\{k - 1, 1\}, \quad \nu^2 = \sum_{t \neq r} \|\hat{\lambda}_t - \lambda_t(c)\|^2.$$

**Lemma 5.2** (Slope perturbation). *Every configuration satisfies*

$$H := \sum_{t \neq r} \|z_{S_t}^{\pi(t)} - z_{S_t}^t\|^2 \leq C_1 Q, \quad |\hat{\Phi} - \Phi| \leq \nu \sqrt{C_1 Q}. \quad (44)$$

Suppose the partitions are graded at width  $h_j$ , the first condition of (29) holds, and  $\nu \leq \zeta \sqrt{N} h_j$ . Then

$$|\hat{\Phi} - \Phi| \leq \frac{\eta}{4} R^2 + K_s N h_j^2, \quad K_s = \sqrt{12 C_1} \zeta + \frac{12 k C_1 \zeta^2 \theta^2}{\eta}. \quad (45)$$

*Proof.* For a separator coordinate, the child point lies in  $D_t$  and the parent point lies in  $B_{\pi(t)}$ . Since their projections meet, their coordinate difference is at most  $b_{\pi(t)} + d_t$ . Hence

$$H \leq 2 \sum_t \sum_{u \in \text{ch}(t)} |S_u| b_t^2 + 2 \sum_{u \neq r} |S_u| d_u^2.$$

Each coordinate of  $V_t$  occurs in at most  $k - 1$  child bags, since the total number of its occurrence bags is at most  $k$ . Therefore  $\sum_{u \in \text{ch}(t)} |S_u| \leq (k - 1) |V_t|$  and

$$H \leq 2p(k - 1) \sum_t b_t^2 + 2p \sum_{t \neq r} d_t^2 \leq C_1 Q.$$

Cauchy–Schwarz applied to the slope-error terms proves the second inequality in (44). By lemma 4.3,  $Q \leq 12 N h_j^2 + 12 k \theta^2 R^2$ , so

$$|\hat{\Phi} - \Phi| \leq \sqrt{12 C_1} \zeta N h_j^2 + \sqrt{12 k C_1} \zeta \theta \sqrt{N} h_j R.$$

Young’s inequality bounds the second term by  $\eta R^2/4 + 12 k C_1 \zeta^2 \theta^2 N h_j^2/\eta$ , giving the claim. When  $k = 1$ , every separator is empty, so  $H = \nu = 0$  directly.  $\square$

**Lemma 5.3** (From bag-gradient errors to slope errors). *Let  $e_t = \hat{g}_t - \nabla a_t(c_{V_t})$  and form  $\hat{\lambda}_t$  by the same subtree sums as in (12), with  $\hat{g}_t$  replacing the exact bag gradient. If the sums are exact, then*

$$\nu^2 \leq G \sum_t \|e_t\|^2, \quad G = \frac{k(k - 1)}{2}. \quad (46)$$

In particular,  $\|e_t\| \leq \gamma h_j$  for each bag ensures the budget of lemma 5.2 with  $\zeta = \sqrt{G} \gamma$ . Component-wise error at most  $\gamma h_j/\sqrt{p}$  is sufficient.

*Proof.* Fix a coordinate  $i$ . Its occurrence bags form a rooted tree  $T_i$  of size  $m_i \leq k$ . For every edge below its top bag, the corresponding slope error is the sum of the errors  $e_{t,i}$  in the occurrence bags below that edge. Write this linear map as  $A_i$ . Its entries are zero or one, and a column corresponding to bag  $t$  has one entry for each ancestor edge, namely  $\text{depth}_{T_i}(t)$  entries. Consequently,

$$\|A_i\|_2^2 \leq \|A_i\|_F^2 = \sum_{t \in T_i} \text{depth}_{T_i}(t) \leq \frac{m_i(m_i - 1)}{2} \leq G.$$

The depth sum bound follows by ordering parents before children: the vertex in position  $j + 1$  has depth at most  $j$ . Applying this operator bound to the coordinate error vector and summing over coordinates gives (46). The per-bag vector bound gives  $\sum_t \|e_t\|^2 \leq N\gamma^2 h_j^2$ . The componentwise condition implies that vector bound because  $|V_t| \leq p$ .  $\square$

There is no branching factor in either lemma. Errors introduced while forming subtree sums must be included in  $\nu$  if those sums are inexact. Exact sums of rational gradient approximations avoid that additional error. No gradient accuracy is needed for slope perturbations when  $k = 1$ .

### 5.3 Contraction and certified termination

**Theorem 5.4** (Inexact regridding). *Use fresh partitions at  $h_j = s_0 2^{-j}$  and center  $x^{(j-1)}$ , with  $\theta$  satisfying (29). Suppose fixed constants  $d, \zeta, \omega \geq 0$  give, at every stage,*

$$\delta_{t,j} \leq d h_j^2, \quad \nu_j \leq \zeta \sqrt{N} h_j, \quad F(x^{(j)}) \leq U_j \leq F(x^{(j)}) + \omega N h_j^2. \quad (47)$$

Here  $x^{(j)}$  is the consistent point of the backtracked configuration. Maintain  $\text{UB}_j = \min_{0 \leq l \leq j} U_l$ , optionally also including a certified feasible initial upper value. Set

$$K = C + K_s + d + \omega, \quad B_{\text{in}} = \max \left\{ p, \frac{8K}{3g} \right\}. \quad (48)$$

Then, starting from any  $x^{(-1)} \in X$ , every stage satisfies

$$\|x^{(j)} - x^*\|^2 \leq B_{\text{in}} N h_j^2, \quad 0 \leq \text{UB}_j - \text{LB}_j \leq g B_{\text{in}} N h_j^2. \quad (49)$$

For  $\varepsilon > 0$ , the gap test stops by stage

$$J = \max \left\{ 0, \left\lceil \log_2 \left( s_0 \sqrt{g B_{\text{in}} N / \varepsilon} \right) \right\rceil \right\}.$$

The final number of leaves and cells is at most  $2N(4/\theta)^p(J+1)$ . Through stage  $J$ , the number of created boxes is at most  $N(4/\theta)^p(J+1)(J+2)$ , and the number of local convex oracle calls is at most

$$N 3^q (4/\theta)^p \frac{(J+1)(J+2)(2J+3)}{6}.$$

*Proof.* Every stage supplies a valid lower certificate by [proposition 5.1](#), and every maintained upper value comes from a feasible point. Thus  $\text{LB}_j \leq f^* \leq \text{UB}_j$ . Apply (32) to the selected configuration with the exact current-center slopes, then use (43) and (45). Since  $\eta = g/20$ , their combined squared-distance allowance is  $\eta + \eta/4 = g/16$ , giving

$$F(x^{(j)}) - \text{LB}_j \leq \frac{g}{16} \|x^{(j)} - x^{(j-1)}\|^2 + (C + K_s + d) N h_j^2.$$

Put  $e_j = \|x^{(j)} - x^*\|^2$ . Quadratic growth, lower-bound validity, and the inequality  $\|x^{(j)} - x^{(j-1)}\|^2 \leq 2(e_j + e_{j-1})$  imply  $ge_j \leq g(e_j + e_{j-1})/8 + KNh_j^2$ . Rearranging gives

$$e_j \leq \frac{1}{7}e_{j-1} + \frac{8K}{7g}Nh_j^2. \quad (50)$$

For  $j \geq 1$ , an induction using  $h_{j-1} = 2h_j$  closes because

$$\frac{4}{7}B_{\text{in}} + \frac{8K}{7g} \leq B_{\text{in}}.$$

For stage zero,  $e_{-1} \leq ns_0^2 \leq pNh_0^2 \leq B_{\text{in}}Nh_0^2$ , and (50) yields the stronger  $e_0 \leq 4B_{\text{in}}Nh_0^2/7$ . This proves the distance bound.

For  $j \geq 1$ , the two distance bounds imply  $\|x^{(j)} - x^{(j-1)}\|^2 \leq 10B_{\text{in}}Nh_j^2$ . At stage zero the bound with coefficient 4 suffices. Since  $\text{UB}_j \leq U_j$ , the evaluation allowance gives

$$\text{UB}_j - \text{LB}_j \leq \left(\frac{5g}{8}B_{\text{in}} + K\right)Nh_j^2 \leq gB_{\text{in}}Nh_j^2,$$

where the last step uses  $K \leq 3gB_{\text{in}}/8$ . The definition of  $J$  makes this upper bound at most  $\varepsilon$  at stage  $J$ . At the stopping stage,  $F(x_{\text{inc}}) - f^* \leq \varepsilon$  for the feasible point associated with the maintained upper value.

The partitions and incidence lists are those of exact regridding. The stage box bound from [theorem 4.5](#) and the incidence bound of [proposition 4.6](#) therefore apply unchanged. Summing  $j+1$  and  $(j+1)^2$  for  $0 \leq j \leq J$  gives the stated created-box and local-call counts. Local numerical iterations are not included in these oracle-call counts.  $\square$

The allocation  $\eta/4$  in the slope estimate preserves the strict margin needed when widths halve: the coefficient  $1/7$  in (50) is smaller than  $1/4$ . Certified lower and upper bounds make the stopping test sound even at grading ratios that fail the convergence conditions.

**Corollary 5.5** (Unknown constants). *Fix positive local, gradient, and evaluation budgets  $d, \gamma, \omega$ , and assume every requested certified oracle call terminates. The operation-budget search of [proposition 4.7](#) applies to inexact regridding without knowledge of  $g, M, A$ : in round  $r$ , run each  $\theta = 2^{-\mu}$ ,  $1 \leq \mu \leq r$ , from the initial point, with a budget of  $2^r$  counted operations per trial and no separate stage cap. Some trial terminates with a certified gap at most  $\varepsilon$ .*

*More precisely, let  $\mu_*$  be a sufficient grading index and let  $W_* \geq 1$  bound the counted work of its completed run. Put  $W = \max\{W_*, 2^{\mu_*}, 2\}$  and  $R = \lceil \log_2 W \rceil$ . The total counted work through termination is at most  $2R2^R$ , hence  $O(W \log W)$ . If box emission is counted, the returned certificate contains at most  $2^R$  boxes.*

*Proof.* Request bag-gradient error  $\gamma h_j$ ; [lemma 5.3](#) supplies  $\zeta = \sqrt{G}\gamma$  for the analysis. None of these requests depends on  $g, M, A$ . A sufficiently small dyadic  $\theta$  satisfies (29), so the corresponding run stops by [theorem 5.4](#). Its counted work is finite. For example, under the same unit-cost bag-oracle and arithmetic accounting as [proposition 4.7](#), it is  $O(pN3^q(4/\theta)^p(J+1)^3)$ ; oracle-internal work is excluded. Enforce each trial budget before its next counted operation. Round  $R$  includes  $\mu_*$  and permits at least  $W_*$  operations for that trial, so that trial completes unless an earlier valid gap test has already stopped the search. The work bound follows from  $\sum_{r=1}^R r2^r \leq 2R2^R$ , and  $2^R < 2W$ .

The first successful trial may have a different grading ratio and stage from the sufficient run. Its guarantee is the certified gap test. Its actual stage has at most  $2N(4/\theta)^p(j+1)$  boxes, and its box emissions also fit within its operation budget, which is at most  $2^R$ . No final-box estimate using the sufficient run's  $\theta$  is asserted for this returned trial.  $\square$

## 5.4 Exact faces and rational reconstruction

The feasibility requirement in (39) also applies when a leaf and a separator cell only touch. A numerical displacement of any size can leave a fixed face. The following construction isolates this issue from objective accuracy.

**Proposition 5.6** (Rational feasible points). *Assume the original box, initial center, widths, and grading ratio have rational representations, and shell endpoints are formed by rational arithmetic. A local domain  $C_{t,B,D}$  is a rational box, possibly lower dimensional. Its fixed coordinates can be removed exactly. Rational local oracle points give a rational consistent point in  $X$ , and therefore a rational center and rational partitions at the next stage.*

*For quantitative reconstruction, suppose  $q = q_{t,B,D}$  is  $L_q$ -Lipschitz on its relative box, and a feasible point  $v_0$  has a certified upper value  $u_0$  with  $q(v_0) \leq u_0$  and  $u_0 - \ell \leq \delta/2$ . Let  $K_q \in \mathbb{Q}$  satisfy  $K_q \geq L_q\sqrt{p}$ . If a rational feasible point  $v$  has  $|v_i - v_{0,i}| \leq \tau$  in every coordinate, then*

$$q(v) \leq u_0 + K_q\tau =: u_{\text{new}}. \quad (51)$$

*For  $\delta > 0$  and  $K_q > 0$ , rational  $\tau > 0$  with  $K_q\tau \leq \delta/2$  ensures  $u_{\text{new}} - \ell \leq \delta$ . If  $K_q = 0$ , reconstruction has no objective penalty.*

*Proof.* For each coordinate the intersection interval is  $[\max\{L_i^B, L_i^D\}, \min\{U_i^B, U_i^D\}]$ , with the own-cell endpoints omitted outside the separator. Exact endpoint comparisons determine emptiness. Equal endpoints fix that coordinate to the common rational value; eliminate it before solving for the remaining coordinates. Restricting the objective preserves continuity and convexity. If no coordinates remain, the task is certified evaluation at the single rational point. The consistent point selects one of the feasible bag coordinates for each original variable, so it is rational and lies in its original interval. Rational shell arithmetic then preserves the claim at the next stage.

To construct  $v$ , obtain a rational approximate vector  $w$  with certified coordinate errors at most  $\tau$  from  $v_0$  and clip each coordinate exactly to its intersection interval. Fixed coordinates are set to their exact endpoint. Since  $v_0$  lies in the interval, clipping cannot increase its coordinate distance from  $v_0$ . Thus  $v$  is rational and feasible, with  $\|v - v_0\| \leq \sqrt{p}\tau$ . The Lipschitz estimate gives  $q(v) \leq q(v_0) + L_q\sqrt{p}\tau \leq u_0 + K_q\tau$ . The remaining assertions follow by adding this allowance to the initial certified gap.  $\square$

Exact clipping certifies feasibility, not the objective gap. Without a supplied quantitative modulus, the oracle must certify the final point and its gap directly. A fresh evaluation after reconstruction must fit within the remaining gap budget. Continuity ensures that rational feasible points approximate a minimum on a rational relative box at every positive tolerance, since such points are dense there. It does not supply a procedure or cost bound for finding and certifying them. Rational gradient approximations and exact rational subtree sums also give rational slopes; rational lower bounds give rational message intercepts.

Zero budgets require exact oracles and an admissible exact representation. They need not admit rational points, even for rational convex polynomial data. On  $[1, 2]$ , the function  $q(x) = (x^2 - 2)^2$  is strictly convex because  $q''(x) = 12x^2 - 8 \geq 4$ . Its unique minimizer is  $\sqrt{2}$ . With  $\delta = 0$ , (39) forces the returned point to equal that minimizer, so no rational point can satisfy the contract. This is why oracle realizability and termination remain assumptions. Irrational fixed domain coordinates likewise require an appropriate exact representation.

For fixed positive budgets, the requested absolute objective accuracy is proportional to  $h_j^2$  and the gradient accuracy to  $h_j$ , so their precision requests grow linearly with  $j$ , apart from scale terms. This observation does not bound the bit lengths of all certificate entries or the internal work of the

oracles. The certificate size counts leaves and cells; a serialized proof may include separate local lower-bound witnesses for every touching pair and may consequently contain more proof objects. Bit-complexity conclusions require an explicit representation and certified routines with their own precision and runtime bounds.

## 6 Exact rational realization and bit complexity

The construction in the preceding sections counts local operations. For an explicit polynomial input, those operations can be performed exactly with rational arithmetic. The endpoint rule used below is essential: the midpoint of a leaf is used to form its model, whereas a backtracked local minimizer is chosen at an endpoint. Consequently, repeated regridding does not repeatedly multiply unrelated coordinate denominators.

We assume that every original interval and the initial center have rational endpoints, and that the bag functions are given explicitly as sparse rational polynomials

$$a_t(v) = \sum_{\alpha \in \mathcal{A}_t} c_{t,\alpha} v^\alpha, \quad c_{t,\alpha} \in \mathbb{Q}, \quad |\alpha| := \sum_i \alpha_i \leq D. \quad (52)$$

The total degree bound  $D$  is fixed. Let  $S = \sum_t |\mathcal{A}_t|$ , and let  $L$  be the binary input length, including the tree, variable incidences, polynomial coefficients and exponents, original intervals, initial center, and the rational target accuracy  $\varepsilon > 0$ . A supplied rational smoothness bound is also included in  $L$ . Rational numbers are represented by signed integer numerators and positive integer denominators. Fixed coordinates have already been removed, as in the model section; the zero-dimensional case requires only one exact evaluation. Thus we take  $p \geq 1$  below. Let  $C_{\text{den}}$  be the least common multiple of all polynomial coefficient denominators; it is 1 when there are no nonzero coefficients.

Bit work uses a random-access store of binary data, charging integer arithmetic, copying, comparisons, and address handling by their bit cost. The polynomial-time conclusions also hold under a standard Turing-machine simulation, with polynomial overhead.

### 6.1 A certified affine model

Put

$$R = \max\{1, |\text{an original endpoint}|\}, \quad H = \max\{1, |c_{t,\alpha}| : t, \alpha\}.$$

Both quantities are rational and can be computed by exact comparisons. They bound coordinate and coefficient magnitudes, respectively.

**Proposition 6.1** (A computable smoothness bound). *For the polynomial input in [equation \(52\)](#), define*

$$M_0 = \max_t \sum_{\substack{\alpha \in \mathcal{A}_t \\ |\alpha| \geq 2}} |c_{t,\alpha}| |\alpha| (|\alpha| - 1) R^{|\alpha|-2}.$$

*Then  $M_0$  is a nonnegative rational number, and  $\|\nabla^2 a_t(v)\|_2 \leq M_0$  on every original bag box. For fixed  $D$ ,  $M_0$  has  $O_D(L)$  bits and is computable in polynomial bit time. One may use  $M = M_0$  or any supplied rational  $M \geq M_0$ ; the latter inequality is checked exactly.*

*Proof.* For a monomial of degree  $d \geq 2$ , differentiation twice and summation over all ordered coordinate pairs gives

$$\sum_{i,j} |\partial_i \partial_j (c v^\alpha)| \leq |c| R^{d-2} \left( \sum_{i \neq j} \alpha_i \alpha_j + \sum_i \alpha_i (\alpha_i - 1) \right) = |c| d(d-1) R^{d-2}.$$

The spectral norm of a matrix is at most the sum of the absolute values of its entries. Applying the triangle inequality to the Hessian proves the claimed bound. Terms of degree zero or one have zero Hessian.

The binary length of  $C_{\text{den}}$  is at most the sum of the coefficient denominator lengths. All terms in the displayed bound can be put over a denominator dividing  $C_{\text{den}}$  times a fixed power of the denominator of  $R$ . Since  $D$  is fixed, this denominator and the summed numerator have  $O_D(L)$  bits. Exact addition, comparison, and the computation of the required least common multiples therefore take polynomial bit time. If desired, a positive  $M_0$  can be rounded upward to a power of two, with a factor of less than two increase;  $M_0 = 0$  is left unchanged.  $\square$

For a rational leaf  $B$ , let  $m_B$  be its coordinatewise midpoint. Use the affine model

$$\ell_{t,B}(v) = a_t(m_B) + \langle \nabla a_t(m_B), v - m_B \rangle - \frac{Mp}{8} \text{width}(B)^2. \quad (53)$$

This notation  $\ell_{t,B}$  denotes a bag model, and is distinct from the separator affine function  $l_{t,D}$  of the certificate definition. Taylor's formula and the Hessian bound give

$$|a_t(v) - a_t(m_B) - \langle \nabla a_t(m_B), v - m_B \rangle| \leq \frac{M}{2} \|v - m_B\|^2 \leq \frac{Mp}{8} \text{width}(B)^2.$$

Consequently

$$0 \leq a_t(v) - \ell_{t,B}(v) \leq \frac{Mp}{4} \text{width}(B)^2, \quad v \in B.$$

Thus [equation \(3\)](#) holds with  $A = Mp$ . The model is affine and hence convex. All its coefficients are obtained by exact rational evaluation of the explicit polynomial and its derivatives. A smaller externally supplied smoothness bound can also be used if it has a separate sound certificate. The size and verification cost of that additional certificate must then be included; the bounds below use  $M \geq M_0$ .

**Proposition 6.2** (Exact local minimization). *With the models in [equation \(53\)](#), every local minimization in the dynamic program of [proposition 2.4](#) is the minimization of an affine function on a rational box. It can be solved exactly without a general optimization oracle. A deterministic minimizer chooses the lower endpoint of each coordinate interval when its affine coefficient is nonnegative, and the upper endpoint when its coefficient is negative.*

*Proof.* The common separator slope  $\lambda_t(c)$  is fixed before the stage. For a child  $u$  and parent leaf  $B$ , the prescribed minorant has the form

$$\psi_{u,B}(s) = \langle \lambda_u(c), s \rangle + b_{u,B}, \quad b_{u,B} = \min\{\beta_{u,E} : E \text{ touches } B_{S_u}\}.$$

It is affine, and its intercept is at most the intercept of every touching child message. The local objective consists of the bag model, these child minorants, and the subtraction of the own separator affine slope. It is therefore affine. Its domain is  $B$  intersected with the cylinder over an own separator cell  $D$ . An intersection of coordinate boxes is a coordinate box; its endpoints are maxima of lower endpoints and minima of upper endpoints. Empty intersections are omitted.

An affine function separates into the sum of one-coordinate affine functions. Each is minimized by the stated endpoint rule, including the choice of the lower endpoint when the coefficient is zero. Exact rational sign comparisons determine all choices. Taking minima over the finite leaf and cell lists then gives the exact message intercepts and the root lower bound. Ties in those finite minima may be resolved by list order.  $\square$

This proposition also supplies exact local certificate inequalities: the minimum on each nonempty leaf–cell intersection is evaluated at the stated corner. An arbitrary rational minimizer in a zero-slope direction would still be mathematically valid, but would provide no bound on its denominator. We use the deterministic endpoint rule throughout the arithmetic realization.

## 6.2 Denominators and finite message sizes

Let  $Q$  be the least common multiple of the denominators of the original endpoints, initial center, and  $s_0$ . The last inclusion is redundant when  $s_0$  is the largest difference of two original endpoints, but makes the invariant explicit. Write  $M = n_M/q_M$  with  $q_M > 0$ , taking  $q_M = 1$  when  $M = 0$ . Use a fixed integer grading exponent  $\mu$  and  $\theta = 2^{-\mu}$  with  $\mu \geq 1$  as in [lemma 4.1](#).

**Lemma 6.3** (Coordinate denominators). *At stage  $j$ , every leaf endpoint, own separator cell endpoint, endpoint of a local intersection, and reconstructed center coordinate has denominator dividing  $Q2^{j+\mu}$ . Every leaf midpoint has denominator dividing  $W_j = Q2^{j+\mu+1}$ . Their numerator and denominator lengths are  $O(L + j + \mu)$ .*

*Proof.* The initial center and original endpoints have denominators dividing  $Q$ . For the induction step, the central partition endpoints are the current center coordinate and its shifts by  $h_j = s_0 2^{-j}$ . A shell at level  $\ell \leq j$  starts at a shift by  $-2^\ell h_j$  and has step  $2^{-\mu} 2^{\ell-1} h_j$ . Hence every untruncated endpoint has the form

$$c_i + s_0 t 2^{-(j+\mu)}, \quad t \in \mathbb{Z}.$$

If  $c_i = x_i^{j-1}$  has denominator dividing  $Q2^{j-1+\mu}$ , this new endpoint has denominator dividing  $Q2^{j+\mu}$ . Clipping to an original interval selects an original endpoint when clipping is active. Projections and local intersections likewise select endpoints already present in the stage. They preserve the same divisibility.

By [proposition 6.2](#), each backtracked local minimizer consists of such endpoints. Reconstruction takes a coordinate from its top occurrence bag, so it also selects one of these endpoints and closes the induction. A midpoint introduces at most one additional factor of two. Midpoints are used only for model evaluation and are not substituted for the reconstructed center. Finally, all these points lie in the original box and have magnitude at most  $R$ . Since  $\log_2 Q$  and  $\log_2 R$  are  $O(L)$ , the numerator bounds follow as well as the denominator bounds. The initial stage obeys the same stated bounds, even though it need not introduce the factor  $2^\mu$ .  $\square$

Define  $d = \max\{D, 2\}$  and the stage denominator

$$Z_j = 8q_M C_{\text{den}} W_j^d. \tag{54}$$

Here and below a denominator “divides  $Z_j$ ” concerns the reduced rational number; the implementation can store its integer numerator over the common denominator  $Z_j$  instead of repeatedly forming products of denominators.

**Lemma 6.4** (Common denominator and magnitude bound). *Every model coefficient, common separator slope, finite message intercept, child-minorant intercept, local minimum, root lower bound, and evaluated incumbent objective at stage  $j$  has denominator dividing  $Z_j$ . Put*

$$\mathcal{H} = (S + 1)HR^D, \quad V = (1 + 2D + 3NpkD)\mathcal{H} + NMpR^2.$$

*The absolute values of finite message intercepts and objective values are bounded by  $V$ , and all scalar quantities used in their calculation have*

$$b_j = O_D(L + j + \mu) \tag{55}$$

bits. In particular, summation through a subtree does not cause exponential growth of numerator or denominator lengths.

*Proof.* A midpoint coordinate has denominator dividing  $W_j$ . Evaluation of a degree- $D$  polynomial has denominator dividing  $C_{\text{den}}W_j^D$ . A derivative has denominator dividing  $C_{\text{den}}W_j^{D-1}$  when  $D \geq 1$ , and is zero when  $D = 0$ . Multiplication of a derivative by a midpoint or an endpoint returns a denominator dividing  $C_{\text{den}}W_j^D$ . The model's constant correction has denominator dividing  $8q_M(Q2^{j+\mu})^2$ . Each of these denominators divides  $Z_j$ . The same argument applies to derivative evaluations at the stage center used in  $\lambda_t(c)$ .

All remaining dynamic-programming operations are addition, subtraction, endpoint evaluation, and selection of a finite minimum. Products of a separator slope with an endpoint have the same polynomial-evaluation denominator just considered. Induction over the tree therefore proves divisibility by the single  $Z_j$  for all finite entries. An empty-domain value is a symbolic  $+\infty$  and requires no rational representation.

For completeness, the magnitude bound follows from the unfolded configuration, rather than from repeated loose bounds on child messages. The sum of absolute polynomial values at bag midpoints is at most  $\mathcal{H}$ , and the sum of absolute gradient-displacement terms is at most  $D\mathcal{H}$ . A bag correction has magnitude at most  $MpR^2/2$ . For any coordinate, a common separator slope sums derivatives from at most  $k$  bags, so its magnitude is at most  $kD\mathcal{H}/R$ . A difference between two coordinate copies has magnitude at most  $2R$ . Thus the separator terms in [equation \(11\)](#) have total absolute value at most  $2NpkD\mathcal{H}$ . The analogous expansion for a message intercept has one additional, unpaired own-separator term, of magnitude at most  $pkD\mathcal{H}$ . The displayed  $V$  bounds these quantities and the finite minima over them. It also bounds the individual polynomial and affine quantities needed to form these entries.

Now  $\log_2 Z_j = O_D(L + j + \mu)$ . The logarithm of  $1 + V$  is  $O_D(L)$ :  $N, p, k, S$  are bounded by the explicit input size, while  $H, R$ , and the supplied or computed  $M$  have  $O_D(L)$  bits. Representing a scalar over  $Z_j$  therefore needs  $O_D(L + j + \mu)$  bits. A sum of contributions from  $N$  bags adds only the logarithm of the corresponding magnitude factor; it does not multiply their common denominators.  $\square$

### 6.3 Bit work and certificate representation

Put  $m = 4/\theta$ , the scalar shell count used in [theorem 4.5](#) and [proposition 4.6](#). We retain its dependence on the numerical grading  $\theta$  explicitly. Stage  $j$  has at most  $m^p(j+1)$  leaves per bag. The incidence construction of [proposition 4.6](#) gives at most  $N3^q m^p(j+1)^2$  nonempty leaf-own-cell pairs, where  $q \leq p$ . We count polynomial terms as part of the input, rather than treating their evaluation as one oracle call. The work bound below uses the coarser factor  $3^p$  in place of  $3^q$ .

**Theorem 6.5** (Bit realization of a sufficiently graded run). *Use the affine models in [equation \(53\)](#) and the deterministic endpoint rule. Suppose the quadratic-growth assumption holds with some  $g > 0$ , and choose a dyadic grading satisfying [equation \(29\)](#). In the constants of [equation \(28\)](#), set*

$$\begin{aligned} A &= Mp, \quad C_0 = k(k-1)p, \quad B_0 = \frac{MC_0}{2} + \frac{Mp}{4}, \\ \eta &= \frac{g}{20}, \quad C = 12B_0 + \frac{6kC_0M^2}{\eta}, \quad B = \max \left\{ p, \frac{2C}{g} \right\}. \end{aligned}$$

Put

$$J = \max \left\{ 0, \left\lceil \log_2 \left( s_0 \sqrt{\frac{gBN}{\varepsilon}} \right) \right\rceil \right\}.$$

The certificate test  $F(x) - \text{LB} \leq \varepsilon$  succeeds by stage  $J$ . There is an exact rational implementation whose total bit work through that stage is

$$\text{poly}_D(L) + O_D\left(p(N + S)3^p m^p (J + 1)^3 \mathfrak{b}_J^2\right), \quad \mathfrak{b}_J = O_D(L + J + \mu + p \log(m + 1) + \log(J + 2)). \quad (56)$$

The logarithmic terms in  $\mathfrak{b}_J$  also allow for table identifiers and counters. Scalar rational values alone obey the smaller bound [equation \(55\)](#).

*Proof.* The analytic certificate and error bounds apply because  $\ell_{t,B}$  satisfies the model contract with  $A = Mp$ , and all local minima are exact. By [theorem 4.5](#) and [equation \(34\)](#), stage  $j$  has certificate gap at most  $gBNh_j^2$ . The chosen  $J$  makes this at most  $\varepsilon$ . Actual incumbent values and the certificate test are evaluated exactly, so no additional rounding allowance is required.

Polynomial and gradient evaluation at all leaf midpoints uses  $O_D(Sm^p(j + 1) + pNm^p(j + 1))$  scalar operations: each explicit monomial has at most  $D$  factors counted with multiplicity. Common slopes at the stage center are computed from the sparse derivatives and the tree. The local affine corner evaluation and the coordinate comparisons for each nonempty pair require  $O_D(p)$  further operations. Touching-cell minima and the remaining message operations use the incidence lists. Thus a conservative stage bound is

$$O_D(p(N + S)3^p m^p (j + 1)^2).$$

This includes the explicit term count and does not assume an uncharged polynomial-evaluation oracle.

For arithmetic, write stage coordinates over  $Q2^{j+\mu}$ , midpoints over  $W_j$ , and scalar costs over  $Z_j$ . Monomial numerators are obtained by integer multiplication and exact scaling to these denominators. No coordinate division or optimization-dependent denominator is introduced. The common-denominator representation also makes a message addition an integer addition and a message comparison an integer comparison. Exact integer division, when used for scaling, and multiplication can be performed in  $O(\mathfrak{b}_j^2)$  bit operations by elementary algorithms. The coordinate tuples and list identifiers have polynomially bounded length included in  $\mathfrak{b}_j$ ; their coordinatewise comparisons are charged to the factor  $p$  in the stage bound. Table addresses fit in  $\mathfrak{b}_j$  bits. Balanced dictionaries keyed by grid-index tuples give  $O(p\mathfrak{b}_j^2)$  bit work per incidence lookup, also covered by this bound. Preprocessing the input denominators and the Hessian bound takes  $\text{poly}_D(L)$  bit operations. Finally, summing  $(j + 1)^2$  for  $0 \leq j \leq J$  gives  $O((J + 1)^3)$  and proves [equation \(56\)](#).  $\square$

The numerical parameters in this theorem are not hidden inside a polynomial in the input length. In particular,  $m$  depends on the sufficient grading, which depends on  $M/g$ ,  $p$ , and  $k$ , and the number of local records has the factor  $m^p$ . The constants in  $J$  likewise retain their stated dependence on  $g$ ,  $M$ ,  $N$ ,  $p$ ,  $k$ ,  $s_0$ , and  $\varepsilon$ . For example,  $F(x_1, x_2) = x_1^2 + 2^{-b}x_2^2$  on  $[-1, 1]^2$  has a rational input of  $O(b)$  bits, Hessian bound  $M = 2$ , and quadratic-growth constant  $g = 2^{-b}$ . The ratio  $M/g$  is exponential in  $b$ . The theorem is a bit realization of the conditional numerical work bound; it does not assert polynomial time in  $L$  when that conditioning or the bag dimension is unrestricted.

**Proposition 6.6** (A compact exact certificate). *The certificate of a sufficiently graded stage  $J$ , together with its incumbent, has a rational serialization of*

$$O_D(L + pNm^p(J + 1)\mathfrak{b}_J)$$

*bits. Given this serialization and the original input, all local certificate inequalities and the final gap can be verified by exact rational operations with the same per-stage type of bound as in [equation \(56\)](#).*

*Proof.* Store the stage index  $J$ , center and grading, leaf endpoint tuples, own separator cell endpoint tuples and intercepts, the shared slope  $\lambda_t(c)$  for each nonroot bag, the intercept  $b_{u,B}$  for each parent-leaf-child edge, the root lower bound, and the retained incumbent. The affine bag model is reconstructed from the midpoint formula, so its coefficients need not be repeated in the serialization. Child minorants use their already stored shared slopes. Likewise, local corner minima and proofs for every leaf-cell pair need not be stored: a verifier recomputes them.

There are at most  $Nm^p(J+1)$  leaves and no more own separator cells under the construction of [lemma 4.1](#). Moreover,

$$\sum_t |\text{ch}(t)| |\mathcal{L}_t| \leq (N-1)m^p(J+1).$$

Thus the endpoint, slope, intercept, and incumbent data comprise  $O(pNm^p(J+1))$  scalar entries. Each has  $O_D(b_J)$  bits by [lemmas 6.3](#) and [6.4](#). List references may be encoded as coordinate-index tuples and fit in the same total bound. Including the original input once gives the stated serialization size.

A verifier first computes  $M_0$ , checks  $M \geq M_0$  and that the stored stage center belongs to  $X$ , then checks the generated partitions, midpoint models, and shared slopes. For each child and parent leaf it checks the minorant intercept against every touching own cell. It then evaluates the local affine minimum at the deterministic corner of every nonempty own-cell intersection and checks the corresponding certificate inequality. The incidence bound controls the number of these checks. Finally, it checks that every incumbent coordinate belongs to its original interval, and evaluates the incumbent objective, the root lower bound, and their difference exactly. Validity then follows from [proposition 2.2](#); the verifier does not need a certificate of the quadratic-growth constant  $g$ .  $\square$

## 6.4 An unknown growth constant

The positive number  $g$  may be a promise used in the convergence analysis rather than part of the rational input. The stopping test does not use it: every correctly formed certificate gives  $\text{LB} \leq f^*$ , and an actual incumbent gives  $F(x) \geq f^*$ . The inequality  $F(x) - \text{LB} \leq \varepsilon$  therefore certifies the required accuracy for every grading, including a grading too coarse for the convergence proof. The smoothness bound is different: it is needed for model validity itself and is computed or certified before any run is accepted.

**Proposition 6.7** (Operation-budget search without a supplied  $g$ ). *For the rational polynomial input, suppose the quadratic-growth assumption holds for some unknown  $g > 0$ . Run the dyadic grading candidates by the following deterministic schedule. In round  $r = 1, 2, \dots$ , start each candidate  $1 \leq \mu \leq r$  from the original center, and give it at most  $2^r$  counted bit operations, including final serialization; accept the first candidate that passes the exact certificate-gap test and finishes writing its certificate within its budget.*

*For any sufficient candidate  $\mu_*$ , let  $T_*$  be a bound of the form [equation \(56\)](#) for its run through the corresponding  $J$ , enlarged if necessary so that  $T_* \geq \max\{2, 2^{\mu_*}\}$ . The search terminates, returns a certified  $\varepsilon$ -accurate incumbent, and uses  $O(T_* \log T_*)$  total bit work. It does not need a stage cap or an estimate of  $g$ .*

*Proof.* Each candidate uses the already valid smoothness bound and exact local arithmetic. Therefore any successful gap test is sound, independently of whether that candidate satisfies the sufficient grading inequalities. The existence of a positive  $g$  guarantees a finite sufficient integer  $\mu_*$  because all inequalities in [equation \(29\)](#) hold when  $\theta$  is sufficiently small. Let  $R_* = \lceil \log_2 T_* \rceil$ . The enlargement of  $T_*$  ensures  $\mu_* \leq R_*$ . Round  $R_*$  includes that candidate and supplies at least  $T_*$  bit operations, so it completes its successful run unless an earlier candidate has already succeeded.

The sufficient-run bound covers serialization: its cost is at most the output length in [proposition 6.6](#), which is bounded by [equation \(56\)](#). The schedule restarts candidates and hence requires no stored suspended states. Charging the bounded runs to their operation budgets gives

$$\sum_{r=1}^{R_*} r2^r = (R_* - 1)2^{R_*+1} + 2 \leq 4T_* \lceil \log_2 T_* \rceil.$$

These limits are elementary-step limits for the fixed exact-arithmetic algorithm. A binary countdown enforces a trial budget: over  $2^r$  decrements, its total digit changes and head movements are  $O(2^r)$ , since the successive digit positions are changed at most  $2^r, 2^{r-1}, \dots, 1$  times, respectively. A maintained end marker detects exhaustion without scanning the full counter at every step. Initializing the counter costs  $O(r)$  per trial, and round enumeration is likewise covered by the displayed sum. Hence scheduling adds only a constant factor, rather than another logarithmic factor. Stopping an unsuccessful trial does not assert any mathematical conclusion about it. The input preprocessing is done once and is already included in  $T_*$ .  $\square$

The logarithmic factor is an upper bound for this particular simple search, not a lower bound on all ways of handling unknown parameters. The first successful candidate may use a different grading and a different stage from  $\mu_*, J$ . Its accuracy follows from its own certificate test, and the preceding proposition bounds its work by the search budget. The compact size bound for a prescribed sufficiently graded stage should therefore not be transferred to that first successful output without its actual grading and stage. A serialization of the actual output is always bounded by the operations used to produce it, in addition to the input it references.

Finally, guessing a smoothness bound and applying the same stopping rule is not justified: an insufficient smoothness bound can make a purported lower bound invalid. The explicit polynomial Hessian bound removes this issue. The search varies only the grading; every candidate uses valid models from the outset.

## 7 Stable nonlinear dynamics

The certificate construction extends to a class of equality-constrained problems with a feasible forward repair. Local affine strips contain the nonlinear graphs. Backward adjoints choose separator slopes whose first-order terms vanish on every coordinate changed by the repair.

### 7.1 The constrained model and graph strips

Consider states  $\sigma_0, \dots, \sigma_T$  and continuous controls  $u_0, \dots, u_{T-1}$ , where  $T \geq 1$ , on the product box

$$X = \prod_{t=0}^T I_t \times \prod_{t=0}^{T-1} U_t.$$

The intervals are compact and nondegenerate. The symbol  $s_0$  retains its earlier meaning as the largest interval width. The feasible set is

$$P = \{x \in X : q_t(x_{V_t}) = 0, \ 0 \leq t < T\}, \quad q_t(\sigma, u, \sigma') = \sigma' - \varphi_t(\sigma, u). \quad (57)$$

There are no additional constraints. The decomposition is the path rooted at bag zero, with

$$V_t = \{\sigma_t, u_t, \sigma_{t+1}\}, \quad S_t = \{\sigma_t\} \quad (1 \leq t < T).$$

Assign  $q_t$  and the bag cost  $a_t$  to  $V_t$ , and put  $F = \sum_t a_t$ . Write  $q(x) = (q_t(x_{V_t}))_{t=0}^{T-1}$  for the residual vector. Here  $N = T$ ,  $p = 3$ , and we use the uniform occurrence bound  $k = 2$  even when  $T = 1$ . The largest separator dimension is one when  $T \geq 2$  and zero when  $T = 1$ .

**Assumption 7.1.** The maps  $\varphi_t$  are continuously differentiable on neighborhoods of their input rectangles, and the following bounds hold throughout those rectangles:

$$\varphi_t(I_t \times U_t) \subseteq I_{t+1}, \quad |\partial_\sigma \varphi_t| \leq a < 1, \quad |\partial_u \varphi_t| \leq b, \quad \text{Lip}(\nabla \varphi_t) \leq H, \quad (58)$$

where  $a, b, H \geq 0$ . The costs and their continuous convex models satisfy (4) and (3) on the full bag boxes. There are  $x^* \in P$  and  $g > 0$  such that

$$F(y) - F(x^*) \geq g \|y - x^*\|^2 \quad (y \in P). \quad (59)$$

At every feasible center the dependent-state objective derivatives satisfy

$$|\partial_{\sigma_j} F(c)| \leq G \quad (c \in P, 1 \leq j \leq T). \quad (60)$$

The growth condition concerns the true feasible set. Stability alone does not imply it. A finite  $G$  exists for each fixed compact instance; a horizon-uniform conclusion requires a horizon-uniform bound. If every bag-gradient component has magnitude at most  $G_b$  on its full box, then  $G = 2G_b$  is valid on all of  $X$ , including infeasible centers. The graph curvature bound  $H$  is used to construct the following valid local domains and must be supplied correctly.

**Lemma 7.2** (Affine graph strips). *Let  $B \subseteq X_{V_t}$  have width  $w_B$ , and let  $\xi_B$  be the midpoint of its projection onto the input coordinates. Define*

$$\begin{aligned} \ell_{t,B}(\sigma, u) &= \varphi_t(\xi_B) + \nabla \varphi_t(\xi_B)^\top ((\sigma, u) - \xi_B), \\ O_t(B) &= \{z \in B : |z_{\sigma_{t+1}} - \ell_{t,B}(z_{\sigma_t}, z_{u_t})| \leq H w_B^2 / 4\}. \end{aligned} \quad (61)$$

*The set  $O_t(B)$  is a compact convex polytope, possibly empty. It contains every true graph point in  $B$ , and every  $z \in O_t(B)$  satisfies*

$$|q_t(z)| \leq K w_B^2, \quad K = H/2. \quad (62)$$

*Proof.* The integral Taylor remainder and the two input-coordinate widths give

$$|\varphi_t(\sigma, u) - \ell_{t,B}(\sigma, u)| \leq \frac{H}{2} \|(\sigma, u) - \xi_B\|^2 \leq \frac{H}{4} w_B^2.$$

This proves graph containment. For a strip point, the triangle inequality adds its strip deviation, at most  $H w_B^2 / 4$ , to this Taylor error and gives (62). All constraints defining the strip are affine.  $\square$

Strips on neighboring leaves may differ on their common faces. Each contains the true graph points of its own leaf, which suffices for certificate validity. If  $H = 0$ , the strip is the exact affine graph and may have smaller dimension.

For this constrained problem, replace the local domains of Definition 2.1 and (10) by

$$C_{t,B,D} = O_t(B) \cap \{z : z_{S_t} \in D\}. \quad (63)$$

At the root omit the separator condition. The objective is still  $m_{t,B}$  plus affine separator terms; it is convex on this polytope. The algorithm does not minimize on the nonlinear graph or construct a convex model of a multiplier-adjusted objective.

Empty domains are represented by infeasibility flags. A flagged separator cell asserts that no feasible subtree assignment has its separator in that cell. A bag leaf is unusable when all intersecting cells of some child are flagged. Otherwise its child intercept is the minimum over the unflagged intersecting cells. An own cell is flagged when every incident leaf is unusable or has empty local domain. Only finite intercepts are stored for unflagged cells.

These assertions and the lower bounds are valid by bottom-up induction. A true feasible assignment selects a local graph point in its leaf strip and own cell. Its child separator value selects an intersecting cell that cannot be flagged by the induction hypothesis. Therefore its leaf is usable and its own local domain is nonempty. On each such assignment the usual affine child-minorant and local inequalities apply, proving the subtree lower bound and, at the root,  $\text{LB} \leq F(x^*)$ . Every true trajectory induces an available consistent configuration, so a finite root configuration exists. Continuity ensures attainment on nonempty compact domains. The unfolding proof of [Proposition 2.4](#) then applies to finite choices with these domains and flags.

The exact local oracle returns either the minimum and an attaining point on a nonempty domain [\(63\)](#), or a certificate that the domain is empty. An empty return counts as an oracle call.

## 7.2 Forward repair and adjoint cancellation

Define  $y = R(x)$  by preserving  $\sigma_0$  and every control and recursively setting

$$y_{\sigma_{t+1}} = \varphi_t(y_{\sigma_t}, x_{u_t}). \quad (64)$$

The invariance promise in [\(58\)](#) proves by induction that  $R : X \rightarrow P$  and that  $R$  fixes every point in  $P$ .

**Lemma 7.3** (Stable repair). *For every  $x \in X$ ,*

$$\|R(x) - x\| \leq \frac{\|q(x)\|}{1-a}. \quad (65)$$

*Proof.* Set  $e_t = R(x)_{\sigma_t} - x_{\sigma_t}$  and  $r_t = q_t(x_{V_t})$ . Both state arguments lie in  $I_t$ , so

$$e_0 = 0, \quad |e_{t+1}| \leq a|e_t| + |r_t|.$$

Extend the residual magnitudes by zero. The error-magnitude vector is bounded coordinatewise by a sum of lag shifts with coefficients  $1, a, a^2, \dots$ . Each shift has Euclidean norm at most one. Taking norms and using the geometric sum proves [\(65\)](#).  $\square$

**Lemma 7.4** (Adjoint at an ambient center). *Let  $c \in X$  satisfy  $|\partial_{\sigma_j} F(c)| \leq G$  for  $1 \leq j \leq T$ , whether or not  $c$  is feasible. Set*

$$d_j = \partial_{\sigma_j} F(c), \quad A_t = \partial_{\sigma} \varphi_t(c_{\sigma_t}, c_{u_t}),$$

*and define the backward recurrence*

$$\mu_{T-1} = -d_T, \quad \mu_{t-1} = A_t \mu_t - d_t \quad (t = T-1, \dots, 1). \quad (66)$$

*For  $a_t^c = a_t + \mu_t q_t$  and  $G_c = \sum_t a_t^c$ ,*

$$|\mu_t| \leq U := \frac{G}{1-a}, \quad \partial_{\sigma_j} G_c(c) = 0 \quad (1 \leq j \leq T), \quad \nabla G_c(c)^\top (x - R(x)) = 0 \quad (x \in X). \quad (67)$$

*Each  $a_t^c$  has gradient-Lipschitz constant at most  $\overline{M} = M + UH$ . Its subtree-gradient separator slopes are*

$$\lambda_t(c) = \partial_{\sigma_t} a_t(c_{V_t}) - \mu_t A_t \quad (1 \leq t < T). \quad (68)$$

*Proof.* The backward recurrence gives  $|\mu_t| \leq G \sum_{r=0}^{T-1-t} a^r \leq U$ . Direct differentiation gives  $\partial_{\sigma_T} G_c(c) = d_T + \mu_{T-1} = 0$  and, for  $1 \leq j < T$ ,

$$\partial_{\sigma_j} G_c(c) = d_j + \mu_{j-1} - A_j \mu_j = 0.$$

This uses no equation  $q(c) = 0$ . Every repair displacement is supported on  $\sigma_1, \dots, \sigma_T$ , proving the final cancellation identity. The gradient of  $q_t$  is  $(-\partial_\sigma \varphi_t, -\partial_u \varphi_t, 1)$ , with Lipschitz constant at most  $H$ ; hence the adjusted gradient has constant at most  $M + UH$ . Within the subtree of bag  $t$ , its separator state  $\sigma_t$  occurs only in that bag, giving (68).  $\square$

The adjusted gradient annihilates the entire coordinate subspace containing all repair displacements. No linearity or differentiability of  $R$  is needed. This stronger cancellation also permits rational, infeasible centers in the next section. The derivative bound is needed because graph curvature magnifies multiplier terms; objective curvature alone does not bound multipliers induced by large affine costs. The obstruction in [Proposition 9.2](#) illustrates the failure of unmodified objective-gradient slopes under constraints.

### 7.3 Aggregate error and constructive bounds

Use the following constants, with  $N = T$ ,  $p = 3$ ,  $k = 2$ :

$$\begin{aligned} C_0 &= k(k-1)p, & U &= G/(1-a), & \overline{M} &= M + UH, \\ K &= H/2, & L_q &= \sqrt{1+a^2+b^2}, & \rho &= \frac{L_q \sqrt{C_0} + K s_0}{1-a}, \\ D &= (\sqrt{C_0} + \sqrt{k}\rho)^2, & B_0 &= \overline{M}D/2 + A/4 + UK, & \eta &= g/20, \\ C &= 12B_0 + 6kD\overline{M}^2/\eta, & B &= \max\{p, 2C/g\}. \end{aligned} \tag{69}$$

Choose a dyadic  $\theta \leq 1/2$  satisfying

$$12D\theta^2 \leq 1, \quad \sqrt{12}k\sqrt{D}\overline{M}\theta \leq \eta/4, \quad 12kB_0\theta^2 \leq \eta/4. \tag{70}$$

Zero coefficients impose no restriction.

**Lemma 7.5** (Configuration error after repair). *Build the shell partitions of [Lemma 4.1](#) around a center  $c \in X$  satisfying the derivative bound of [Lemma 7.4](#), at scale  $h$  and ratio  $\theta$  satisfying (70). Use slopes (68). For every finite configuration with consistent top-copy point  $x$ , forward repair  $y = R(x)$ , and configuration value  $\Phi$ ,*

$$|F(y) - \Phi| \leq \frac{g}{20} \|y - c\|^2 + CNh^2. \tag{71}$$

*Proof.* Let  $b_t = \text{width}(B_t)$ ,  $d_t = \text{width}(D_t)$ , and put

$$Q = \sum_t b_t^2 + \sum_{t=1}^{T-1} d_t^2, \quad E_x = \sum_t \|z^t - x_{V_t}\|^2, \quad E_y = \sum_t \|z^t - y_{V_t}\|^2.$$

The incidence-based argument of [Lemma 4.2](#) gives  $E_x \leq C_0 Q$  independently of the strip constraints. Each residual  $q_t$  is  $L_q$ -Lipschitz on its full bag box. By (62) and  $b_t \leq s_0$ ,

$$\|q(x)\| \leq L_q \sqrt{E_x} + K \sqrt{\sum_t b_t^4} \leq (L_q \sqrt{C_0} + K s_0) \sqrt{Q}.$$

[Lemma 7.3](#) gives  $\|y - x\| \leq \rho\sqrt{Q}$ . Stacking the bag vectors and using coordinate multiplicity yields

$$\sqrt{E_y} \leq \sqrt{E_x} + \sqrt{k} \|y - x\|, \quad E_y \leq DQ. \quad (72)$$

The term  $Ks_0$  remains necessary even for one bag: a strip point may require repair although it has no copy mismatch.

Write  $\text{err}_t = a_t(z^t) - m_{t,B_t}(z^t)$ . Apply [Lemma 2.5](#) to the adjusted gradients to obtain

$$\Phi = \sum_t a_t(z^t) - \sum_t \text{err}_t - \sum_t \nabla a_t^c(c_{V_t})^\top (z^t - x_{V_t}).$$

[Lemma 7.4](#) permits replacing  $x$  by  $y$  in the last sum. Because  $q_t(y_{V_t}) = 0$ , rearrangement gives the exact identity

$$\Phi = F(y) + \sum_t T_t - \sum_t \text{err}_t - \sum_t \mu_t q_t(z^t), \quad (73)$$

where

$$T_t = a_t^c(z^t) - a_t^c(y_{V_t}) - \nabla a_t^c(c_{V_t})^\top (z^t - y_{V_t}).$$

The multiplier residual in (73) accounts for the relaxed graphs and has absolute value at most  $UKQ$ . The integral gradient remainder bounds

$$|T_t| \leq \overline{M} \|y_{V_t} - c_{V_t}\| \|z^t - y_{V_t}\| + \frac{\overline{M}}{2} \|z^t - y_{V_t}\|^2.$$

Set  $r = \|y - c\|$ . By Cauchy–Schwarz, (3), and (72),

$$|F(y) - \Phi| \leq \overline{M}\sqrt{kD} r \sqrt{Q} + B_0 Q. \quad (74)$$

For a leaf containing  $z^t$ , the shell width bound gives  $b_t \leq h + \theta \|y_{V_t} - c_{V_t}\| + \theta \|z^t - y_{V_t}\|$ . The same holds for its own cell with coordinates restricted to  $S_t$ . Each coordinate occurs in at most  $k$  bags and  $k - 1$  separators. The bag-plus-separator copy-error sum is at most  $2E_y$ . It need not equal  $2E_y$ , since repair may change private top-bag states. Summing  $(v_1 + v_2 + v_3)^2 \leq 3 \sum_i v_i^2$  therefore gives

$$Q \leq 6Nh^2 + 3(2k - 1)\theta^2 r^2 + 6\theta^2 E_y.$$

By (72) and the first grading condition, absorption yields

$$Q \leq 12Nh^2 + 12k\theta^2 r^2. \quad (75)$$

Insert this bound into (74), using  $\sqrt{v+w} \leq \sqrt{v} + \sqrt{w}$ . The result is

$$|F(y) - \Phi| \leq \sqrt{12kD} \overline{M} \sqrt{N} hr + [\sqrt{12k} \sqrt{D} \overline{M} \theta + 12kB_0\theta^2] r^2 + 12B_0Nh^2.$$

Young's inequality bounds the mixed term by  $\eta r^2/2 + 6kD\overline{M}^2Nh^2/\eta$ ; the remaining grading conditions bound the bracket by  $\eta/2$ . This proves (71).  $\square$

**Theorem 7.6** (Constructive certificates for stable dynamics). *Under [Assumption 7.1](#), fix a dyadic ratio satisfying (70). Start with any  $y_{-1} \in P$ , obtained by forward repair if needed, and initialize  $\text{UB} = F(y_{-1})$ . At stage  $j$ , build fresh shell partitions at  $c = y_{j-1}$  and  $h_j = s_0 2^{-j}$ , construct the strips, compute (66) and (68), and run the flagged convex dynamic program. Backtrack a minimizing configuration, form its top-copy point  $x_j$ , and set  $y_j = R(x_j)$ . Update the incumbent with  $F(y_j)$  and stop when  $\text{UB} - \text{LB}_j \leq \varepsilon$ .*

Every stopping output is valid. With the constants (69), termination occurs by

$$J = \max\{0, \left\lceil \log_2(s_0 \sqrt{gBN/\varepsilon}) \right\rceil\}. \quad (76)$$

For  $m = 4/\theta$  and largest actual separator dimension  $q$ , the counts are bounded by

$$\begin{aligned} \text{final leaves and cells} &\leq 2Nm^p(J+1), \\ \text{total created boxes} &\leq Nm^p(J+1)(J+2), \\ \text{local convex calls} &\leq N3^q m^p(J+1)(J+2)(2J+3)/6. \end{aligned} \quad (77)$$

Every local problem has at most three variables and affine constraints.

*Proof.* The local-domain induction already proves  $\text{LB}_j \leq F(x^*)$  and the existence of a finite minimizing configuration. Put  $e_j = \|y_j - x^*\|^2$ . Feasible growth and Lemma 7.5 give

$$ge_j \leq F(y_j) - \text{LB}_j \leq \frac{g}{20} \|y_j - y_{j-1}\|^2 + CNh_j^2 \leq \frac{g}{10}(e_j + e_{j-1}) + CNh_j^2.$$

Hence  $e_j \leq e_{j-1}/9 + (10C/(9g))Nh_j^2$ . Initially  $e_{-1} \leq ns_0^2 \leq pNh_0^2$ . Since  $B \geq p$  and  $B \geq 2C/g$ , the stage-zero estimate implies  $e_0 \leq BNh_0^2$ . For  $j \geq 1$ , the induction closes because  $h_{j-1} = 2h_j$  and  $4B/9 + 10C/(9g) \leq B$ . Thus  $e_j \leq BNh_j^2$  throughout.

For  $j \geq 1$ ,  $\|y_j - y_{j-1}\|^2 \leq 10BNh_j^2$ ; at stage zero the bound  $4BNh_0^2$  suffices. Therefore

$$0 \leq \text{UB} - \text{LB}_j \leq F(y_j) - \text{LB}_j \leq (gB/2 + C)Nh_j^2 \leq gBNh_j^2.$$

This proves (76). The stopping rule remains valid at ratios outside the contraction conditions, since each lower certificate and feasible upper value remains valid independently of those conditions.

The shell counts and Proposition 4.6 give (77). Graph strips preserve the incidence lists, and empty local domains still count as calls. Each strip requires one value and gradient evaluation of its map; each stage also requires  $O(T)$  evaluations and arithmetic for objective state derivatives, adjoints, and forward repair. These costs fit the displayed work bounds when charged as exact oracle operations. Costs of evaluating arbitrary lists of factors assigned to a bag must be charged separately.  $\square$

For uniformly bounded numerical constants  $a, b, H, G, M, A, g, s_0$ , with  $1 - a$  and  $g$  uniformly positive, the theorem gives  $O(T(1 + \log_+(T/\varepsilon)))$  final boxes and cells, the squared logarithm for all created boxes, and the cubed logarithm for local oracle calls, where  $\log_+ z = \max\{0, \log z\}$ . These are exact-real oracle counts. Nonlinear simulation may produce long rational coordinates or transcendental states; the theorem does not bound their expanded representation or the internal cost of convex minimization. The next section supplies a finite-precision result for fixed-degree rational polynomials with compressed trajectory output.

**Example 7.7** (Uniform nonlinear conditioning). Let every interval be  $[-1, 1]$  and set

$$\varphi_t(\sigma, u) = \sigma/4 + u/2 + \sigma^2/8, \quad F = \sigma_0^2 + \sum_{t=0}^{T-1} [\sigma_{t+1}^2 + u_t^2 - u_t^4/4].$$

Assign the initial square to bag zero. The map image is  $[-5/8, 7/8]$ , and  $a = b = 1/2$ ,  $H = 1/4$ ,  $M = 2$ , and  $G = 2$  are valid. Since  $u^2 - u^4/4 \geq 3u^2/4$  on  $[-1, 1]$ , the zero trajectory is the unique minimizer and  $g = 3/4$  works even on the full box. Keep square factors exact and, on a

control interval  $[l, r]$ , replace its quartic cost by  $u^2 - u^4/4 - (u - l)(r - u)/2$ . Its second derivative is  $3 - 3u^2 \geq 0$  and its model error is at most  $(r - l)^2/8$ , so  $A = 1/2$ . All these constants are independent of  $T$ .

The reduced objective is nonconvex: set the initial state and all earlier controls to zero and vary only the final control  $v$ . Then  $\sigma_T = v/2$  and the second derivative of the restricted objective is  $5/2 - 3v^2$ , which is negative at  $v = 15/16$ . Keeping the state variables retains three-variable bags and one-variable separators under nonlinear dynamics.

The invariance promise ensures feasibility for every retained initial state and control sequence in their intervals. A terminal target or additional state restriction requires a separate repair argument. Enumerating discrete controls can require exponential work and is outside the horizon counts above.

## 8 Rational certificates and compressed feasible trajectories

For polynomial dynamics, the exact local optimization problems in [section 7](#) can be replaced by rational linear programs. Exact forward simulation requires a different representation: polynomial composition can produce rational states with exponentially long expanded denominators. We retain the rational initial state and controls, and let the original recurrence define the exact trajectory. Contraction provides short rational enclosures for its objective value. Rounding every coordinate of the next center then prevents denominator growth across regridding stages.

**Assumption 8.1** (Rational polynomial data). Assume [assumption 7.1](#). Every interval endpoint and every coefficient of  $\varphi_t$  and  $a_t$  is rational. Their numerical degrees are bounded by a fixed constant. The local dimensions are fixed, with  $N = T$ ,  $p = 3$ ,  $k = 2$ , and maximum separator dimension  $q \leq 1$  ( $q = 0$  when  $T = 1$ ). Valid rational bounds  $a, b, H, M$ , with  $0 \leq a < 1$ , are supplied for the derivatives in [assumption 7.1](#). The box-invariance assertion and feasible quadratic growth are promises. The positive growth constant  $g$  need not be supplied.

Choose a rational bound  $G_b \geq 0$  on the absolute value of every component of every bag-cost gradient on its full box. At fixed degree this can be done in polynomial bit work: bound each differentiated monomial by its coefficient magnitude times the appropriate powers of the maximum coordinate magnitudes, and sum these bounds. Set  $G = 2G_b$ . Since each state occurs in at most two bags,

$$|\partial_{\sigma_j} F(c)| \leq G, \quad j = 1, \dots, T,$$

for every ambient-box center  $c$ , whether or not it satisfies the dynamics. Thus [lemmas 7.4](#) and [7.5](#) apply to the rational centers used below. The bounds  $M, H, a, G_b$  and box invariance determine certificate soundness. Growth is used only to prove a rate.

### 8.1 Exact rational local optimization

**Lemma 8.2** (Affine models and exact local linear programs). *For a bag leaf  $B$  of width  $w_B$ , let  $v_B$  be its full midpoint and use*

$$m_{t,B}(z) = a_t(v_B) + \langle \nabla a_t(v_B), z - v_B \rangle - \frac{Mp}{8} w_B^2. \quad (78)$$

*This rational affine model satisfies [equation \(3\)](#) with  $A = pM$ . Use the rational graph strip of [lemma 7.2](#). Every finite local task in the certificate dynamic program is a rational linear program with at most three variables and eight inequalities. Its minimum and an attaining point, or its emptiness, can be found exactly by a constant number of rational basis computations.*

*Proof.* Taylor's estimate on the box gives

$$|a_t(z) - a_t(v_B) - \langle \nabla a_t(v_B), z - v_B \rangle| \leq \frac{M}{2} \|z - v_B\|^2 \leq \frac{Mp}{8} w_B^2.$$

Subtracting the last bound proves  $0 \leq a_t(z) - m_{t,B}(z) \leq Mp w_B^2/4$ . No vertex exactness is needed. The graph strip has the form

$$|\sigma' - \varphi_t(v) - \langle \nabla \varphi_t(v), (\sigma, u) - v \rangle| \leq \frac{H}{4} w_B^2,$$

where  $v$  is the input midpoint. All its coefficients are rational. It contains every true graph point in  $B$ , and every point of the strip has residual at most  $H w_B^2/2$ .

Intersecting  $B$  with its own separator cell only changes coordinate bounds. Merge these bounds by taking the maximum lower endpoint and minimum upper endpoint. There are then at most six coordinate inequalities and two strip inequalities. The model and separator terms are affine. Child-message intercepts are constants, so they can be added after minimizing the nonconstant part.

For an unreduced three-dimensional task, enumerate the at most  $\binom{8}{3} = 56$  triples of constraint rows. Discard dependent triples, solve each remaining rational linear system, and check all inequalities exactly. Select a feasible candidate of least objective value. A nonempty bounded polyhedron has a vertex. At a vertex its active normals span the ambient space: otherwise a nonzero common null direction and its negative give feasible small displacements, contradicting extremality. Thus a vertex appears in this enumeration, and finding no feasible candidate proves emptiness. This argument also covers lower-dimensional polyhedra. Alternatively, eliminate coordinates fixed by the intersected bounds and enumerate bases in the remaining dimension; in dimension zero, check the unique point directly. Touching faces and the exact affine graph when  $H = 0$  therefore require no numerical tolerance.

The infeasibility flags and bottom-up rules in [section 7](#) remain valid. A true feasible trajectory supplies an available configuration, so a finite root minimum exists. Exact backtracking returns rational bag points and hence a rational top-copy point, rational controls, and a rational initial state.  $\square$

## 8.2 Enclosing an exact feasible trajectory

Fix the rational initial state and controls obtained by backtracking. The equations  $\sigma_{t+1} = \varphi_t(\sigma_t, u_t)$  define the exact states without expanding their numerators and denominators. Box invariance makes this trajectory feasible.

**Lemma 8.3** (Contraction enclosures and an objective upper bound). *For any dyadic mesh  $\delta > 0$ , rational state intervals can be computed with midpoints  $\hat{\sigma}_t$  and radii  $r_t$  such that they contain the exact states, lie in the prescribed state intervals, and satisfy*

$$r_0 = 0, \quad r_{t+1} \leq ar_t + \delta, \quad r_t \leq \frac{\delta}{1-a}. \quad (79)$$

The rational number

$$U_{\text{enc}} = \sum_{t=0}^{T-1} a_t(\hat{\sigma}_t, u_t, \hat{\sigma}_{t+1}) + G_b \sum_{t=0}^{T-1} (r_t + r_{t+1}) \quad (80)$$

satisfies

$$F(y) \leq U_{\text{enc}} \leq F(y) + \frac{4G_b N \delta}{1-a}.$$

*Proof.* Start from the singleton interval containing the exact rational initial state. Given its current midpoint and radius, evaluate  $\varphi_t(\hat{\sigma}_t, u_t)$  exactly as a rational number. The derivative bound gives the enclosure

$$[\varphi_t(\hat{\sigma}_t, u_t) - ar_t, \varphi_t(\hat{\sigma}_t, u_t) + ar_t].$$

Round its lower endpoint down and its upper endpoint up to the dyadic mesh, and intersect it with the prescribed next-state interval. Box invariance preserves containment under this intersection. Outward rounding increases the radius by at most  $\delta$ , and intersection cannot increase it. This proves the recurrence and its geometric-sum bound. The intersection also keeps every subsequent midpoint in the box where the derivative promise holds. Ordinary interval substitution in a polynomial does not establish this width recurrence: repeated occurrences of the state can overestimate variation.

Both the exact bag point and its midpoint approximation lie in the full bag box. Integrating the componentwise gradient bound along their segment gives

$$|a_t(\hat{\sigma}_t, u_t, \hat{\sigma}_{t+1}) - a_t(y_{V_t})| \leq G_b(r_t + r_{t+1}).$$

Adding these inequalities proves that  $U_{\text{enc}}$  is an upper bound and that its excess is at most  $2G_b \sum_t (r_t + r_{t+1})$ , hence at most the stated  $4G_b N \delta / (1 - a)$ . No independence between adjacent state intervals is required.  $\square$

Fix a positive rational  $\omega$ ; one may take  $\omega = 1$  throughout. At stage  $j$ , choose the largest reciprocal power-of-two mesh satisfying

$$\delta_j \leq \min \left\{ 1, \frac{\omega(1-a)h_j^2}{4 \max\{1, G_b\}}, \frac{(1-a)h_{j+1}}{4} \right\}. \quad (81)$$

The computed upper bound  $U_j$  then exceeds  $F(y_j)$  by at most  $\omega N h_j^2$ . Keep the least such upper bound, together with the rational initial state and controls that produced it.

### 8.3 Rational center resetting

The feasible incumbent and the next center serve different purposes. The incumbent retains its exact rational initial state and controls. The center retains only newly rounded coordinates and may be infeasible.

**Lemma 8.4** (Resetting and the rounded-center rate). *At stage  $j$ , use the rational center  $c_j$ , the models of lemma 8.2, and the exact adjoints of lemma 7.4. Let  $y_j$  be the forward repair of the backtracked top-copy point. Round its enclosure midpoints and its exact input coordinates to the largest reciprocal power-of-two mesh of size at most  $\min\{1, h_{j+1}/2\}$ , then clip each rounded coordinate to its exact interval. The resulting center satisfies*

$$\|c_{j+1} - y_j\|^2 \leq p N h_{j+1}^2. \quad (82)$$

Suppose  $\theta$  satisfies equation (70), with  $G = 2G_b$  and  $A = pM$ . Write  $C$  for the corresponding constant in equation (69), and put

$$B_{\text{round}} = \max \left\{ p, \frac{4(C + \omega)}{g} + \frac{3p}{5} \right\}. \quad (83)$$

Starting from the rational box midpoint, the successive repaired points obey  $e_j = \|y_j - x^*\|^2 \leq B_{\text{round}} N h_j^2$  and

$$0 \leq \text{UB} - \text{LB}_j \leq g B_{\text{round}} N h_j^2. \quad (84)$$

*Proof.* By [equation \(81\)](#), each state midpoint has error at most  $h_{j+1}/4$ . Rounding to the prescribed mesh adds at most  $h_{j+1}/2$ . The initial state and controls have only the latter error. Clipping cannot increase distance from a coordinate in the interval. Every coordinate error is therefore at most  $h_{j+1}$ . There are  $2T + 1 \leq pN$  coordinates, proving [equation \(82\)](#).

For initialization, let  $c_0$  be the rational box midpoint and choose any feasible reference trajectory  $y_{-1}$ , for example the recurrence from its retained input coordinates. Both  $c_0$  and  $y_{-1}$  lie in the box, so  $\|c_0 - y_{-1}\|^2 \leq pNh_0^2$  and  $e_{-1} \leq pNh_0^2$ . This reference is needed only for the proof; its expanded coordinates and objective value are not computed.

The arbitrary-center estimate [equation \(71\)](#) gives

$$F(y_j) - \text{LB}_j \leq \frac{g}{20} \|y_j - c_j\|^2 + CNh_j^2.$$

The three-vector inequality and the reset bound give  $\|y_j - c_j\|^2 \leq 3(e_j + e_{j-1} + pNh_j^2)$ , including stage zero by the initialization bound. Since  $\text{LB}_j \leq f^*$ , feasible quadratic growth yields

$$e_j \leq \frac{3}{17}e_{j-1} + \frac{20C/g + 3p}{17}Nh_j^2. \quad (85)$$

In particular the contraction factor  $3/17$  is smaller than  $1/4$ . For  $j \geq 1$ , induction gives  $e_{j-1} \leq 4B_{\text{round}}Nh_j^2$ ; this bound also holds at  $j = 0$  because  $B_{\text{round}} \geq p$ . The induction closes because

$$12B_{\text{round}} + 20C/g + 3p \leq 17B_{\text{round}}.$$

Adding the upper-bound evaluation error and using these distance bounds gives

$$\begin{aligned} 0 \leq \text{UB} - \text{LB}_j &\leq \left( \frac{3gB_{\text{round}}}{4} + \frac{3gp}{20} + C + \omega \right) Nh_j^2 \\ &\leq gB_{\text{round}}Nh_j^2, \end{aligned}$$

where the last inequality follows from  $gB_{\text{round}}/4 \geq C + \omega + 3gp/20$ .  $\square$

The center reset includes the controls and initial state. Retaining their arbitrary LP denominators in successive centers would defeat the bit-length argument even if all dependent states were rounded. At an infeasible center we still use the original cost models and the adjusted gradients only for separator slopes. A model of the full adjusted function would also need its nonzero constant  $\sum_t \mu_t q_t(c)$ ; the present construction uses the exact identity in [lemma 7.5](#) directly.

## 8.4 Bit lengths and conditioned complexity

Let  $I$  be the total binary length of the rational input, including supplied bounds and tolerance. The fixed rational  $\omega$  is included when it is not set to one. Rational numbers are stored as reduced numerator-denominator pairs. Write  $\theta = 2^{-\mu}$ ,  $\mu \geq 1$ .

**Lemma 8.5** (Polynomial numerical representations). *At stage  $j$ , let  $b_j = O(I + j + \mu + \log(T + 1))$ . Reset-center coordinates have  $O(I + j)$  bits, fresh shell endpoints have  $O(I + j + \mu)$  bits, adjoints and local affine contributions have  $O(Tb_j)$  bits, and every finite dynamic-programming message has  $O(T^2b_j)$  bits. Forward enclosures and their objective upper bounds also have polynomial bit length. Each local LP and each arithmetic operation can therefore be performed in polynomial bit work in  $T, I, j, \mu$ .*

*Proof.* A reset coordinate is a dyadic rational at the prescribed mesh or a clipped input endpoint; its magnitude is bounded by the input box. The required mesh exponent is  $O(I + j)$ . Fresh shell boundaries are formed from these centers,  $h_j = s_0 2^{-j}$ , and the dyadic grading ratio, so their bit lengths are  $O(I + j + \mu)$ . They do not inherit denominators from the previous stage's LP solutions.

At fixed degree and arity, rational polynomial values and derivatives at points of this size have  $O(b_j)$ -bit representations. In the adjoint recurrence  $\mu_{t-1} = A_t \mu_t - d_t$ , each coefficient  $A_t, d_t$  has  $O(b_j)$  bits. Multiplication by  $A_t$  and addition of  $d_t$  increase length by only  $O(b_j)$  per step. Across  $T$  steps this gives  $O(Tb_j)$  bits. Fixed-size vertex determinants and local objective evaluation consequently have polynomial bit length; the bound  $O(Tb_j)$  suffices for both local points and nonconstant local values.

A finite message is a sum of chosen local contributions over its subtree. To see this directly, write a node's contribution as its affine model plus its child slopes evaluated at the node point, minus its own separator slope evaluated there. The dynamic-programming recurrence adds selected child intercepts. Induction expresses the resulting message as a sum with one such contribution per selected bag, hence at most  $T$  terms. Taking a minimum only selects one of these sums. No two messages are multiplied. A sum of  $T$  rational numbers of length  $O(Tb_j)$  has length  $O(T^2 b_j)$  after reduction, regardless of how many grid tasks were compared. Exact comparisons and rational reduction take polynomial work at these lengths.

One can choose [equation \(81\)](#) with

$$\log(1/\delta_j) = O\left(I + j + \log(1 + G_b) + \log \frac{1}{1-a} + \log(1 + 1/\omega)\right).$$

The scale terms have input-sized binary encodings. At each forward step, polynomial evaluation uses the current bounded-length midpoint and fixed rational control. Outward rounding then returns to the prescribed dyadic precision; clipping can introduce only input endpoint denominators. There is no nonlinear composition of unrounded rational state values. The initial-state singleton and controls have the polynomial lengths already proved for LP points. Finally, summing the  $T$  rational cost values and radii in [equation \(80\)](#) still gives polynomial length.  $\square$

If a rational growth lower bound is supplied, all grading choices can be made by rational arithmetic. For  $p = 3$ ,  $k = 2$ , use the rational majorants

$$\hat{L}_q = 1 + a + b, \quad \hat{\rho} = \frac{3(1 + a + b) + (H/2)s_0}{1 - a}, \quad \hat{D} = (3 + 2\hat{\rho})^2.$$

They dominate the corresponding  $L_q, \rho, D$  in [equation \(69\)](#). Substituting  $\hat{D}$  for  $D$  in  $B_0, C$  and the grading conditions preserves every estimate. Square the nonnegative sides of the condition containing  $\sqrt{D}$ , and repeatedly halve a dyadic ratio until all the resulting rational inequalities hold. The stopping-stage estimate can likewise be computed by comparing  $gB_{\text{round}} N s_0^2 4^{-J}$  with  $\varepsilon$ .

**Theorem 8.6** (Conditioned rational approximation). *Under [assumption 8.1](#), fix a sufficient dyadic ratio and run the rational local optimization, enclosure, and center-reset procedure above. For every rational  $\varepsilon > 0$ , it returns a finite rational lower certificate, a rational feasible-objective upper bound, and rational initial state and controls in their exact intervals, with  $\text{UB} - \text{LB} \leq \varepsilon$ . The original recurrence is part of the output and defines the exact feasible trajectory from these retained inputs. No expanded state numerators or denominators are required.*

The algorithm stops by

$$J = \max \left\{ 0, \left\lceil \log_2 \left( s_0 \sqrt{gB_{\text{round}} N / \varepsilon} \right) \right\rceil \right\},$$

has the sharp shell counts of [equation \(77\)](#) with this  $J$ , and uses at most

$$N(4/\theta)^3(J+1)^3 \text{poly}(T, I, J, \mu) \quad (86)$$

bit operations. Choosing the first sufficient dyadic ratio, or the first one satisfying the conservative rational tests above, makes this work polynomial in input size, horizon,  $\log_+(1/\varepsilon)$ , and the numerical conditioning parameters. One may take  $0 < g \leq 1$  without loss and, with  $\omega = 1$ , use

$$\mathcal{K} = 1 + b + H + G_b + M + s_0 + \frac{1}{1-a} + \frac{1}{g}$$

as a single conditioning bound. The assertion is polynomial in these numerical quantities, rather than uniformly polynomial in their binary encoding lengths.

*Proof.* Each stage has a valid lower certificate by [section 7](#) and [lemma 8.2](#). Its compressed repaired trajectory is feasible by box invariance, and its upper bound is sound by [lemma 8.3](#). Thus the rational stopping test proves the claimed objective gap. [Lemma 8.4](#) gives the stated stage bound. The incumbent stores the retained input coordinates associated with its least upper bound, so its trajectory also satisfies the returned gap.

The fresh partitions and incidence lists have the counts established in [proposition 4.6](#) and [equation \(77\)](#). Here the actual separator dimension is at most one. Empty domains only remove available choices, and each local task now requires a constant number of rational basis solves. [Lemma 8.5](#) bounds their arithmetic costs. Grid generation, gradients, adjoints, forward enclosures, resetting, and output construction also take polynomial work in the same sizes, proving [equation \(86\)](#).

The constants  $D, \overline{M}, B_0, C$  are bounded by polynomials in  $\mathcal{K}$  after replacing  $g$  by  $\min\{g, 1\}$ . The first sufficient dyadic reciprocal ratio is also polynomially bounded in  $\mathcal{K}$  by [equation \(70\)](#). Hence  $B_{\text{round}}$  is polynomially bounded and  $J$  grows only logarithmically with the numerical scale, horizon, and reciprocal tolerance. This proves the conditioned complexity claim.

The certificate's numerical entries and flags have the bit lengths in [lemma 8.5](#). A checker can reproduce each constant-size LP basis enumeration, verify finite message recurrences and empty-state flags, and run the upper-bound enclosure recurrence. It trusts, or separately verifies, the supplied global derivative bounds and box-invariance promise. Thus proof serialization and these local checks fit the polynomial work bound. For any further rational requested state precision  $\tau > 0$ , rerunning [lemma 8.3](#) with  $\delta \leq (1-a)\tau$  encloses every retained trajectory state with radius at most  $\tau$ , in bit work polynomial in the preceding parameters and the requested precision length.  $\square$

## 8.5 Searching without a supplied growth constant

**Theorem 8.7** (Unknown-growth search with bit budgets). *Under [assumption 8.1](#), there is a sound algorithm that uses no supplied value of  $g$  and has polynomial bit complexity in the parameters of [theorem 8.6](#). Its returned certificate and compressed feasible trajectory have polynomial output length. The sharp shell-count conclusion of [theorem 8.6](#) concerns the sufficient fixed-ratio run; the search algorithm's output bound follows from its total bit work.*

*Proof.* Trial  $\mu \geq 1$  uses  $\theta = 2^{-\mu}$  and starts at the rational box midpoint. It executes stages  $j = 0, 1, \dots$  using the rational LPs, exact adjoints, enclosure mesh, and reset procedure above. These operations use only the input bounds,  $h_j$ , and the fixed  $\omega$ . They use none of  $g, C, B_{\text{round}}$ , or  $J$ . A trial succeeds only when its computed rational upper and lower bounds have gap at most  $\varepsilon$ . Every such success is sound, whether or not the ratio satisfies the convergence inequalities.

In round  $r = 1, 2, \dots$ , restart each trial  $1 \leq \mu \leq r$  and simulate at most  $2^r$  elementary bit operations of its algorithm. Include setup, grid generation, arithmetic, control logic, and completed

output in this trial budget; no trial-specific preprocessing occurs outside it. Stop an unsuccessful simulation when its budget expires, even within an arithmetic routine. A bit-operation budget imposes no separate stage cap and therefore also handles trials whose late stages are expensive. A binary clock and time-bounded simulation add at most a polynomial logarithmic overhead to the simulated work. Completed output is part of the successful trial's work.

Let  $\mu^*$  be the smallest sufficient dyadic index. With  $\eta = g/20$ , it satisfies  $2^{\mu^*} \leq 2R$ , where

$$R = \max \left\{ 2, \sqrt{12D}, \frac{4\sqrt{12} k \sqrt{D} \overline{M}}{\eta}, \sqrt{\frac{48k B_0}{\eta}} \right\}.$$

Indeed, every reciprocal ratio at least  $R$  satisfies each condition in [equation \(70\)](#), and the next power of two is at most  $2R$ . Thus  $2^{\mu^*}$  is polynomially bounded in the conditioning parameters. Let  $W^* \geq 1$  bound the elementary bit work of this trial through its successful output, as supplied by [theorem 8.6](#). Set

$$r^* = \max\{\mu^*, \lceil \log_2 W^* \rceil\}.$$

Round  $r^*$  contains the sufficient trial and supplies enough work for it to succeed, unless another sound trial has already returned. Moreover,

$$\sum_{r=1}^{r^*} r 2^r \leq 2r^* 2^{r^*} \leq 4r^* \max\{2^{\mu^*}, W^*\}. \quad (87)$$

Since  $r^*$  is logarithmic in the maximum on the right, accounting for the clock and simulation overhead preserves polynomial bit complexity. Writing the returned output was budgeted, so its length is bounded by the same total work.

A coarse-ratio trial can succeed before the sufficient trial and may do so at a different stage. Its stage index has not been bounded by the sufficient trial's  $J$ . Consequently the search gives a polynomial output bound, while the sharper final leaf-and-cell count applies to the fixed sufficient ratio. No growth test or estimate of the best growth constant is needed.  $\square$

This result uses rational polynomial data of fixed numerical degree and the compressed feasible-solution representation. It does not provide an expanded rational coordinate vector of polynomial length. For example,  $\varphi(\sigma) = \sigma^2/4$  on  $[0, 1]$ , starting at  $\sigma_0 = 1/2$ , has derivative bounded by  $1/2$  but gives  $\sigma_t = 2^{-(3 \cdot 2^t - 2)}$ . Its reduced denominator is exponentially long, whereas its recurrence and rational initial state are short. Additional constraints require a repair that preserves them; the constraint obstruction in [proposition 9.2](#) lies outside the present box-invariant model.

## 9 Scope and obstructions

The aggregate argument controls the sum of squared copy errors. It does not assert that every bag copy is within a constant number of core widths of the optimizer. We first show that such a uniform localization assertion fails on branching trees, even when all structural and conditioning constants are fixed. We then show why a geometric feasible repair alone does not extend the objective-gradient slope rule to coupled constraints.

### 9.1 Uniform localization fails on branching trees

A possible alternative to aggregate contraction would bound every copy of every minimizing configuration by  $KW$  whenever all selected boxes have width at most  $W$ . The following example

rules out that assertion with  $K$  depending only on occurrence, width, branching, and conditioning. The configuration minimum is the one in [proposition 2.4](#), with all nonempty closed intersections retained.

**Theorem 9.1** (Failure of uniform copy localization). *There is a family satisfying [equations \(3\)](#) and [\(5\)](#) with  $p = 2$ ,  $q = 1$ ,  $k = 3$ , at most two children per bag,  $M = 3$ ,  $A = 1/2$ , and a fixed  $g > 0$ , whose optimizer is  $x^* = 0$  and whose exact optimizer-based separator slopes are zero, with the following property. For every  $K > 0$ , one member admits finite dyadic bag and separator partitions of width at most  $W$  such that every minimizing configuration has its root coordinate greater than  $KW$ . The objective is a strongly convex quadratic with a spectral condition number bounded independently of the number of bags.*

*Proof.* Fix  $a = 7/20$  and let  $\mathcal{G}_d$  be a complete binary tree of depth  $d \geq 1$ , with  $n = 2^{d+1} - 1$  vertices, root  $o$ , adjacency matrix  $A_d$ , and graph-leaf set  $\mathcal{V}_d$ . Set

$$H = I - aA_d, \quad \delta = 1 - 2\sqrt{2}a > 0, \quad F(x, y) = \frac{1}{2}x^\top Hx + \frac{1}{2} \sum_{v \in \mathcal{V}_d} y_v^2 \quad \text{on } [-1, 1]^{n+|\mathcal{V}_d|}.$$

To bound the adjacency norm, orient each edge from its parent to its child and apply  $2|uv| \leq u^2/\sqrt{2} + \sqrt{2}v^2$ . Each vertex is a parent on at most two edges and a child on at most one, so  $|x^\top A_d x| \leq 2\sqrt{2} \|x\|_2^2$ . Thus

$$\delta I \preceq H \preceq (1 + 2\sqrt{2}a)I. \quad (88)$$

The unique optimizer is zero and  $g = \delta/2$  is valid in every dimension. The full objective Hessian has condition number at most  $(1 + 2\sqrt{2}a)/\delta$ .

Use a root bag  $\{x_o\}$ . For each nonroot graph vertex  $v$ , use the bag  $\{x_{\text{par}(v)}, x_v\}$  and attach it to the bag of its graph parent. At every graph leaf  $v$ , attach a final bag  $\{x_v, y_v\}$ . The running-intersection property holds. Internal graph coordinates occur in their own bag and their two children's bags; graph-leaf coordinates occur in their own bag and the final bag. Hence  $k = 3$ ,  $p = 2$ ,  $q = 1$ , and branching is at most two.

We next specify a factorization and its local models. Put

$$\sigma = \sqrt{1 - 8a^2}, \quad \lambda_+ = (1 + \sigma)/2, \quad P_v = \begin{cases} 2, & v \in \mathcal{V}_d, \\ 1 - \sum_{u \text{ child of } v} a^2/P_u, & v \notin \mathcal{V}_d. \end{cases}$$

All pivots lie in  $[\lambda_+, 2]$ : the map  $P \mapsto 1 - 2a^2/P$  preserves that interval and fixes  $\lambda_+$ . Expanding squares gives the identity

$$F(x, y) = \frac{P_o}{2}x_o^2 + \sum_{v \neq o} \frac{P_v}{2} \left( x_v - \frac{a}{P_v} x_{\text{par}(v)} \right)^2 + \sum_{v \in \mathcal{V}_d} \left( -\frac{x_v^2}{2} + \frac{y_v^2}{2} \right). \quad (89)$$

Indeed, every off-diagonal coefficient is  $-a$ , every nonleaf diagonal coefficient is  $(P_v + \sum_u a^2/P_u)/2 = 1/2$ , and every graph-leaf diagonal coefficient is  $(2 - 1)/2 = 1/2$ . Assign each displayed summand to its corresponding bag. Keep the root and edge squares exact, and keep  $y_v^2/2$  exact. On an interval  $[l, u]$ , replace  $-z^2/2$  by its chord

$$-\frac{z^2}{2} - \frac{1}{2}(z - l)(u - z).$$

This is an affine valid lower model. Its error is at most  $(u - l)^2/8$ , so the bag contract holds with  $A = 1/2$ . The edge-square Hessian norm is  $P_v + a^2/P_v < 3$ ; the root norm is at most 2 and the

final-bag norm is 1. Thus  $M = 3$  is valid. Every assigned bag gradient vanishes at zero, so all slopes in [equation \(12\)](#) at zero vanish.

For a dyadic  $0 < W \leq 1$ , define

$$h_W(z) = \begin{cases} z(W - z), & 0 \leq z \leq W, \\ 0, & \text{otherwise,} \end{cases} \quad G_W(x, y) = F(x, y) - \frac{1}{2} \sum_{v \in \mathcal{V}_d} h_W(x_v).$$

We compute its unique global minimizer before constructing the finite partitions. Partition the graph coordinates into interior vertices  $I$  and leaves, write  $z = x_{\mathcal{V}_d}$ , and set

$$B = -H_{II}^{-1}H_{I\mathcal{V}_d}, \quad S = H_{\mathcal{V}_d\mathcal{V}_d} - H_{\mathcal{V}_dI}H_{II}^{-1}H_{I\mathcal{V}_d}.$$

Completing the square yields

$$G_W = \frac{1}{2}(x_I - Bz)^\top H_{II}(x_I - Bz) + \frac{1}{2}z^\top Sz - \frac{1}{2} \sum_v h_W(z_v) + \frac{1}{2} \|y\|_2^2. \quad (90)$$

The convergent Neumann series for  $H_{II}^{-1}$  has nonnegative entries. Thus  $S$  has nonpositive off-diagonal entries, and symmetry makes its row sums equal, say to  $q_d$ .

To compute  $q_d$ , harmonically extend the constant leaf vector 1. With

$$r_\pm = \frac{1 \pm \sigma}{4a}, \quad \gamma = r_-/r_+,$$

its value at graph level  $j$  is

$$v_j = \frac{r_+^{j+1} - r_-^{j+1}}{r_+^{d+1} - r_-^{d+1}}.$$

The roots satisfy  $2ar^2 - r + a = 0$ , so  $v_0 = 2av_1$ ,  $v_j = av_{j-1} + 2av_{j+1}$  for  $0 < j < d$ , and  $v_d = 1$ . These are exactly the interior equations, whose solution is unique by [equation \(88\)](#). The residual in a leaf row is therefore

$$q_d = 1 - av_{d-1} = \lambda_+ \frac{1 - \gamma^{d+2}}{1 - \gamma^{d+1}}, \quad \lambda_+ < q_d \leq 1.$$

The last upper bound also follows from  $v_{d-1} > 0$ .

Write  $b_{ij} = -S_{ij} \geq 0$  for distinct leaves. The reduced part of [equation \(90\)](#) is

$$\sum_i \psi(z_i) + \frac{1}{2} \sum_{i < j} b_{ij}(z_i - z_j)^2, \quad \psi(z) = \frac{q_d}{2} z^2 - \frac{1}{2} h_W(z).$$

Its scalar minimizer is  $t_d = W/[2(q_d + 1)]$ . Inside  $[0, W]$ ,  $\psi(z) - \psi(t_d) = (q_d + 1)(z - t_d)^2/2$ . Outside that interval,

$$\psi(z) - \psi(t_d) - \frac{q_d}{4}(z - t_d)^2 = \frac{q_d}{4}(z + t_d)^2 + \frac{t_d^2}{2} \geq 0.$$

Consequently the unrestricted auxiliary minimizer is unique: every graph leaf equals  $t_d$ ,  $y = 0$ , and the interior vector is its harmonic extension. Its coordinate at level  $j$  is

$$\bar{x}_j = t_d \frac{r_+^{j+1} - r_-^{j+1}}{r_+^{d+1} - r_-^{d+1}}.$$

Since  $a > 1/3$ ,  $r_+ < 1$ , and hence

$$\frac{\bar{x}_0}{W} = \frac{(1-\gamma)r_+^{-d}}{2(q_d+1)(1-\gamma^{d+1})} \rightarrow \infty. \quad (91)$$

At each depth choose  $W$  dyadic and small enough that every  $\bar{x}_j$  lies strictly inside  $[-1/2, 1/2]$ . The ratio in [equation \(91\)](#) is unchanged, and this unrestricted minimizer is also the unique minimizer on the domain box.

We need a growth bound for the auxiliary objective. By [equation \(88\)](#),  $\|B\|_2 \leq b_0 := 2\sqrt{2}a/\delta$ . The preceding scalar inequality and [equation \(90\)](#) show

$$E := G_W(x, y) - G_W(\bar{x}, 0) \geq \frac{\delta}{2} \|x_I - Bz\|_2^2 + \frac{q_d}{4} \|z - t_d \mathbf{1}\|_2^2 + \frac{1}{2} \|y\|_2^2.$$

Using  $\bar{x}_I = B(t_d \mathbf{1})$  and

$$\|x_I - \bar{x}_I\|_2^2 \leq 2 \|x_I - Bz\|_2^2 + 2b_0^2 \|z - t_d \mathbf{1}\|_2^2,$$

we obtain, for the depth-independent constant

$$C_* := \frac{4}{\delta} + \frac{4(2b_0^2 + 1)}{\lambda_+}, \quad \|(x, y) - (\bar{x}, 0)\|_2^2 \leq C_* E.$$

Here  $C_* > 2$ , so it also covers the  $y$  coordinates.

Choose a dyadic  $0 < h \leq W$  dividing  $W$ . Partition all original root and edge bags into uniform cubes of width  $h$ , and every separator into intervals of width  $h$ . In each final bag  $\{x_v, y_v\}$ , use squares of width  $W$  on the slab  $[0, W] \times [-1, 1]$  and squares of width  $h$  on its complement. These are finite dyadic partitions: start with the width- $W$  grid and refine the squares outside the slab. Every leaf and cell has width at most  $W$ . Use zero slopes and the maximal finite intercepts of [proposition 2.4](#).

For any configuration  $c$ , form its top-copy point  $(x(c), y(c))$ . The parent coordinate in an original edge bag and the graph-leaf coordinate in a final bag differ from their top copies by at most  $2h$ : the relevant separator interval has width  $h$  and meets a parent interval of width  $h$ . All other coordinates of those bags already are top copies. On  $[-1, 1]^2$ , the gradient norm of an edge square is at most  $3\sqrt{2}$ , so changing its parent copy alters its value by at most  $12h$ .

Put  $\varphi_W(z) = -z^2/2 - h_W(z)/2$ . It is 1-Lipschitz on  $[-1, 1]$ : its derivative is  $-z$  outside  $[0, W]$  and  $-W/2$  inside. On a coarse final square, the chord model is exactly  $\varphi_W$ . On a fine final square it is  $\varphi_W - e$ , with  $0 \leq e \leq h^2/8$ . Its discrepancy from evaluation at the top copy is thus at most  $2h + h^2/8$ . The root and dummy-coordinate factors require no change. Summing, since  $h \leq 1$  and  $|\mathcal{V}_d| \leq n$ , gives the uniform configuration estimate

$$|\Phi(c) - G_W(x(c), y(c))| \leq 15nh. \quad (92)$$

This estimate includes every configuration using boundary-only intersections.

There is also a consistent configuration at  $(\bar{x}, 0)$ . Its graph leaves lie strictly in  $[0, W]$ , so choose their coarse final squares. All convex factors are exact and the concave chords supply precisely the auxiliary wells. Hence  $\min_c \Phi(c) \leq G_W(\bar{x}, 0)$ . For every minimizing configuration, [equation \(92\)](#) and auxiliary growth imply

$$\|(x(c), y(c)) - (\bar{x}, 0)\|_2^2 \leq 15C_*nh.$$

Given  $K$ , first choose  $d$  with  $\bar{x}_0/W > 2K$ , then choose the domain-compatible dyadic  $W$ , and finally choose  $h$  dividing  $W$  so small that  $15C_*nh < \bar{x}_0^2/4$ . Every minimizing configuration then has root coordinate greater than  $\bar{x}_0/2 > KW$ , as required.  $\square$

The example satisfies the stronger grading condition  $\text{width}(Y) \leq W + \theta \text{dist}_\infty(Y, x^*)$  with  $\theta = 0$ , for every leaf and cell. Exact slopes therefore do not repair unrestricted tree localization. The true objective is strongly convex, but its supplied factorization contains concave leaf factors. Keeping another convex factorization exact would change the local relaxation.

This theorem concerns arbitrary allowed partitions. It does not show that a specified nested-refinement algorithm reaches these partitions, give a lower bound on the smallest certificate, or preclude a localization theorem under an additional influence-contraction assumption. It also does not contradict aggregate contraction: a root error much larger than  $W$  can still be compatible with a bound on total squared error that depends on the number of bags.

## 9.2 Feasible repair does not cancel first-order error

**Proposition 9.2** (An affine-constraint obstruction). *Let  $L > 0$  and minimize*

$$F(x, s, y) = \frac{x^2 + s^2 + y^2}{2} + Lx - Ly \quad \text{on } [-1, 1]^3, \quad x = s, \quad y = s.$$

*Use root bag  $\{x, s\}$  with objective  $a_r = (x^2 + s^2)/2 + Lx$  and child bag  $\{s, y\}$  with objective  $a_t = y^2/2 - Ly$ , assigning each equality to its corresponding bag. Take exact local objective models and enforce the local equalities in the convex subproblems. The feasible problem has point growth  $g = 1/2$ , bag curvature  $M = 1$ , zero objective-model error, and a box-preserving feasible repair with Euclidean Lipschitz constant one. Nevertheless, any valid certificate with constant child cell minorants proving tolerance  $0 < \varepsilon \leq 3/2$  requires*

$$|\mathcal{P}_t| \geq \sqrt{\frac{2}{3}} \frac{L}{\sqrt{\varepsilon}}. \quad (93)$$

*The unmodified objective-gradient slope is zero at every center. A multiplier-adjusted slope gives an exact certificate with one leaf per bag and one separator cell.*

*Proof.* The feasible set is  $\{(v, v, v) : -1 \leq v \leq 1\}$ , on which  $F = 3v^2/2 = \|(v, v, v)\|_2^2/2$ . Thus zero is the unique optimizer and  $g = 1/2$ . Both bag Hessian norms are at most one. The orthogonal projection

$$R(z) = \frac{z_x + z_s + z_y}{3}(1, 1, 1)$$

maps the box into the feasible set and is 1-Lipschitz. It also obeys

$$\|z - R(z)\|_2^2 = \frac{(z_x - z_s)^2 + (z_s - z_y)^2 + (z_x - z_y)^2}{3} \leq (z_x - z_s)^2 + (z_y - z_s)^2.$$

In particular even residual-to-repair distance has a fixed constant. The prescribed child slope is  $\partial_s a_t = 0$ , independently of the center.

Let  $D$  be any separator interval and let  $u \leq v$  belong to  $D$ . The child certificate inequality at its locally feasible point  $(v, v)$  forces its constant intercept to satisfy  $\beta_D \leq v^2/2 - Lv$ . At the root point  $(u, u)$ , choose any leaf containing that point. Its convex child bound is at most  $\beta_D$  at  $u$  by the child-minorant test. Consequently

$$\text{LB} \leq u^2 + Lu + \beta_D \leq u^2 + v^2/2 - L(v - u). \quad (94)$$

This argument permits arbitrary finite bag partitions and arbitrary valid convex child bounds. Weaker objective models can only lower the right-hand comparison.

If the certificate proves tolerance  $\varepsilon$ , then  $\text{LB} \geq -\varepsilon$ . Fix  $0 < r \leq 1$ , intersect each cell with  $[-r, r]$ , and use the endpoints of that intersection for  $u, v$  in [equation \(94\)](#). It follows that

$$L \text{ length}(D \cap [-r, r]) \leq \varepsilon + 3r^2/2.$$

The cell interiors are disjoint and cover the separator domain, so their intersection lengths sum to  $2r$ . Thus  $|\mathcal{P}_t| \geq 2rL/(\varepsilon + 3r^2/2)$ . Choosing  $r = \sqrt{2\varepsilon/3} \leq 1$  proves [equation \(93\)](#).

For comparison, take the single cell  $[-1, 1]$  with minorant  $\ell_t(s) = -Ls$ . It is below the exact child objective on  $y = s$ , because their difference is  $s^2/2$ . Adding it at the root on  $x = s$  gives  $s^2 \geq 0$ . Thus the resulting root bound is exactly zero. The equality multipliers for the convention  $F + \mu_1(x - s) + \mu_2(y - s)$  at zero are  $\mu_1 = -L$  and  $\mu_2 = L$ ; the adjusted child separator derivative is  $-L$ .  $\square$

For example, a cell  $[0, h]$  admits copies  $(0, 0)$  and  $(h, h)$  whose configuration value is  $h^2/2 - Lh$ . Repairing the top-copy point  $(0, 0, h)$  produces objective  $h^2/6$ , so its difference from that configuration value is  $Lh - h^2/3$ . The first-order coefficient is not controlled by objective curvature or a geometric repair constant. Constraint-aware slopes must supply the cancellation used by the dynamics construction. In nonlinear dynamics, bounding the resulting multipliers also requires control of objective first derivatives.

### 9.3 Parameters, output, and other optimal sets

The point-growth assumption [equation \(5\)](#) forces a unique minimizer: applying it at any other minimizer gives zero distance from  $x^*$ . Growth toward an entire optimal set is a different hypothesis. The contraction proof here does not discover or retain the multiple centers needed to handle separated optimal components. An extension could retain several centers or exploit a description of optimal coordinate projections, but it would require a new progress argument and a bound on the additional representation. The lower allocation result in [appendix B](#) requires an additional lower bound on local model error; it does not supply a matching upper construction for general degenerate optimal sets.

Occurrence and conditioning enter the construction through the supplied bag functions. Neither treewidth nor a bound on the full objective Hessian controls the number of occurrences or the bag-gradient constants. Changing factor assignments, adding cancelling bag terms, or retaining useful convex terms exactly can change the local models and their errors. Thus the parameters describe the supplied representation. They are not invariant properties of the objective alone. A small growth constant caused by a distant nearly optimal point also remains a substantive limitation; the algorithm does not verify the global growth promise.

Fresh regridding produces a new certificate at each stage. Its final member count, total number of created boxes, and local convex-oracle count are different quantities. Internal oracle work, factor evaluation costs, and the length of serialized local proofs require their own representation and precision bounds. The arithmetic specializations establish such bounds only for their stated data and output contracts. In particular the nonlinear output is an exact recurrence from rational initial state and controls, together with certified objective bounds; it need not list expanded rational state coordinates.

Unknown-conditioning searches preserve validity by stopping only at a computed certificate gap and enforcing budgets before work is performed. Their work guarantee compares total search with a sufficient trial. The first successful trial may use a larger grading ratio and reach a later stage than that sufficient trial. The sharp final-size formula for a sufficient fixed-ratio run therefore must be

distinguished from the total work and output-budget guarantee of the search. A box-only budget does not by itself bound incidence processing, oracle-internal work, or bit operations.

The general result is for product boxes. The dynamics extension uses a specific box-invariant stable simulation and valid local affine outer strips, with no extra terminal or state constraints. Arbitrary coupled constraints, merely approximate local feasibility, or a repair controlled only in distance are not covered. A curvature bound used to build a graph enclosure is part of certificate soundness; decreasing the grading ratio cannot make an underestimated enclosure valid. These restrictions separate the validity of an emitted certificate from the assumptions that give the stated rate and computational bound.

## 10 Consequences and remaining limits

Affine separator certificates can be constructed without a known optimizer by alternating a lower-bound calculation and a new graded partition. The decisive estimate controls the sum of squared disagreements between coordinate copies. It gives a contraction on branching trees while preserving a final partition count that is logarithmic in accuracy. Certified local errors can be incorporated without charging an error for every tested pair of boxes. For polynomial objectives, affine Taylor models turn the abstract construction into a rational algorithm with explicit local checks.

The nonlinear dynamics extension shows how a coupling constraint changes the construction. Forward repair controls feasibility, but it must be paired with adjoint-adjusted slopes to cancel first-order state errors. Contractivity bounds the repair and adjoint constants uniformly in the horizon. The finite-precision analysis separates an exactly feasible recurrence representation from the potentially much larger expansion of all rational states.

These results concern certified accuracy under stated growth and conditioning assumptions. They give neither a general complexity bound for all sparse nonconvex problems nor a performance comparison with existing solvers. Several questions remain beyond the present theorems: how to exploit several minimizers without imposing point growth, whether weaker stability conditions suffice for nonlinear dynamics, and whether pointwise projected covering lower bounds have matching constructive upper bounds in more than one separator dimension. Resolving these questions requires additional arguments; none is used in the guarantees proved here.

## A Covering bounds for scalar separators

This appendix bounds the number of separator cells in the exact-bag split relaxation of [definition 3.1](#). It does not count local leaves or replace the bag relaxations in a decomposition certificate. We use the value functions  $U_t, V_t, L_t$ , the separator margin  $w_t = U_t - L_t$ , and the graded split  $\psi_t = U_t - \theta_t w_t$  from [equation \(16\)](#) and [theorem 3.8](#).

### A.1 Concentration and affine approximation

A finite-valued function  $U$  on an interval is  $M$ -semiconcave if  $U(s) - Ms^2/2$  is concave. A function  $L$  is  $M$ -semiconvex if  $L(s) + Ms^2/2$  is convex. For a nondegenerate closed interval  $J = [a, b]$  with finite one-sided endpoint derivatives, put

$$\Delta_U(J) = U'(a+) - U'(b-) + M(b - a), \quad (95)$$

$$\Delta_L(J) = L'(b-) - L'(a+) + M(b - a). \quad (96)$$

These quantities are nonnegative and increase when  $J$  is enlarged. They are the slope drop of  $U(s) - Ms^2/2$  and the slope rise of  $L(s) + Ms^2/2$ , respectively.

**Lemma A.1** (Scalar concentration). *Let  $M \geq 0$ , let  $U$  be  $M$ -semiconcave and  $L$  be  $M$ -semiconvex on an interval  $I$ , and suppose  $L \leq U$ . Write  $w = U - L$ . If  $h > 0$  and  $[c - h, c + h] \subset I$ , then*

$$\Delta_U([c - h/2, c + h/2]) + \Delta_L([c - h/2, c + h/2]) \leq \frac{4w(c)}{h} + 4Mh. \quad (97)$$

*Proof.* Set

$$U_c(s) = U(s) - \frac{M}{2}(s - c)^2, \quad L_v(s) = L(s) + \frac{M}{2}(s - c)^2, \quad \varphi = L_v - U_c.$$

The function  $\varphi$  is convex, satisfies  $\varphi(s) \leq M(s - c)^2$ , and has  $\varphi(c) = -w(c)$ . For  $a = c - h/2$  and  $b = c + h/2$ , the left side of (97) is  $\varphi'(b-) - \varphi'(a+)$ . Convexity gives

$$\varphi'(b-) \leq \frac{2}{h}(\varphi(c + h) - \varphi(b)), \quad \varphi'(a+) \geq \frac{2}{h}(\varphi(a) - \varphi(c - h)).$$

Also  $\varphi(a) + \varphi(b) \geq 2\varphi(c)$ . Therefore

$$\begin{aligned} \varphi'(b-) - \varphi'(a+) &\leq \frac{2}{h}(\varphi(c + h) + \varphi(c - h) - 2\varphi(c)) \\ &\leq \frac{2}{h}(2Mh^2 + 2w(c)). \end{aligned}$$

The one-sided derivatives used here are finite because  $a$  and  $b$  lie in the interior of  $[c - h, c + h]$ .  $\square$

**Lemma A.2** (Chord oscillation). *Let  $D = [c - r, c + r]$  with  $r > 0$ . Suppose  $U$  is  $M$ -semiconcave,  $L$  is  $M$ -semiconvex, and their one-sided endpoint derivatives on  $D$  are finite. For every  $\theta \in [0, 1]$ , the function  $\psi = (1 - \theta)U + \theta L$  has an affine approximation  $l$  satisfying*

$$\text{osc}_D(\psi - l) \leq \frac{r}{2}(\Delta_U(D) + \Delta_L(D)) + \frac{M}{2}r^2. \quad (98)$$

*Proof.* First let  $f$  be concave on  $[a, b]$ , with finite endpoint slopes  $p = f'(a+)$  and  $q = f'(b-)$ , and let  $l_f$  be its chord with slope  $\mu$ . Then  $q \leq \mu \leq p$ , and the endpoint supporting lines give

$$0 \leq f(x) - l_f(x) \leq \min\{(p - \mu)(x - a), (\mu - q)(b - x)\}.$$

If  $p = q$ , then  $f = l_f$ . Otherwise the maximum of the right side is

$$(b - a) \frac{(p - \mu)(\mu - q)}{p - q} \leq \frac{b - a}{4}(p - q).$$

Applying the same argument to  $-f$  proves the corresponding bound for the distance of a convex function below its chord.

Now  $U_c = U - M(s - c)^2/2$  is concave and  $L_v = L + M(s - c)^2/2$  is convex. Let  $l_U, l_L$  be their chords on  $D$ , and take

$$l = (1 - \theta)l_U + \theta l_L + (1 - 2\theta)\frac{M}{2}r^2.$$

The residuals  $U_c - l_U$  and  $L_v - l_L$  have oscillations at most  $r\Delta_U(D)/2$  and  $r\Delta_L(D)/2$ , respectively. The remaining quadratic residual is  $(1 - 2\theta)M((s - c)^2 - r^2)/2$ , whose oscillation is at most  $Mr^2/2$ . Oscillation is subadditive, so

$$\text{osc}_D(\psi - l) \leq \frac{r}{2}((1 - \theta)\Delta_U(D) + \theta\Delta_L(D)) + \frac{M}{2}r^2,$$

which implies (98).  $\square$

## A.2 The scalar covering theorem

Assume there are  $N$  bags and  $n_T = N - 1 \geq 1$  edges. Identify each edge with its nonroot bag  $t$ . Suppose every separator is one coordinate with nondegenerate compact range  $I_t$  of length  $\ell_t > 0$ . The bag functions are continuous on their compact boxes. For each  $t \neq r$ , choose strictly positive constants  $M_t, G_t$  such that  $U_t$  is  $M_t$ -semiconcave,  $L_t$  is  $M_t$ -semiconvex, and both are  $G_t$ -Lipschitz on  $I_t$ . A zero regularity constant may always be replaced by a positive upper bound.

**Proposition A.3** (Regularity from smooth bag data). *If each bag objective  $a_u$  is  $C^1$  on a neighborhood of its compact product box and has a Lipschitz gradient there, then the preceding semiconcavity, semiconvexity, and Lipschitz assumptions hold with finite positive constants.*

*Proof.* Fix a scalar separator  $S_t$  and write its coordinate as  $s$ . The objective on either side of the edge is a sum of bag objectives. For every fixed choice of the other coordinates, this sum has a derivative in  $s$  whose Lipschitz constant is bounded uniformly by the sum of the bag gradient Lipschitz constants on that side. Its derivative is also uniformly bounded, because the bag gradients are continuous on compact boxes. Thus each side objective is uniformly semiconcave and uniformly Lipschitz in  $s$ , with finite constants.

The remaining coordinates range over a product box independent of  $s$ . If  $H(s, y)$  is one of the side objectives and  $M$  is its uniform semiconcavity constant, then

$$\inf_y H(s, y) - \frac{M}{2}s^2 = \inf_y \left( H(s, y) - \frac{M}{2}s^2 \right)$$

is concave: a pointwise infimum of concave functions is concave. The same infimum is Lipschitz with the uniform Lipschitz constant, since  $H(s, y) \leq H(s', y) + G|s - s'|$  for every  $y$ , and one can take infima and then interchange  $s, s'$ . Applying these observations to the child side gives semiconcavity and Lipschitz continuity of  $U_t$ , and applying them to the other side gives the same properties for  $V_t$ . Consequently  $L_t = f^* - V_t$  is semiconvex and Lipschitz. Taking positive upper bounds for the two sides supplies  $M_t, G_t$ .  $\square$

Fix  $\varepsilon > 0$  and  $\tau = 1/(2n_T)$ . For a partition  $P_t$  into intervals, let  $\text{PA}(P_t)$  be the class of independent affine pieces on its cells, with a fixed convention assigning every shared endpoint to one cell. Write  $\Phi(P) = \prod_{t \neq r} \text{PA}(P_t)$ . All cell suprema below are taken on the closed cell, so they also bound this endpoint convention. Define the closed-cell bracket

$$g_{t,D} = \inf_{l \text{ affine}} \left\{ \sup_D (l - \psi_t - \tau w_t) + \sup_D (\psi_t - \tau w_t - l) \right\}. \quad (99)$$

The cell sandwich (24) gives  $\text{gap}(\Phi(P)) \leq \sum_{t \neq r} \max_{D \in P_t} g_{t,D}$ .

Starting from  $I_t$ , bisect a cell  $D$  of radius  $r_D$  while  $B_t(D) > \varepsilon/n_T$ , where

$$B_t(D) = \begin{cases} (8n_T + \frac{1}{2})M_t r_D^2 - \frac{\min_D w_t}{2n_T}, & \text{dist}(D, \partial I_t) \geq 4n_T r_D, \\ 2G_t r_D + \frac{5}{2}M_t r_D^2 - \frac{\min_D w_t}{n_T}, & \text{dist}(D, \partial I_t) < 4n_T r_D. \end{cases} \quad (100)$$

On each final cell use a minimizer of (99), with its intercept shifted so that the two suprema are equal.

For  $A \subset \mathbb{R}$  and  $d > 0$ , let  $N_\infty(A, d)$  be the least number of closed intervals of length  $d$  covering  $A$ , with  $N_\infty(\emptyset, d) = 0$ . Put

$$E(\eta) = \{x \in X : F(x) - f^* \leq \eta\}, \quad (101)$$

$$\mathcal{N}_t = \sup_{\eta \geq 0} N_\infty\left(\pi_{S_t} E(\eta), 2\sqrt{\frac{\varepsilon + \eta}{M_t}}\right), \quad (102)$$

$$J_t = \max \left\{ 0, \left\lceil \log_2 \left( \ell_t \sqrt{\frac{(16n_T^2 + n_T)M_t}{8\varepsilon}} \right) \right\rceil \right\}, \quad (103)$$

$$\varrho_t = \min \left\{ \frac{\varepsilon}{4n_T G_t}, \sqrt{\frac{\varepsilon}{5n_T M_t}} \right\}, \quad (104)$$

$$J'_t = \max \left\{ 0, \left\lceil \log_2 \left( \frac{\ell_t}{2\varrho_t} \right) \right\rceil \right\}. \quad (105)$$

The covering number  $\mathcal{N}_t$  is finite, since all covering intervals have length at least  $2\sqrt{\varepsilon/M_t}$ ; it is at least one because  $E(0)$  is nonempty.

**Theorem A.4** (Covering bound for scalar separators). *Under the preceding assumptions, the rule (100) terminates and produces an exact-bag split with gap at most  $\varepsilon$ . Its separator partitions satisfy*

$$|P_t| \leq 1 + (4n_T + 2)J_t \mathcal{N}_t + 4n_T J'_t \quad (t \neq r). \quad (106)$$

In particular,  $\text{gap}(\Phi(P)) \leq \varepsilon$ .

*Proof.* We first prove  $g_{t,D} \leq B_t(D)$  for every dyadic cell. For a cell in the first case of (100), choose  $c_m \in D$  minimizing  $w_t$  and put  $h = 4n_T r_D$ . Then  $[c_m - h, c_m + h] \subset I_t$  and  $D \subset [c_m - h/2, c_m + h/2]$ . Monotonicity of the slope masses and lemmas A.1 and A.2 give an affine  $l$  with

$$\text{osc}_D(\psi_t - l) \leq \frac{\min_D w_t}{2n_T} + (8n_T + \frac{1}{2})M_t r_D^2.$$

For every affine  $l$ , the bracket in (99) is at most  $\text{osc}_D(\psi_t - l) - 2\tau \min_D w_t$ . This proves the required bound for the first case.

For any cell, Lipschitz continuity bounds every one-sided derivative of  $U_t$  and  $L_t$  by  $G_t$  in absolute value. Consequently

$$\Delta_{U_t}(D) + \Delta_{L_t}(D) \leq 4G_t + 4M_t r_D.$$

Lemma A.2 now bounds the oscillation by  $2G_t r_D + \frac{5}{2}M_t r_D^2$ , proving the second case. Since  $w_t \geq 0$ , both upper bounds tend uniformly to zero as the cell radius tends to zero. Thus the rule terminates after finitely many levels, and every final cell has  $g_{t,D} \leq \varepsilon/n_T$ .

The affine minimizers used by the rule exist. Indeed, after writing  $l(s) = as + b$ , the intercept cancels in the bracket. The resulting function of  $a$  is continuous. Evaluating its two suprema at opposite endpoints of  $D$  bounds it below by  $2r_D |a|$  minus a fixed constant, so it is coercive and attains its minimum. Shifting  $b$  balances the two suprema. The proof of the cell sandwich then bounds the gap of the resulting split by  $\sum_{t \neq r} \max_{D \in P_t} g_{t,D} \leq \varepsilon$ .

It remains to count the split cells. Fix  $t$  and abbreviate  $M = M_t$ ,  $G = G_t$ , and  $\ell = \ell_t$ . At level  $j \geq 0$ , all dyadic cells have radius  $r_j = \ell 2^{-j-1}$ . A split cell in the first case must satisfy

$$\min_D w_t < \eta_j := (16n_T^2 + n_T)M r_j^2 - 2\varepsilon.$$

In particular  $\eta_j > 0$ . Such a cell meets  $\{s : w_t(s) \leq \eta_j\} = \pi_{S_t} E(\eta_j)$ ; the equality follows from the attained separator-margin minimum on a compact box. There are at most  $J_t$  levels with  $\eta_j > 0$ , since this condition is equivalent to

$$2^j < \ell \sqrt{\frac{(16n_T^2 + n_T)M}{8\varepsilon}}.$$

At each such level, cover  $\pi_{S_t} E(\eta_j)$  by at most  $\mathcal{N}_t$  intervals of length  $d_j = 2\sqrt{(\varepsilon + \eta_j)/M}$ . Their length relative to a cell is bounded by

$$\frac{d_j}{2r_j} = \sqrt{16n_T^2 + n_T - \frac{\varepsilon}{Mr_j^2}} < \sqrt{16n_T^2 + n_T} < 4n_T + 1.$$

A closed interval of length  $d_j$  meets at most  $\lfloor d_j/(2r_j) \rfloor + 2 \leq 4n_T + 2$  closed dyadic cells. There are therefore at most  $(4n_T + 2)J_t\mathcal{N}_t$  split cells in the first case.

At any level, fewer than  $2n_T$  cell lengths separate a cell in the second case from one of the two endpoints of  $I_t$ . There are at most  $2n_T$  such cells at each end, and hence at most  $4n_T$  in total. A cell in this case can split only if

$$2Gr_j + \frac{5}{2}Mr_j^2 > \varepsilon/n_T.$$

If  $r_j \leq \varrho_t$ , each term on the left is at most  $\varepsilon/(2n_T)$ , so the cell cannot split. The condition  $r_j > \varrho_t$  holds at at most  $J'_t$  levels. Thus the second case contributes at most  $4n_T J'_t$  split cells. Each bisection increases the final cell count by one, which proves (106).  $\square$

**Corollary A.5** (Refinement by the cell bracket). *Under the assumptions of [theorem A.4](#), start from  $I_t$  and bisect a dyadic cell  $D$  while  $g_{t,D} > \varepsilon/n_T$ . Use balanced bracket minimizers on the final cells. This rule terminates, gives an exact-bag split with gap at most  $\varepsilon$ , and produces a coarsening of the partition in [theorem A.4](#). Its cell count therefore satisfies (106).*

*Proof.* The proof of [theorem A.4](#) establishes  $g_{t,D} \leq B_t(D)$  for every dyadic cell. Hence any cell split by the bracket rule is also split by (100). Induction on the level shows that every ancestor needed to reach such a cell is split by both rules. The bracket rule therefore uses a subset of the finitely many bisections in [theorem A.4](#); its final partition is a coarsening. Its final brackets are at most  $\varepsilon/n_T$ , so the cell sandwich gives the asserted gap bound.  $\square$

## B Pointwise allocation of model error

The construction uses an upper bound on local model error. A necessary partition-size bound requires additional information: the model must lose a specified amount in the interior of a box. This appendix gives such a bound. It also explains why a single tolerance share for each bag can be too restrictive when different near-optimal points activate different bags. No quadratic-growth assumption is used here.

### B.1 A lower gap contract and variable-radius coverings

For a bag box  $B = \prod_{i \in V_t} [\ell_i^B, u_i^B]$ , write

$$q_{B,K}(z) = \sum_{i \in K} (z_i - \ell_i^B)(u_i^B - z_i) \quad (K \subseteq V_t, z \in B).$$

Fix  $\alpha > 0$  and coordinate sets  $K_t \subseteq V_t$ . The additional *lower gap contract* is

$$a_t(z) - m_{t,B}(z) \geq \alpha q_{B,K_t}(z) \quad (z \in B). \quad (107)$$

It is imposed on every model used by a certificate under consideration. It does not follow from (3): an exact convex model has zero error even in the interior. Thus the results below do not apply to the main construction solely on the basis of its upper error bound.

Let

$$E(\eta) = \{x \in X : F(x) - f^* \leq \eta\}, \quad \eta \geq 0.$$

For a nonnegative function  $e : X_{V_t} \rightarrow [0, \infty)$  and a set  $A \subseteq X_{V_t}$ , let  $\mathcal{N}_{t,2}(e; A)$  be the least cardinality of a finite set  $C \subseteq \mathbb{R}^{K_t}$  such that

$$\text{for every } z \in A \text{ there is } v \in C \text{ with } \|z_{K_t} - v\| \leq \sqrt{e(z)/\alpha}.$$

If no finite set exists, the number is  $+\infty$ . Define  $\mathcal{N}_{t,\infty}(e; A)$  by replacing the Euclidean norm with the sup norm. Empty coordinate sets have the usual one-point space. Since a Euclidean ball is contained in the sup-norm ball of the same radius,  $\mathcal{N}_{t,2} \geq \mathcal{N}_{t,\infty}$ .

A family of nonnegative functions  $(e_t)$  is *admissible at  $\eta$*  if

$$\sum_t e_t(x_{V_t}) \leq \varepsilon + \eta \quad (x \in E(\eta)). \quad (108)$$

For  $\nu \in \{2, \infty\}$ , set

$$\Phi_\nu(\varepsilon, \eta) = \inf_{(e_t) \text{ admissible at } \eta} \sum_t 2^{-|K_t|} \mathcal{N}_{t,\nu}(e_t; \text{proj}_{V_t} E(\eta)). \quad (109)$$

The functions can depend on the point in the bag. The infimum imposes no regularity or computability requirement; it defines a necessary size bound, rather than an algorithm for locating a partition.

**Theorem B.1** (Pointwise Euclidean covering bound). *Let a separator certificate satisfy (107) and  $f^* - \text{LB} \leq \varepsilon$ . For every bag define*

$$Q_t(z) = \min_{B \in \mathcal{L}_t: z \in B} q_{B,K_t}(z).$$

*Then  $(\alpha Q_t)$  is admissible at every  $\eta \geq 0$ , and*

$$|\mathcal{L}_t| \geq 2^{-|K_t|} \mathcal{N}_{t,2}(\alpha Q_t; \text{proj}_{V_t} E(\eta)) \geq 2^{-|K_t|} \mathcal{N}_{t,\infty}(\alpha Q_t; \text{proj}_{V_t} E(\eta)). \quad (110)$$

*Consequently the certificate size satisfies*

$$\mathcal{S} \geq \sup_{\eta \geq 0} \Phi_2(\varepsilon, \eta) \geq \sup_{\eta \geq 0} \Phi_\infty(\varepsilon, \eta). \quad (111)$$

*Proof.* The minimum defining  $Q_t$  exists because the leaf family is a finite cover. Fix  $x \in E(\eta)$  and, at each bag, choose a containing leaf  $B_t$  attaining  $Q_t(x_{V_t})$ . Cancellation along the consistent point, (8), and the lower gap contract give

$$\alpha \sum_t Q_t(x_{V_t}) \leq \sum_t [a_t(x_{V_t}) - m_{t,B_t}(x_{V_t})] \leq F(x) - \text{LB} \leq \eta + \varepsilon.$$

This proves admissibility, including at points on partition boundaries.

For  $z \in X_{V_t}$  choose a leaf  $B$  attaining  $Q_t(z)$ . If  $d_i$  is the distance from  $z_i$  to the nearer endpoint of  $[\ell_i^B, u_i^B]$ , then  $d_i^2 \leq (z_i - \ell_i^B)(u_i^B - z_i)$ : the smaller of two nonnegative numbers has square at most their product. Choosing the nearer endpoint in every coordinate of  $K_t$  gives a vertex  $v$  of  $B_{K_t}$  with

$$\|z_{K_t} - v\|^2 = \sum_{i \in K_t} d_i^2 \leq Q_t(z).$$

The union of the projected vertices of all leaves has at most  $2^{|K_t|} |\mathcal{L}_t|$  points and is therefore a permissible covering set. This proves (110). Sum over bags, use  $\mathcal{S} \geq \sum_t |\mathcal{L}_t|$ , and take the infimum in (109) and then the supremum over  $\eta$ .  $\square$

For each fixed  $\eta$ , admissibility also implies  $e_t(z) \leq \varepsilon + \eta$  on  $\text{proj}_{V_t} E(\eta)$ : lift  $z$  to a point of  $E(\eta)$  and use nonnegativity of the other terms. Thus (111) contains the corresponding constant-radius projected covering bound. It preserves the stronger pointwise restriction on the sum of the radii squared.

## B.2 Different points can use different bags

The following finite-state example isolates the allocation issue. Its separator variables are binary; it is not an instance of the continuous box model used in the regridding theorem. The certificate inequalities retain the same meaning on each continuous slice, with singleton cells for binary separator values. Affine functions on these singleton cells can be constants.

For a bounded set  $A$  in a coordinate space, let  $\mathcal{C}(A, r)$  be the least number of closed axis-parallel cubes of side length  $r$  covering  $A$ . Unlike the radius convention above,  $r$  here is the full side length. A measure based on constant positive shares is

$$\Psi(\varepsilon) = \inf_{\substack{\varepsilon_t > 0 \\ \sum_t \varepsilon_t = \varepsilon}} \sum_t \sup_{\eta \geq 0} \mathcal{C} \left( \text{proj}_{V_t} E(\eta), \sqrt{(\varepsilon_t + \eta)/\alpha} \right). \quad (112)$$

**Proposition B.2** (A staircase of flat slices). *Fix  $K \geq 2$ ,  $d \geq 1$ ,  $0 < \alpha \leq 1$ ,  $0 < \varepsilon < 1$ , and  $P \geq 1 + \alpha d/4$ . Let  $s_1, \dots, s_{K+1} \in \{0, 1\}$  and  $y_t \in [0, 1]^d$ , and use the path bags  $V_t = \{s_t, y_t, s_{t+1}\}$  with costs*

$$a_t = (1 - s_{t+1} + s_t) \|y_t\|^2 + P \max\{s_t - s_{t+1}, 0\}^2.$$

*On a box  $B$  in a fixed binary slice supply the convex model  $m_{t,B} = a_t - \alpha q_{B_y, \{1, \dots, d\}}$ . Then  $f^* = 0$  and*

$$\Psi(\varepsilon) \geq K(\alpha K/\varepsilon)^{d/2}. \quad (113)$$

*There is a finite-state separator certificate, using zero slopes, with  $\text{UB} = 0$ ,  $\text{LB} \geq -\varepsilon$ , and at most*

$$K \left[ \left[ \sqrt{\alpha d/(2\varepsilon)} \right]^d + 2(J+1)(4/\theta)^d + 1 \right] + 2(K-1) \quad (114)$$

*leaves and cells, where*

$$h_0 = \sqrt{\frac{2\varepsilon}{Kd\alpha^2}}, \quad J = \max\{0, \lceil \log_2(1/h_0) \rceil\}, \quad \theta = \max\{2^{-j} : j \in \mathbb{N} \cup \{0\}, 2^{-j} \leq \min(1, 2/\sqrt{\alpha d})\}.$$

*For fixed  $d, \alpha, \varepsilon$ , the ratio of (113) to the size in (114) is at least  $cK^{d/2}/(1 + \log K)$  for a constant  $c > 0$  independent of  $K$ .*

*Proof.* Every cost is nonnegative on the binary domain. A zero-cost point has no descent in  $s$ , has  $y_t = 0$  at every equal pair, and can have arbitrary  $y_t$  at its ascent, if any. In particular, for each  $t$ ,  $E(0)$  contains the slice with  $s_1 = \dots = s_t = 0$ ,  $s_{t+1} = \dots = s_{K+1} = 1$ ,  $y_t \in [0, 1]^d$ , and all other  $y_j = 0$ . A cube of side  $r$  meets the projection of this slice in  $d$ -dimensional volume at most  $r^d$ . Consequently the  $t$ th term of (112) is at least  $(\alpha/\varepsilon_t)^{d/2}$ , by taking  $\eta = 0$ . Convexity of  $u \mapsto u^{-d/2}$  and  $\sum_t \varepsilon_t = \varepsilon$  prove (113).

Partition each bag separately on its four binary slices. On the ascent slice  $(0, 1)$  use a uniform grid with  $n_1 = \lceil \sqrt{\alpha d / (2\varepsilon)} \rceil$  intervals per continuous coordinate. Its  $n_1^d$  leaves have model minimum at least  $-\alpha d / (4n_1^2) \geq -\varepsilon/2$ .

On each equal slice use the following partition. Its central cube is  $[0, 2^{-J}]^d$ . For  $j = 0, \dots, J-1$ , partition  $[0, 2^{-j}]^d \setminus [0, 2^{-(j+1)}]^d$  by retaining the cubes outside the inner cube from the uniform grid of side  $\theta 2^{-(j+1)}$  on the outer cube. The dyadic choice of  $\theta$  makes both boundaries grid boundaries. These closed cubes and the central cube cover  $[0, 1]^d$  and have disjoint interiors. There are at most  $1 + J(2/\theta)^d \leq (J+1)(4/\theta)^d$  cubes. Each noncentral cube  $B$  satisfies  $\text{width}(B) \leq \theta \text{dist}_\infty(B, 0)$ , since at least one of its lower coordinates is at least  $2^{-(j+1)}$ . Its model is therefore bounded below by

$$\text{dist}_\infty(B, 0)^2 - \frac{\alpha d}{4} \text{width}(B)^2 \geq \text{dist}_\infty(B, 0)^2 (1 - \alpha d \theta^2 / 4) \geq 0.$$

On the central cube of side  $h = 2^{-J} \leq \min(h_0, 1)$ , the minimum is

$$\min_{[0, h]^d} \left[ (1 + \alpha) \|y\|^2 - \alpha h \sum_i y_i \right] = -\frac{d\alpha^2 h^2}{4(1 + \alpha)} \geq -\frac{\varepsilon}{2K}.$$

On the descent slice  $(1, 0)$  use one leaf. Its model minimum is at least  $P - \alpha d / 4 \geq 1$ . The models are convex on all four slices: the coefficient of  $\|y\|^2$  is  $1 - \sigma' + \sigma + \alpha > 0$ .

To construct the affine certificate, let  $\mu_t(\sigma, \sigma')$  be the least model minimum among the leaves in slice  $(\sigma, \sigma')$ . Put  $b_{K+1}(\sigma) = 0$  and recursively set

$$b_t(\sigma) = \min_{\sigma' \in \{0, 1\}} [\mu_t(\sigma, \sigma') + b_{t+1}(\sigma')].$$

For each nonroot separator use its two singleton cells, with constant minorants  $b_t(0), b_t(1)$ . On a parent leaf the child bound is the constant  $b_{t+1}(\sigma')$  determined by its fixed right state. The child and local inequalities hold by this recurrence; set  $\text{LB} = \min_{\sigma \in \{0, 1\}} b_1(\sigma)$  at the root. Equivalently,  $\text{LB}$  is the minimum of  $\sum_t \mu_t(s_t, s_{t+1})$  over binary sequences. If a sequence has  $A_s$  ascents and  $D_s$  descents, then  $A_s - D_s = s_{K+1} - s_1 \leq 1$ . Its value is at least

$$-A_s \varepsilon / 2 - K \varepsilon / (2K) + D_s \geq -\varepsilon + D_s (1 - \varepsilon / 2) \geq -\varepsilon.$$

The all-zero point gives  $\text{UB} = 0$ . Counting the four slice partitions and the two cells at each separator yields (114). For fixed  $d, \alpha, \varepsilon$ , its size is  $O(K(1 + \log K))$ , while (113) grows as  $K^{1+d/2}$ , proving the last assertion.  $\square$

Here one near-optimal point can use a flat slice in one bag and another point can use a flat slice in another bag. Constant shares must reserve tolerance for every bag at once. In this finite-state class, a lower size estimate based on  $\Psi$  cannot have a polynomial loss in  $K$  whose degree is independent of  $d$ : fixing  $d$  larger than twice that degree makes the ratio above diverge. This does not establish a counterexample for the continuous growth model, nor a converse to [theorem B.1](#).

### B.3 Why the Euclidean radius matters

**Proposition B.3** (Dimensional loss for a flat bag). *Let  $d \geq 1$ ,  $\alpha > 0$ , and  $\varepsilon > 0$ . Let there be one bag with  $K_r = V_r = \{1, \dots, d\}$ ,  $X = [0, 1]^d$ ,  $F = 0$ , and  $m_B = -\alpha q_{B, \{1, \dots, d\}}$ . Write  $v_d = \text{vol}\{z \in \mathbb{R}^d : \|z\| \leq 1\}$ . Every certificate with  $\text{LB} \geq -\varepsilon$  has at least  $(\alpha/\varepsilon)^{d/2}/v_d$  leaves. If  $0 < \varepsilon \leq \alpha/4$ , then*

$$\sup_{\eta \geq 0} \Phi_\infty(\varepsilon, \eta) \leq 2^{-d} \left\lceil \sqrt{\alpha/(4\varepsilon)} \right\rceil^d \leq 2^{-d} (\alpha/\varepsilon)^{d/2}.$$

Thus the ratio of the necessary leaf count to this sup-norm allocation bound is at least  $2^d/v_d$ .

*Proof.* The root inequality on a leaf  $B$  implies  $\alpha q_B(z) \leq \varepsilon$  for every  $z \in B$ . Set  $R = \sqrt{\varepsilon/\alpha}$  and divide  $B$  into its  $2^d$  nearest-vertex regions, cutting each coordinate at its midpoint. The nearest-endpoint argument in [theorem B.1](#) puts each region in one orthant of a Euclidean ball of radius  $R$  centered at its vertex. Each has volume at most  $v_d R^d/2^d$ , so  $\text{vol } B \leq v_d R^d$ . The leaves cover a cube of volume one, which proves the leaf bound.

For every  $\eta$ ,  $E(\eta) = X$ , and the constant function  $e = \varepsilon$  is admissible. Partition each coordinate interval into  $n = \lceil 1/(2R) \rceil$  intervals of length at most  $2R$ . Their product centers cover  $X$  by  $n^d$  sup-norm balls of radius  $R$ . This proves the first upper bound. Since  $1/(2R) \geq 1$ , the inequality  $\lceil u \rceil \leq 2u$  gives the second.  $\square$

The distinction is within a single bag, without separator inconsistency: the lower gap adds squared coordinate distances, whereas a sup-norm radius bounds only their maximum. The pointwise Euclidean bound retains that sum. These necessary bounds give no general two-sided characterization of the smallest affine separator certificate.

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